

Inheriting Hardware Without the System: A Sociotechnical Knowledge and Capability Transfer Framework for IOC Asset Divestments

by

Ikenna Ekekwe

B. Tech., Rivers State University of Science and Technology, 2008

Submitted to the System Design and Management Program
in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE IN ENGINEERING AND MANAGEMENT

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

September 2026

© 2026 Ikenna Ekekwe. All rights reserved.

The author hereby grants to MIT a nonexclusive, worldwide, irrevocable, royalty-free license to exercise any and all rights under copyright, including to reproduce, preserve, distribute and publicly display copies of the thesis, or release the thesis under an open-access license.

Authored by:	Ikenna Ekekwe System Design and Management Program August 7, 2026
Certified by:	George Lordos, PhD Research Scientist and Lecturer, System Design and Management Program and Department of Aeronautics and Astronautics Thesis Supervisor
Accepted by:	Joan Rubin Executive Director MIT System Design and Management Program

Inheriting Hardware Without the System: A Sociotechnical Knowledge and Capability Transfer Framework for IOC Asset Divestments

by

Ikenna Ekekwe

Submitted to the System Design and Management Program
on August 7, 2026 in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE IN ENGINEERING AND MANAGEMENT

ABSTRACT

When an international oil company divests an aging producing asset to an indigenous operator in Nigeria, the wells, facilities, and data transfer at completion, but the operating organization that held the asset within safe and productive bounds does not. Outcomes diverged sharply: some acquirers sustained production and integrity, others suffered failures, including an uncontrolled blowout. That divergence is most consistently explained by capability-related factors.

The evidence is six divestment cases in a discriminating-case design. The critical experiment is the OML 29 transfer to Aiteo, a well-funded operator that raised production and then could not contain a major well-control event; the remaining transfers provide cross-case validation. Each case is read on two outcome dimensions, production performance and operational integrity, against five competing explanations.

Capital constraints are rejected as a sufficient cause, fiscal regime effects on grounds of timing; regulatory inadequacy is an enabling condition, and political economy explains access and commercial survival but not operational integrity. Capability-related factors are the explanation most consistent with the combined pattern. A within-asset comparison on OML 30, where operatorship changed while the asset was held fixed and performance improved, strengthens that reading against an asset-quality account.

Such a divestment is a sociotechnical control failure that no voluntary transfer corrects. No party holds both the knowledge of what is lost and the incentive to transfer it, and much of the cost of failure falls on host communities and the state. The contribution is a Capability Assurance Framework, proposed as a mandatory condition on high-risk producing-asset transfers in which seller disclosure, buyer absorption, and regulator verification are mutually constraining, operationalized as six recommendations that re-specify instruments Nigeria already holds. The framework is warranted by the documented conditions rather than proven by implementation. It makes the operating capability a transfer requires observable, testable, and enforceable at the point of transfer.

Thesis supervisor: George Lordos, PhD

Title: Research Scientist and Lecturer, System Design and Management Program and
Department of Aeronautics and Astronautics

Contents

Title page	1
Abstract	3
List of Figures	9
List of Tables	11
List of Acronyms	13
1 Introduction	15
1.1 Background and Context	15
1.2 The Problem	16
1.3 Research Gap	19
1.4 Research Objectives	20
1.5 Research Questions	21
1.6 Research Approach	21
1.7 Thesis Structure	23
1.8 Personal Motivation	23
2 Theoretical Foundation	25
2.1 Introduction	25
2.2 The Anatomy of an Upstream Asset Transaction	26
2.3 Five Competing Explanations	28
2.3.1 Capital Constraints	28
2.3.2 Fiscal Regime Effects	29
2.3.3 Regulatory Inadequacy	30
2.3.4 Political Economy	31
2.3.5 Capability-Related Factors	32
2.4 Coordination Failure	35
2.5 Cross-Cutting Conditions	36
2.5.1 Information Asymmetry	36
2.5.2 Absorptive Capacity	37
2.6 Synthesis: Discrimination Logic and the Central Empirical Question	37
2.7 How This Analysis Advances the Literature	41

3	Research Methodology	43
3.1	Research Design	43
3.2	Case Selection and Justification	44
3.3	Evidence Architecture	46
3.4	Analytical Framework	48
3.5	Interview Protocol	50
3.6	Validity, Reliability, and Limitations	51
4	Empirical Analysis	55
4.1	Introduction	55
4.2	Outcome Differentiation	56
4.3	The Critical Experiment: Aiteo on OML 29	63
4.3.1	The Selective Failure Pattern	63
4.3.2	Fiscal Regime Effects	65
4.3.3	Regulatory Inadequacy	66
4.3.4	Political Economy at the Case Level	66
4.3.5	Coordination Failure	67
4.3.6	The Aiteo Verdict	67
4.4	Cross-Case Validation: Failure Replication and Regulatory Screening	70
4.4.1	Eroton on OML 18: A Partial Replication Complicated by a Shared Trunk Line	70
4.4.2	First Hydrocarbon on OML 26: A Test of Regulatory Screening	74
4.5	Theoretical Replication: Operators That Differed on Capability	75
4.5.1	Seplat on OMLs 4, 38, and 41: A Founding Team's Technical Depth	76
4.5.2	OML 30: A Within-Asset Operator Change	77
4.5.3	NPDC: A State Operator and the Limits of Political Access	78
4.6	The Marginal Field Program: A Structural Precedent	80
4.7	Cross-Case Synthesis and Discrimination	83
4.7.1	Consolidated Verdicts	83
4.7.2	Discrimination Among the Surviving Explanations	85
4.7.3	Substitutability and the Distribution of Failure Costs	88
4.8	Limitations and Unresolved Ambiguities	89
5	A Capability Assurance Framework for IOC Divestments	93
5.1	From Explanation to Design	93
5.2	The Embedded Nature of Operating Capability	95
5.3	The Market and Regulatory Failure to Transfer Capability	97
5.3.1	The Pre-Transfer Condition and Disclosure Gap	98
5.3.2	Information Asymmetry and Adverse Selection	98
5.3.3	The Buyer's Absorptive Capacity	99
5.3.4	The Externality	99
5.3.5	The Wrong-Variable Diagnosis	101
5.3.6	The Same Gap Across Acquisition Pathways	102
5.4	The Capability Assurance Framework	102
5.4.1	Three Roles in a Closed Loop	103

5.4.2	Where Each Obligation Attaches Across the Transaction	105
5.4.3	The Bundle, by Capability Domain	107
5.5	Calibration: Existing Nigerian Rules and International Models	110
5.5.1	The Domestic Foundation	110
5.5.2	Demonstrated Capability in International Practice	110
5.5.3	The Limits of Voluntary Standards	113
5.6	Proportionality and Structural Levers	113
5.6.1	Requirements for the Buyer's Organization	113
5.6.2	Scaling Requirements to Risk	114
5.6.3	The Structural Levers	117
5.7	Design Risks and the Limits of the Warrant	118
6	Conclusions	121
6.1	Answering the Research Questions	121
6.2	The Structural Case for Mandatory Intervention	123
6.3	Prioritized Recommendations	124
6.4	Limitations	127
6.5	Boundaries of Generalizability and Future Research	129
6.6	Contribution and Conclusion	130

Appendices

Appendix A: Case Study Protocol (COUHES Protocol E-7658)	133
A.1 Study Overview	133
A.2 Participant Groups and Sample Design	133
A.3 Informed Consent Procedures	134
A.4 Data Handling	134
A.5 Pre-specified Coding Categories	134
A.6 Pretest Design	135
Appendix B: Discrimination Matrix	137
Appendix C: Interview Guides	143
C.1 Group A Interview Guide (IOC Technical Staff and Management)	143
C.2 Group B Interview Guide (Domestic Oil Company Operators and Managers)	147
C.3 Group C Interview Guide (Regulatory Officials and Industry Observers)	151
Appendix D: Case-by-Case Evidence Table	155
D.1 Operator Evidence Inventory	156
D.1.1 Aiteo, OML 29	159
D.1.2 Eroton, OML 18	159
D.1.3 First Hydrocarbon, Afren, OML 26	160
D.1.4 Seplat, OMLs 4, 38, 41	160
D.1.5 Shoreline, Heritage Oil, OML 30	160
D.1.6 NPDC, NEPL Portfolio	161

D.1.7	Neconde, OML 42	161
D.1.8	Elcrest, OML 40	161
D.1.9	Renaissance, SPDC JV	161
D.1.10	Shebah FPSO	162
D.2	Marginal Field Program Evidence	162
D.2.1	Program Overview	162
D.2.2	Performance Outcomes	162
D.2.3	2020 Bid Round	163
D.2.4	Academic Evidence on Marginal Field Capability	163
Appendix E: Verdict Traceability Matrix		165
E.1	Aiteo (OML 29)	166
E.2	Eroton (OML 18)	168
E.3	First Hydrocarbon (OML 26)	170
E.4	Seplat (OMLs 4, 38, 41)	172
E.5	OML 30: Within-Asset Operator Change	174
E.6	NPDC	176
E.7	Marginal Field Program	178
E.8	Cross-References Not Scored as Separate Verdicts	180
Appendix F: Survey Response Data		181
Appendix G: Capability Element Traceability Matrix		187
G.1	How the Matrix Is Built	187
G.2	Value-Dimension Legend	188
G.3	Criticality Tiers of the Transfer-Process Elements	188
G.4	Pillar 1: Commit to Process Safety	190
G.5	Pillar 2: Understand Hazards and Risks	192
G.6	Pillar 3: Manage Risk	193
G.7	Pillar 4: Learn from Experience	196
G.8	Reading the Bridge Backward	199
References		201

List of Figures

2.1	Divestment failure equilibrium	41
4.1	Post-divestment production trajectories of the formal case...	63
5.1	The capability assurance loop	104
5.2	Breaking the failure equilibrium	105

List of Tables

2.1	Key concepts used in the discrimination analysis	34
2.2	The binding constraint and key prediction of each competing explanation . .	38
3.1	The decisive falsification condition and discriminating test for each explanation	49
3.2	Design-quality tests applied to the study, after Yin (2009)	52
4.1	Post-divestment outcomes across the six formal cases	59
4.2	Structural precedent and design contrasts beyond the formal cases	62
4.3	Verdicts for the Aiteo critical experiment on OML 29	68
4.4	Two transfer mechanisms sharing one capability-assessment gap	81
4.5	Consolidated verdicts on the five explanations across the cases	85
4.6	Buyer background against the two outcome dimensions	87
4.7	Where political access or capital can substitute for operating capability . . .	88
5.1	The value dimensions an operating-capability gap puts at risk	94
5.2	The four structural gaps behind an unassisted-divestment capability gap . .	100
5.3	The capability bundle by domain: what is disclosed, absorbed, and verified .	108
5.4	Capability as a condition of transfer across four petroleum regimes	112
5.5	Risk-tiered divestment assurance scaled to demonstrated capability	116
B.1	The discrimination matrix: evidence patterns and their strength-coded impli- cations	138
B.2	The five-step decision protocol for applying the discrimination matrix	141
D.1	Case-by-case evidence summary across the discrimination logic	156
E.1	Verdict matrix for the Aiteo case (OML 29)	166
E.2	Verdict matrix for the Eroton case (OML 18)	168
E.3	Verdict matrix for the First Hydrocarbon case (OML 26)	170
E.4	Verdict matrix for the Seplat case (OMLs 4, 38, 41)	172
E.5	Verdict matrix for the OML 30 within-asset operator change	174
E.6	Verdict matrix for the NPDC case	176
E.7	Verdict matrix for the Marginal Field Program	178
F.1	Profile of the three completed survey respondents	182
F.2	Driver rankings by the three completed respondents	183
F.3	Counterfactual, policy, and overall-adequacy survey responses	184

G.1	Criticality tiers for the transfer-process elements	188
G.2	RBPS Pillar 1 (Commit to Process Safety): elements and transfer factors . .	190
G.3	RBPS Pillar 2 (Understand Hazards and Risks): elements and transfer factors	192
G.4	RBPS Pillar 3 (Manage Risk): elements and transfer factors	193
G.5	RBPS Pillar 4 (Learn from Experience): elements and transfer factors	196

List of Acronyms

Acronym	Expansion
AER	Alberta Energy Regulator
COUHES	Committee on the Use of Humans as Experimental Subjects
DOC	domestic operating company
DPR	Department of Petroleum Resources
HEOSL	Heritage Energy Operational Services Limited
IOC	international oil company
MPNU	Mobil Producing Nigeria Unlimited
NAOC	Nigerian Agip Oil Company
NCTL	Nembe Creek Trunk Line
NEITI	Nigeria Extractive Industries Transparency Initiative
NNPC	Nigerian National Petroleum Corporation
NOGICD	Nigerian Oil and Gas Industry Content Development Act
NOSDRA	National Oil Spill Detection and Response Agency
NPDC	Nigerian Petroleum Development Company
NSTA	North Sea Transition Authority
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Oil Mining Lease
OPEC	Organization of the Petroleum Exporting Countries
PIA	Petroleum Industry Act
SPDC	Shell Petroleum Development Company of Nigeria
bpd	barrels per day

Chapter 1

Introduction

1.1 Background and Context

Nigeria is Africa's largest oil producer. The country has participated in global petroleum markets since crude oil was first discovered in commercial quantities at Oloibiri, Bayelsa State, in 1956. It joined the Organization of the Petroleum Exporting Countries (OPEC) in 1971, and by 2005 had reached an all-time production high of approximately 2.5 million barrels per day (OPEC, 2006). For over six decades, international oil companies, principally Shell, ExxonMobil, Chevron, Total, and Eni, built and operated the upstream infrastructure that sustained this output: wells, flowstations, gathering systems, trunk lines, and export terminals, all spread across the onshore and shallow-water Niger Delta. In deeper waters offshore, the same companies operated floating production, storage, and offloading vessels under a separate contractual agreement with the Nigerian Government.

These onshore and shallow-water operations were organized under joint venture agreements in which the Nigerian National Petroleum Corporation (NNPC) held the majority participating interest, typically 55 to 60 percent, while the international oil company held the remainder and served as technical operator. Deepwater operations were structured differently, under Production Sharing Contracts with distinct fiscal terms, and are not part of the divestment wave examined in this thesis. Under the joint venture arrangement, the IOCs supplied the engineering personnel, the management systems, the health, safety, and environment protocols, and the organizational processes used to explore, produce, and export crude oil. That operating environment was marked by mangrove terrain, armed militancy, crude oil theft, pipeline vandalism, and overlapping community, state, and federal jurisdictional claims. The physical assets belonged to the joint venture. Whether and how the operational capability required to run those assets would transfer to a new operator when the IOCs departed was a question that the institutional design of the sector left unaddressed.

From around 2010 onward, and at an accelerating pace through 2024, the major IOCs initiated a systematic program of divesting their onshore and shallow-water assets in Nigeria. Shell, the operator of the SPDC joint venture in which the NNPC held the majority interest alongside Total and Eni (Agip) as co-venturers on several blocks, sold participating interests in Oil Mining Leases 4, 18, 24, 25, 29, 30, 38, 40, 41, and 42. Shell then sold its entire onshore subsidiary (Shell Petroleum Development Company of Nigeria, SPDC) to a consortium called

Renaissance in 2024. ExxonMobil sold Mobil Producing Nigeria to Seplat Energy. Total and Eni entered similar divestment processes. The buyers were predominantly Nigerian domestic oil companies (also referred to as indigenous oil companies in Nigerian legislation and industry practice). This pattern was driven in part by the Nigerian Oil and Gas Industry Content Development Act (NOGICD) of 2010, which legislated preference for indigenous firms in all upstream asset transactions (NOGICD, 2010). In 2021, the Petroleum Industry Act (PIA) restructured the fiscal and regulatory architecture of the upstream sector and established the Nigerian Upstream Petroleum Regulatory Commission (NUPRC), replacing the former Department of Petroleum Resources (DPR), as the lead regulatory agency. However, neither piece of legislation addressed the question of whether operational capability would survive these transactions.

The drivers of this divestment wave were well documented (SDN, 2022; Gaughran et al., 2024). The IOCs cited the rising costs of onshore operations, the security threats to personnel and infrastructure, the escalation of oil theft and pipeline vandalism. They also cited the increasing likelihood of litigation for environmental pollution, and the global pressure to decarbonize and shift investment toward deepwater and liquefied natural gas. These factors made the Nigerian onshore portfolio increasingly unattractive relative to the IOCs' global alternatives.

1.2 The Problem

Between 2010 and 2024, at least 26 Oil Mining Leases changed hands from international to domestic operators, representing transactions valued at over \$21 billion. For the first time in the history of Nigeria's petroleum industry, domestic companies controlled a larger share of onshore production than international operators.

The Nigeria Extractive Industries Transparency Initiative (NEITI) announced in 2024 that it would conduct a review of all 26 divestments (NEITI, 2024).

The operational outcomes of these transactions, measured along two dimensions, production performance and operational integrity (the latter spanning safety records, environmental compliance, and organizational sustainability), were not uniform. They were sharply differentiated, and the pattern of differentiation is the empirical puzzle that motivates this thesis.

Some acquirers experienced significant operational failures. Aiteo Eastern Exploration and Production acquired a 45 percent interest in OML 29 from the SPDC joint venture in 2015 for \$1.7 billion, or \$2.562 billion including the Nembe Creek Trunk Line, and grew production substantially in its first two years. In November 2021, an uncontrolled blowout occurred at the Santa Barbara Well 1, a well that had been non-producing since 2015 and was neither decommissioned by Shell before the sale nor by Aiteo after it. The blowout spilled an estimated 200,000 barrels over 33 days and affected 41 communities (SDN, 2022). The National Oil Spill Detection and Response Agency (NOSDRA) stated that the incident exceeded Aiteo's internal capacity (NOSDRA, 2021). The well had experienced prior spill incidents in 2018 and 2019, neither of which had been remediated.

Eroton Exploration and Production acquired OML 18 from the SPDC joint venture in 2015, grew production substantially in its early years, and was recognized by the Nigerian

National Petroleum Corporation as one of two operators with the lowest technical cost per barrel (Wood Mackenzie, 2026). By the fourth quarter of 2021, Eroton's production had fallen to zero, driven by the unavailability of the Nembe Creek Trunk Line (operated by Aiteo on OML 29) and by internal governance failures that a subsequent NNPC forensic audit identified (Nairametrics, 2023). The JV funding had been maintained throughout the period.

The case of OML 26 exposed a different failure mode. First Hydrocarbon Nigeria, a special purpose vehicle created by Afren Plc, a FTSE 250 company valued at \$2.6 billion at its peak, acquired a 45 percent interest from the SPDC joint venture in 2011. Afren's CEO and COO were subsequently convicted by the UK Serious Fraud Office for fraud and money laundering (UK Serious Fraud Office, 2017). Afren entered administration in July 2015. No regulatory mechanism had assessed the fitness of the buyer at the point of transfer.

These were not isolated incidents. In 2022, a nearly 50-year-old floating production vessel operated by Shebah Exploration and Production exploded, killing at least 10 crew members; most major trading companies had reportedly abandoned the vessel years earlier. The Stakeholder Democracy Network cited the incident as part of a broader concern that some domestic operators were running aging infrastructure with weak integrity controls (SDN, 2022).

Other acquirers achieved markedly different results. Seplat Energy acquired OMLs 4, 38, and 41 from the SPDC joint venture in 2010. It sustained production growth across its portfolio, nearly doubled its reserves, accumulated two million man-hours without a recordable safety incident, and won the AfriSAFE Energy Company of the Year award in 2024 (Seplat Energy, 2024). Seplat's founding team brought decades of industry experience, its partnership with Maurel et Prom (a French exploration and production company) provided international technical capacity, and its dual listing on the London and Nigerian Stock Exchanges imposed governance discipline. The reviewed record shows no structured capability-transfer provisions in the 2010 transaction.

Shoreline Natural Resources acquired a 45 percent interest in OML 30 from the SPDC joint venture in 2012. The Nigerian Petroleum Development Company (NPDC), the government's own exploration and production subsidiary, held 55 percent and initially operated the asset. In 2017, the government approved the transfer of operatorship to Heritage Energy Operational Services Limited (HEOSL), the technical partner. HEOSL embarked on a rehabilitation program that included pressure vessel inspection, wellhead and pipeline inspection, gas compressor installation, and well optimization. Gross production recovered under the new operator. The same asset, under the same regulatory regime and fiscal environment, produced different results under different operators.

In 2024, a consortium called Renaissance agreed to acquire 100 percent of Shell's onshore subsidiary for \$2.4 billion, a transaction that completed in 2025 (Shell, 2024). Renaissance's leadership, comprised of former Shell staff (Shell, 2024; BusinessDay, 2025), had studied why previous acquirers of IOC assets had seen production decline after acquisition and had deliberately chosen staff retention over the typical approach of replacing IOC workers (BusinessDay, 2025). Shell committed to preserving the full range of SPDC's operating capabilities, including technical expertise, management systems, and processes (Shell, 2024). By April 2025, Renaissance reported production above 200,000 barrels per day with pipelines operational, an early continuity signal rather than proof of medium-term integrity. The Africa Oil and Gas Report noted that this capability-preservation commitment was the first

of its kind in 15 years of IOC divestments in Nigeria (Africa Oil and Gas Report, 2024).

The government's own entity followed a different trajectory. NPDC, holding 55 percent equity in eight or more Shell-divested OMLs, targeted production growth from 180,000 to 500,000 barrels per day by 2020 (Vanguard, 2017). It did not achieve the target. An academic study measured NPDC's technical efficiency at 23 percent, the lowest of all operator categories (Adekunle et al., 2019). In July 2020, an explosion at the Benin River Valve Station on OML 40, operated by NPDC, killed seven contractor workers during a ladder installation (NNPC, 2020).

This pattern of differentiated outcomes was not new. A decade before the IOC divestment wave, the Federal Government of Nigeria had conducted a marginal field bid round in 2003, awarding 24 fields to 31 indigenous companies with the stated objective of building indigenous technical capacity in the upstream sector (DPR, 2003). By the early 2020s, many of the awarded fields had still not reached production, and eleven operators had lost their licenses for failure to meet government criteria (NUPRC, 2022). The program contributed just two percent to national oil and gas reserves (Sweet Crude Reports, 2020). The Department of Petroleum Resources acknowledged that Nigeria had not benefited from the 2003 round (Sweet Crude Reports, 2020). The marginal field experience constitutes a decade-old precedent for the same structural condition that characterized the subsequent IOC divestments. The comparable failure rate across two different transfer mechanisms (government licensing and bilateral IOC sale) suggests that the pattern is institutional rather than peculiar to any single transaction type. That this precedent did not produce institutional learning, despite the marginal field record, is itself evidence of the design gap examined here.

At the national level, total crude oil production across all operators declined from its all-time high of approximately 2.5 million barrels per day in 2005 to a low near one million barrels per day in 2022, recovering to roughly 1.5 million by 2024, still short of the 2 million target (OPEC, 2022; NUPRC, 2024). This national total is distinct from the divested-operator subset this thesis examines, whose aggregate output rose even as the national figure fell. The country repeatedly failed to meet its OPEC production quota. At the sector level, NEITI reported outstanding government receivables from oil and gas operations exceeding \$6 billion (NEITI, 2024). Population-level data showed that indigenous operators recorded 35 percent more oil spills than IOCs (Stakeholder Democracy Network, 2022), flared ten times more gas per barrel of oil produced (Stakeholder Democracy Network, 2022). They scored 44 percent on safety awareness assessments compared with higher scores for IOC-operated facilities (Anumadu et al., 2014). A peer-reviewed quantitative study of 314 Nigerian oilfields using panel data spanning 2006 to 2016 found that divestment to indigenous operators was associated with a 33 percent increase in oil production. The same divestment brought 10 percent more oil spills attributable to operational failure and 41 percent more gas flaring (Rexer, 2025).

The costs of these operational failures did not fall on the transacting parties. The IOCs had exited. The acquiring companies that failed bore financial losses, but the environmental damage, the safety incidents and fatalities, and the lost production revenue were absorbed by host communities and the Nigerian state (SDN, 2022; Gaughran et al., 2024; SOMO, 2024; Steiner, 2023). The costs landed on the 41 communities affected by the Aiteo spill, the families of the workers killed at the Benin River Valve Station and on the Shebah FPSO, and the Federation budget that lost billions in unrealized production revenue. None of these

groups stood inside the divestment transactions that produced these outcomes.

The pattern is not simply that some operators failed and others succeeded. It is that the failures appeared where operating capability had to be exercised, not merely where assets had to be owned. The coexistence of rising production and declining operational quality across the oilfield population, the sharp divergence in outcomes among individual operators, the marginal field precedent, and the national production decline together constitute an empirical pattern. That pattern raises a central question: what explains this systematic differentiation in post-divestment outcomes? The answer has direct policy consequences: it determines whether the appropriate intervention is improved financing, fiscal reform, regulatory enforcement, institutional redesign of the divestment process, or some combination of these. The competing explanations have not been systematically tested against the available evidence.

1.3 Research Gap

Several explanations have been proposed for post-divestment operational challenges in Nigeria's upstream sector, but none has been systematically tested against the full range of observed outcomes. The pattern they must account for is demanding: well-funded operators such as Aiteo and Eroton grew production after acquisition and then suffered severe operational failures. A \$2.6 billion company, Afren, collapsed entirely, and operators working under the same regulatory and fiscal terms diverged sharply in outcome.

Five candidate explanations recur in the literature and the public record: capital constraints, fiscal regime effects from the Petroleum Industry Act of 2021, weak regulatory enforcement, political economy, and capability-related factors. This last factor is the operational knowledge embedded in decades of IOC practice that may not transfer with the physical assets. The theoretical foundation (Chapter 2) develops each one, with its testable predictions and the conditions under which the evidence would reject it. Two features of the existing accounts define the gap this thesis addresses: none has been tested head-to-head against case-level evidence in a design built to discriminate among them, and none has been paired with an institutional remedy for the failure it identifies.

Every unsuccessful divestment involves coordination failures of some kind: poorly managed transitions, delayed contractor mobilization, inadequate interface management. Coordination failure is not a sixth competing root-cause explanation. Coordination failure is an observable outcome across all unsuccessful cases and therefore does not discriminate between explanations on its own. The analytical question is not whether coordination failed but why it failed. It failed either because the operator lacked financial resources to coordinate effectively (capital constraints), because the regulatory framework did not require coordination (regulatory inadequacy), or because the knowledge being coordinated did not exist on the buyer's side regardless of coordination quality (capability-related factors). The empirical analysis evaluates the evidence on this distinction through a persistence test: coordination failures that are independently caused should self-correct as the operator gains experience, while failures rooted in absent knowledge should persist or worsen over time. The test is indicative rather than decisive, because organizational learning curves vary and the timeline for self-correction is not universal. Persistence weakens coordination failure as an independent explanation but does not eliminate it; late-stage governance decay or operator complacency could also

account for the persistence pattern.

The closest existing study to this thesis is Rexer (2025), whose quantitative analysis of 314 oilfields in the American Economic Review establishes that divestment to indigenous operators is associated with measurable declines in operational quality even as production increases. However, Rexer’s study measures the aggregate statistical effect without testing the competing explanations against case-level evidence and without proposing an institutional framework to address the pattern. The inter-organizational knowledge transfer literature addresses facility handovers in general terms (Ferreira et al., 2024), and the absorptive capacity literature addresses the conditions under which receiving organizations can assimilate new knowledge (Cohen and Levinthal, 1990). Neither body of work has been applied to the specific institutional context of IOC-to-DOC asset transfers in a petroleum-producing developing country.

The evidence base for the study consists of two tiers. Observable outcomes, documentary records, regulatory framework texts, corporate disclosures, court filings, production data, spill records, and independently compiled datasets constitute the primary evidence layer. These sources are sufficient to evaluate competing explanations at the pattern level, including falsification of inconsistent explanations and identification of the most consistent explanation. Semi-structured interviews with industry participants, where obtainable, constitute a second tier of evidence that provides additional insight into mechanisms that are not fully observable from documentary sources. Specifically, participant testimony can upgrade the analysis in two ways: it can resolve whether coordination failure in a given case was independently caused or structurally produced (a distinction that documentary evidence can narrow but not always close). It can also trace the specific channels through which capability-related factors operated, a level of mechanism detail that strengthens the finding. Without interviews, the verdicts stand at the pattern level: which explanations are falsified, which are consistent, and which is most consistent. With interviews, specific verdicts can be upgraded from “consistent with the evidence” to “supported by converging testimony and documentary evidence,” and can specify the mechanism through which the most consistent explanation operated. The analytical conclusions do not change; the confidence level and mechanism resolution increase.

This investigation asks whether the differentiated outcomes reflect a system-theoretic failure in the institutional design of the divestment process, including but not limited to the absence of mandatory mechanisms for sociotechnical capability transfer. It also asks whether such a systems-perspective explanation accounts for the observed pattern more fully than the alternatives. The evidence may instead support one of the competing explanations, a combination of factors, or a mechanism not yet identified. This thesis goes beyond Rexer’s quantitative finding by examining the mechanisms at the operator level through case analysis and, if the findings warrant it, by proposing a mitigation framework grounded in the empirical results.

1.4 Research Objectives

This thesis pursues three objectives:

First, to test competing explanations (capital constraints, fiscal regime effects, regulatory inadequacy, political economy, and capability-related factors) for differentiated post-

divestment outcomes against observable evidence drawn from multiple operator cases. The test determines which are contradicted by the evidence, which are insufficient on their own, and which are most consistent with the observed pattern.

Second, to investigate what primarily drove the differentiated operational outcomes observed following IOC asset divestments in Nigeria’s upstream petroleum sector between 2010 and 2024. This examines whether the binding constraint is best located in capability-related factors, in another of the proposed explanations, or in an interaction among them at the level of the divestment as a sociotechnical system.

Third, to develop, if the findings indicate that institutional design plays a significant role in shaping outcomes, a framework specifying obligations for the seller, readiness requirements for the buyer, and verification responsibilities for the regulator, calibrated to the institutional context of petroleum-producing developing countries.

1.5 Research Questions

The following research questions guide the investigation:

1. What explains the differentiated operational outcomes observed across acquiring operators following IOC asset divestments in Nigeria, and which competing explanations are most consistent with the observable evidence?
2. Which of the competing explanations, capital constraints, fiscal regime effects, regulatory inadequacy, political economy, and capability-related factors, does the evidence contradict, which are insufficient on their own, and which are jointly necessary?
3. If the findings indicate that institutional design plays a significant role in shaping outcomes, what framework design would address the identified deficiencies, and how should responsibilities be allocated across the seller, the buyer, and the regulator?

Each question admits genuine alternative answers. The first could be answered by any of the competing explanations or by a combination. Where explanations cannot be distinguished, the thesis reports the ambiguity. The second could find that any single explanation is sufficient, that several are individually insufficient but jointly necessary, or that the evidence contradicts explanations that initially appeared plausible. The third could produce a voluntary rather than mandatory framework, or no framework at all, if the evidence showed that market forces or existing regulation were sufficient. Each possibility is tested against the evidence.

1.6 Research Approach

The study uses a critical experiment case study design with cross-case validation, designed to be supplemented, where obtainable, by participant testimony through interviews or structured surveys, and grounded in sociotechnical systems theory.

The critical experiment is conducted on Aiteo’s acquisition and operation of OML 29. This case was selected because its observable trajectory, which includes both sustained production

growth and a subsequent prolonged crisis failure, creates conditions for discriminating between the competing explanations. Those explanations generate distinct predictions about whether and how these two outcomes can co-occur in the same operator over the same period. The specific predictions and their confrontation with the evidence are presented in the empirical analysis. The case was selected on its analytical discriminating power between theories, not on personal familiarity or evidence availability.

Cross-case validation extends the critical experiment’s findings to additional operator outcomes using replication logic (Yin, 2009). Literal replications test whether the pattern established by the Aiteo case holds in other cases with similar characteristics: Eroton (production growth followed by operational collapse despite maintained funding) and First Hydrocarbon (acquisition by a buyer whose fitness was never assessed). Theoretical replications test whether outcomes differ in cases where the conditions are different. These are Seplat (buyer with prior industry experience and international technical partnership), Shoreline/HEOSL (indigenous technical operator taking over from NPDC on the same asset), and NPDC (government entity with political connections but limited operational capability, testing the pattern beyond commercial acquirers). The Shoreline/HEOSL case on OML 30 provides a within-OML comparison that holds regulatory and fiscal variables constant, isolating operator-level factors as the source of variation.

The marginal field program (2003 to 2020) is a parallel case: an earlier instance of a similar institutional pattern, occurring through a different mechanism (government licensing rather than bilateral IOC sale) but producing a comparable outcome (majority failure among operators awarded upstream assets).

The Renaissance transaction (2024) provides a design contrast, not a formal case analyzed through the full decision sequence. Its analytical role is specific: it demonstrates what a capability-preserving transaction design looks like in practice, providing a natural contrast with earlier transactions where no such structure existed. Because the Renaissance deal closed recently and its long-term operational trajectory is not yet established, it is not equivalent evidence to the primary and validation cases. It is examined as a real-world instance of the proposed framework design.

Semi-structured interviews with industry participants span three groups: IOC technical staff involved in pre-divestment operations, DOC operators and managers who took over the assets, and regulatory officials responsible for divestment oversight. The interview protocol received a human-subjects exemption (COUHES Protocol E-7658); the full research design is specified in Chapter 3.

A limitation inherent in this research design is that the internal organizational decisions made during and after divestment transactions are not fully observable from outside the organization. The study relies primarily on inference from observable indicators, with participant testimony used where obtainable to contextualize and triangulate documentary findings. These observability constraints, and the validity safeguards employed to manage them, are addressed in Chapter 3.

The theoretical grounding draws on three bodies of literature: sociotechnical systems theory, which holds that operational knowledge is embedded in the interaction between people, processes, and organizational structures (Baxter and Sommerville, 2011). The second is organizational capability as combinative knowledge (Kogut and Zander, 1992). The third is the absorptive capacity framework, which posits that the ability to recognize, assimilate,

and apply new knowledge depends on prior related knowledge (Cohen and Levinthal, 1990). The methodology provides the full justification for the research design.

1.7 Thesis Structure

The thesis is organized in six chapters, each building on the preceding chapter and enabling the analysis that follows.

This chapter has established the context of the Nigerian IOC divestment wave, documented the differentiated outcomes that constitute the empirical puzzle, identified the competing explanations that have been proposed, and stated the research objectives, questions, and approach.

The analytical framework is constructed (Chapter 2): operational capability in upstream petroleum is defined, the components that transfer through standard commercial terms are distinguished from those that do not, and the five competing explanations are cast as testable predictions with falsification conditions. Coordination failure is treated separately, observable in every unsuccessful case but not discriminating among the explanations on its own, and two cross-cutting conditions, information asymmetry and absorptive capacity, shape how the explanations interact.

The research methodology is set out (Chapter 3): the critical-experiment design, the case-selection criteria, the cross-case validation logic, the evidence tiers ordered by observability, and the validity and reliability safeguards, with the methodological limits stated directly.

The empirical analysis is carried out (Chapter 4): the critical experiment is run on Aiteo and OML 29, each explanation is tested against the documentary evidence, and the pattern is examined across the replication cases and the marginal-field program. The data pattern expected under each explanation is specified in advance and checked against what the record shows.

The policy implications are drawn in Chapter 5. If capability-related factors are found to be the explanation most consistent with the evidence, a three-actor capability assurance framework follows: disclosure obligations for the seller, readiness requirements for the buyer, and verification responsibilities for the regulator. Each component is grounded in a documented failure pattern and calibrated against the NUPRC's 2024 framework, the Renaissance design, and the UK North Sea Transition Authority's capability requirement.

The conclusions, limitations, and directions for future research are presented (Chapter 6), calibrated to what the evidence supports and bounded in their generalizability.

1.8 Personal Motivation

I am an Engineering Manager at Chevron Nigeria with direct professional experience in upstream operations in the Niger Delta. This proximity to the phenomenon under investigation provides domain expertise in the technical, organizational, and institutional dimensions of IOC operations that would be difficult to acquire through secondary research alone. It also creates an obligation of transparency: the research design, case selection, and analytical

conclusions in this thesis stand on their own merits, independent of my professional position or employer.

I was motivated to pursue this thesis by observing the consequences of the divestment wave at close range. I watched acquiring companies struggle with well integrity management systems their technical teams had not been trained on. I saw environmental incidents follow transfers where the buyer had inherited the hardware but not the institutional knowledge to maintain it. And I recognized that no mechanism existed to ensure that the sociotechnical capability embedded in decades of IOC operations would survive the transition. The System Design and Management program at MIT, with its emphasis on sociotechnical systems analysis and the interaction between organizational design and engineering outcomes, provided the lens to investigate these observations as a question of institutional design rather than individual operator performance.

This thesis is not sponsored by, reviewed by, or subject to approval by Chevron or any other oil company. The views expressed are my own.

Chapter 2

Theoretical Foundation

2.1 Introduction

The transfer of upstream petroleum assets from international to domestic operators in Nigeria between 2010 and 2024 produced sharply differentiated outcomes. While some acquirers grew production and maintained operational integrity, others experienced severe safety failures, environmental disasters, or in some cases, organizational collapse. Across the divested operators, aggregate production rose marginally while operational quality declined. Why the split? Five explanations have been proposed, but none has been tested against the others using the full range of observed outcomes.

The analytical framework developed here tests whether the differentiated outcomes reflect a system-theoretic failure in the institutional design of the divestment process, or whether simpler, single-factor explanations account for them. Five competing explanations are examined: capital constraints, fiscal regime effects, regulatory inadequacy, political economy, and capability-related factors. Each is grounded in a distinct intellectual tradition, generates testable predictions about what should be observable in the evidence, and has conditions under which the evidence would reject it.

The five explanations enter the analysis with the same structure and falsification conditions. The evidence determines which explanation carries the most weight. The empirical investigation may find that several explanations are jointly operative, each accounting for a different dimension of the observed pattern, or that one explanation is most consistent with the evidence while others capture secondary effects. Where the evidence does not permit discrimination between two explanations, the analysis reports the ambiguity.

Coordination failure appears in every unsuccessful case, but the analytical question is whether it is an independent cause or a consequence of the deeper conditions the five explanations describe. Two cross-cutting conditions modify how the five explanations interact: information asymmetry, which explains why the capability gap persists without market self-correction, and absorptive capacity, which conditions whether a buyer can embed whatever capability is made available.

2.2 The Anatomy of an Upstream Asset Transaction

A divestment transaction transfers some components of an upstream petroleum operation and not others. The components that transfer and those that do not are where the five explanations disagree.

A typical IOC-operated onshore or shallow-water asset in the Niger Delta comprises wells drilled into multiple reservoir formations, connected by flowlines to gathering stations where crude oil is separated from water and gas. The crude is routed through trunk lines to export terminals, and loaded onto vessels for shipment. The physical infrastructure is substantial: wellheads, flowline networks, separators, compressors, tank farms, loading buoys, and associated electrical and instrumentation systems. Operating an asset of this scale requires sustained expenditure on maintenance, well intervention, corrosion management, community relations, security, and logistics.

What changes hands in a divestment transaction falls into three categories, and the five explanations disagree on which category matters most. Physical assets and legal rights transfer through the sale agreement: wells, flowlines, terminals, licenses, contracts, and associated infrastructure. Documented knowledge transfers through the data room: geological surveys, production records, engineering standards, maintenance logs, equipment specifications, and regulatory correspondence. Embedded organizational knowledge resides in forms whose transferability is contested: tacit operational practices accumulated through decades of operating specific equipment under specific conditions. These forms include crisis response judgment developed through experience with well control events and pipeline ruptures, and community relationships built through years of engagement with specific host communities. It also resides in organizational routines through which individual expertise is integrated into collective capability.

This third category is what distinguishes an operating asset from its hardware. A producing upstream asset is a sociotechnical system: the physical facilities and the operating organization that runs them form one working whole, and safe, productive operation is an emergent property of the coupling between them rather than a property of the equipment alone. The founding studies in this tradition showed that reorganizing the social arrangement of work degraded output even when the technology was unchanged (Trist and Bamforth, 1951). The same principle carries into contemporary systems engineering, where the social and technical elements of a system are designed and sustained together rather than separately (Baxter and Sommerville, 2011). The operational boundary of the asset therefore extends past the licensed hardware to the operating organization through which that hardware is run. A transaction that conveys the wells, the facilities, and the data room while leaving the operating organization behind divides the system along the interface between its technical and social subsystems, conveying the first and separating it from the second. What the transaction does not convey is the capability that emerged from that coupling. This thesis refers to that emergent capability as operating capability: the domain-specific knowledge of how a particular asset is run safely and productively. That knowledge is embedded in the operating workforce, the operating routines, the integrity and emergency systems, the structure of decision rights, the asset's data and history, and the contractor and technical-support architecture. It is specified element by element in Section 5.2. Whether an acquiring

operator can reconstitute it, and within what period, is the question on which the competing explanations divide and which the evidence must settle.

The knowledge-based view of the firm explains why the social subsystem resists transfer through an arm's-length sale. The coordination mechanisms through which specialist knowledge is integrated into productive activity are specific to the organization that developed them (Grant, 1996; Kogut and Zander, 1992). Knowledge ranges from fully explicit to fully tacit, and the tacit portion resists codification (Nonaka, 1994). Organizational routines encode accumulated experience in patterns that guide behavior under both routine and non-routine conditions (Nelson and Winter, 1982). Multinational corporations act on this directly: they operate their foreign assets through wholly owned subsidiaries rather than licensing their technology, because the knowledge required to run those assets is too context-specific to move through arm's-length mechanisms (Kogut and Zander, 1993). A divestment is the arm's-length mechanism that this organizational form exists to avoid.

A quantitative analysis of 65 Nigerian oil and gas projects found that subsurface complexity and technology challenges were not the major causes of poor performance. Non-technical factors, including local content, community relations, security, and partnership dynamics, played a more significant role (Rui et al., 2018). If non-technical factors dominate performance outcomes in Nigerian upstream operations, then a concept of what transfers during a divestment that includes only physical assets and documented knowledge is incomplete.

This view of what transfers has a direct precedent in the enterprise architecture literature, which treats the organizational architecture as the enterprise that delivers value and the physical plant as only one element within it (Nightingale and Rhodes, 2015). On that reading, a divestment that conveys the plant while leaving the organizational architecture behind moves the hardware and strands the system that runs it. This is the same separation the sociotechnical tradition draws between a technical and a social subsystem (Trist and Bamforth, 1951). The process safety literature gives this abstraction operational content: the risk-based process safety framework specifies the management systems, competency programs, and operating discipline through which an organization holds a hazardous facility within safe bounds (CCPS, 2007). The inter-organizational transfer literature explains which of these elements degrade under which transfer conditions (Ferreira et al., 2024). The decommissioning literature closes the loop on cost: where an operator lacks the means to sustain these systems, the liabilities of late-life remediation and abandonment settle on host communities and the state ahead of the transacting parties (Stakeholder Democracy Network, 2022).

The five competing explanations disagree on which of these three categories is the binding constraint on post-divestment performance. Capital constraints treat physical assets plus financial resources as sufficient: a buyer with adequate funding can hire engineers, purchase training, and rebuild capability from documented sources. Fiscal regime effects point to the external fiscal environment, which degrades returns and makes previously viable investments uneconomic. Regulatory inadequacy locates the problem in the institutional framework, which does not assess whether embedded organizational knowledge transfers, permitting operators without it to acquire assets. Political economy argues that political connections can substitute for some components of embedded knowledge, particularly security management and community relations, though not for others. Capability-related factors predict that embedded organizational knowledge is non-substitutable within operationally relevant timeframes, and that its absence determines the quality outcome. Each is an empirical claim that the evidence

must evaluate.

2.3 Five Competing Explanations

Each explanation that follows generates testable predictions, specifies observable evidence that would test them, and has conditions under which the evidence would reject it.

2.3.1 Capital Constraints

Indigenous operators may fail because they lack the financial resources to maintain the operational standard the IOC sustained. If this explanation is correct, financial capacity should correlate with operational outcome: better-funded operators should perform better, and operators with comparable financial resources should not diverge sharply on operational quality. Failure should manifest as progressive, cumulative degradation as the operator defers expenditures it cannot afford. If an operator fails suddenly in one domain while routine operations continue successfully, that pattern is inconsistent with capital constraints.

The mechanism producing this degradation is prioritized deferral. Operators with insufficient capital cut expenditures whose absence does not produce immediate visible consequences (HSE training, emergency preparedness, corrosion management, environmental monitoring) while maintaining expenditures whose absence produces immediate production loss (well intervention, flowline repairs, export terminal operations). Deferred maintenance accumulates, equipment degrades, contractor relationships deteriorate through payment delays, and qualified personnel leave for better-funded operators.

Resource dependency theory grounds this explanation: organizations depend on their external environment for critical resources, and the inability to secure those resources constrains organizational behavior (Pfeffer and Salancik, 1978). The pecking order theory of capital structure adds a prediction about the form the constraint takes: firms prefer internal financing over external debt, and external debt over equity, because information asymmetry between managers and external investors makes external financing more expensive (Myers and Majluf, 1984). For Nigerian indigenous operators without substantial retained earnings, the pecking order pushes them toward expensive debt, which constrains the operational investments the asset requires. Research on the capital structure of producing marginal fields in Nigeria found that both long-term and short-term debt had an inverse and significant impact on operator performance (Spiff et al., 2022). While Spiff et al. studied marginal fields (smaller, lower-capitalization operators), the direction of effect (debt constrains performance) is generalizable, though the magnitude may differ for the full-scale OMLs examined here.

The test therefore looks to the operator’s financing structure at acquisition (deal records, corporate disclosures), production trajectory after acquisition (growth or decline), JV funding status (audit statements), and the sector-wide cost environment. Data presented at the NAPE pre-conference workshop show that Nigeria’s operational expenditure is approximately 40 percent higher than comparable basins, with 30 to 50 percent capital project overruns, establishing that cost pressure is a system-level condition affecting all operators (Guardian Nigeria, November 2025).

Two conditions would reject capital constraints as sufficient. An operator that demonstrably funded sustained production growth but subsequently failed in a domain requiring organizational knowledge rather than capital expenditure would be inconsistent with capital as the binding constraint, because funding production growth demonstrates capital adequacy for major operational spending. An operator whose JV funding was maintained throughout a period of operational decline, confirmed by independent audit, would also reject the explanation for that case. A weaker but relevant condition is two operators with comparable financial resources producing divergent outcomes, which would weaken the prediction that capital determines performance.

High leverage may create financial fragility that constrains operational investment even when total capital appears adequate. The test is whether capital constraints, given observed funding levels and sector-wide cost conditions, best explain the pattern of variation across operators. If well-funded operators fail in specific domains while underfunded operators succeed, capital constraints cannot be the primary differentiating factor.

Capital may have been adequate in aggregate but deliberately sub-allocated away from emergency preparedness toward production growth. If the record indicates that the operator chose to underfund crisis readiness despite having resources, this is a management decision, not a problem of capital constraints. Capital constraints as an explanation require that the operator lacked the resources to fund both production and crisis preparedness simultaneously, not that the operator chose one over the other.

2.3.2 Fiscal Regime Effects

The Petroleum Industry Act, enacted on August 16, 2021, restructured Nigeria’s upstream fiscal terms. The temporal relationship between the PIA and the observed failures is the first analytical constraint: the primary transactions examined here closed between 2010 and 2015, and the marginal field program failures between 2003 and 2020 predate the PIA entirely. The cause cannot postdate the effect.

Where the PIA could have effects on future transactions, the mechanism operates through the same deferral pathway as capital constraints but is triggered by external fiscal conditions rather than operator-specific financial weakness. Changed fiscal terms reduce net revenue per barrel, making previously viable operational investments uneconomic, and producing cumulative reduction in operational quality.

Fiscal regime design has measurable effects on operator behavior. Comparative analysis has found that Nigeria’s fiscal terms under the PIA are regressive relative to comparable producing countries (Mohamed et al., 2024). The PIA reduced headline tax rates from 85 percent to 45 percent (comprising Hydrocarbon Tax of 15 percent and Company Income Tax of 30 percent), while introducing a cost-price ratio ceiling of 65 percent and new regulatory levies (Andersen, 2022). The direction and magnitude of the net fiscal effect on operator investment behavior depend on the specific asset and production profile.

If fiscal regime effects are the primary explanation, three patterns should be observable. A temporal break: operational outcomes for pre-PIA divestments should differ from post-PIA divestments. A uniform fiscal effect: all operators under the same fiscal regime should be affected similarly. And fiscal sensitivity: operational failures should correlate with fiscal pressure periods rather than with capability-specific events unrelated to investment deferral.

Here the evidence is the specific fiscal rates and their institutional implications for operator economics (Andersen, 2022; Nwuke, 2021); pre-PIA versus post-PIA operator performance comparisons; and the marginal field program outcomes, which unfolded entirely under pre-PIA fiscal terms. Operators who succeed under post-PIA terms while others fail under the same terms would weaken the uniform fiscal effect prediction.

Fiscal regime effects could, in principle, interact with operator-specific characteristics such as production profile or debt structure. For fiscal regime to be a primary explanation for differentiated outcomes, there would need to be evidence that the operators who failed were systematically more exposed to fiscal pressure than those who succeeded. The empirical analysis tests whether such systematic exposure exists.

The pre-PIA fiscal terms themselves also merit examination. The Joint Venture and Production Sharing Contract frameworks, royalty rates, and Petroleum Profits Tax that governed the observed period applied uniformly to all operators holding the same contract type. Cross-operator variation under those identical terms is inconsistent with fiscal regime as a sufficient explanation: if the fiscal terms are the same for everyone, they cannot explain why some operators failed and others succeeded. The empirical investigation focuses on the observed period, where pre-PIA and marginal field evidence dominates. Post-PIA cases are also examined for completeness, though the weight of evidence rests on the pre-PIA transactions and the marginal field precedent.

2.3.3 Regulatory Inadequacy

Weak regulatory oversight offers a different account of the pattern. This explanation locates the cause of post-divestment failure in the institutions responsible for overseeing asset transfers and monitoring operator performance, and contains two distinct mechanisms that generate different predictions and call for different interventions.

Design failure holds that the regulatory framework was structurally incomplete. It did not include operational capability assessment as a criterion for divestment approval. The mandate did not exist. The regulator assessed what the rules required, specifically financial capacity, legal compliance, and equity structure, and did not assess what the rules did not cover: embedded operational knowledge, crisis response capability, and organizational readiness. Institutional economics grounds this claim: institutions defined as the rules structuring human interaction determine the constraints within which organizations operate (North, 1990). If the rules do not require capability assessment, no amount of enforcement effort will produce it.

Execution failure holds that the regulator had the legal mandate and institutional framework to assess buyer capability but lacked the institutional capacity, resources, or political independence to exercise that mandate effectively. The rules existed on paper; they were not enforced in practice. Regulatory capture theory predicts that regulatory agencies in concentrated industries will come to serve the interests of the firms they regulate (Stigler, 1971).

The two sub-mechanisms generate different predictions. Under design failure, strengthening enforcement of existing rules would not improve outcomes, because the existing rules do not cover capability; only redesigning the framework would address the gap. Under execution failure, strengthening enforcement would improve outcomes, because the rules exist but are

not applied. Both sub-mechanisms predict that cross-operator variation under the same regulatory framework should exist, because the regulation either permits variation by design or fails to prevent it through weak enforcement.

The record here is statutory: the Petroleum Act text (does Section 29 include capability criteria?), the PIA text (do Sections 232-233 mandate capability transfer assessment?). It also includes the NUPRC’s institutional response to post-divestment failures (did it create new criteria or enforce existing ones more vigorously?), and cross-operator variation in outcomes under the same regulatory framework.

The NUPRC’s response is diagnostically important. The Commission created a seven-pillar divestment framework in 2024 that it described as “established for the first time in the 68-year history of Nigeria’s exploration and production” (NUPRC, 2024). It initially withheld consent on the ground that the consortium had not demonstrated the requisite technical capacity, before approving it after due diligence (The Energy Year, 2025; BusinessDay, 2025). These actions are consistent with design failure: the institution recognized that the framework was incomplete and created new assessment criteria, rather than enforcing existing ones more vigorously. The NUPRC’s response is evidence that the design gap existed, not that the gap caused the specific failures observed. Establishing causation requires showing that the failures occurred in domains where capability assessment at the point of transaction would have identified the deficit and where the deficit was not compensable through post-acquisition market mechanisms. Both conditions are tested in the empirical analysis.

Regulatory inadequacy functions as an enabling condition rather than a differentiating cause. It explains why the system permitted failures to occur at scale but cannot explain why some operators failed while others succeeded under the same permissive framework. If operators exposed to the same pre-2024 regulatory framework produced sharply different outcomes, regulatory inadequacy would explain why failure was permitted but not which operators failed. That differentiation would require operator-level explanations. The distinction between regulatory design failure and execution failure has direct implications for the framework proposed in subsequent chapters: if the evidence supports design failure, the appropriate intervention is institutional redesign.

2.3.4 Political Economy

The political economy explanation reframes the question. Rather than asking why some operators fail operationally, it asks why some operators survive commercially despite lower operational quality. Rexer (2025), in the *American Economic Review*, analyzed panel data covering 314 Nigerian oilfields from 2006 to 2016 and proposed a formal bargaining mechanism. Indigenous operators appoint political and military elites as shareholders and directors, making them residual claimants on production revenue, which aligns their incentives to protect the asset from theft rather than to tolerate or participate in it. The result is that indigenous operators reduce oil theft losses, increase net production revenue, and achieve commercial viability despite lower operational quality as measured by spill rates and gas flaring.

Rexer established three empirical results that bear directly on this analysis. Divestment to indigenous operators was associated with increased production. Divestment was simultaneously associated with declining operational quality: divested fields experienced higher rates of oil spills attributable to operational failure and higher rates of gas flaring. The production

gains were driven by a reduction in oil theft through the political connections mechanism. Rexer's findings are population-level, derived from 314 oilfields. Translating that pattern to individual operators requires case-level evidence: board composition, shareholding structure, and theft rate changes specific to each operator examined.

Iwuoha (2021), in a qualitative study of patron-client politics in Nigerian oil block allocation published in the *Review of African Political Economy*, found that patron-client relations contribute to poor development of the upstream oil sector by indigenous operators, to defaults in oil remittances. The same relations drive a consistent decline in crude oil production levels. Akpomera (2015), in the same journal, documented elite predatory behavior in the oil theft economy.

If political connections were sufficient for operational quality, connected operators would show better quality indicators. Rexer's data show the opposite: connected operators achieve higher output "despite lower quality." Political economy therefore cannot be a sufficient explanation for differentiated operational quality outcomes, though it may well explain differentiated commercial survival. Production should increase after divestment while quality indicators decline, and operators with strong political connections should survive commercially even with lower operational quality.

Political connections and operational capability relate to each other in a specific way. They may be substitutable in some dimensions: connections can reduce theft through patronage networks and manage community relations through elite-mediated access. They are non-substitutable in others: connections cannot substitute for well control knowledge, blowout contingency planning, or equipment-specific operational expertise. Where the two are substitutable, political economy provides a genuine alternative to capability as an explanation for commercial outcomes. Where they are non-substitutable, the two explanations operate on different outcome dimensions and are complementary rather than competing. The critical experiment case provides the empirical test of this boundary. Oil theft reduction is a substitutable dimension, where connections perform a function that operational capability also performs. Well control is non-substitutable, where only well control expertise and practiced crisis response apply. If political connections reduced theft on the asset but could not have prevented or contained a well blowout, the within-case split confirms that the two explanations operate on different outcome dimensions.

This explanation is tested through board composition and shareholding structure of acquiring companies (for the subset of cases where ownership and board data are publicly available), oil theft rates before and after operator change, and production trajectory versus spill and flaring trajectory over time. An operator without political connections that survives and performs well would suggest operational capability rather than connections drives viability. If operational quality does not decline in cases where capability was effectively transferred, quality decline is contingent on whether capability transfers, not on indigenization itself.

2.3.5 Capability-Related Factors

Capability-related factors predict a distinctive pattern that the other four explanations do not, a selective failure signature: routine success in capital-intensive, documented-knowledge-dependent operations coexisting with crisis-specific failure in knowledge-intensive domains that require tacit expertise. The proposition underlying this prediction is that the divestment

transaction failed to transfer embedded organizational knowledge that the buyer needed to operate the asset under all conditions.

The mechanism draws on the knowledge-based view of the firm. Organizations exist as institutions for integrating specialist knowledge into productive activity, and the coordination mechanisms through which this integration occurs are specific to the organization that developed them (Grant, 1996; Kogut and Zander, 1992). A divestment transaction dissolves the organizational form that integrates operational knowledge. Physical assets, legal rights, and documented knowledge transfer through the sale agreement and data room. Tacit operational knowledge, organizational routines, and the organizational architecture that supported continuous knowledge creation depart with the IOC personnel who held them. An empirical study of knowledge transfer across multinational subsidiaries found that codifiability and teachability significantly predicted transfer speed, and that across firm boundaries the transfer was slower still (Zander and Kogut, 1995).

The Nigerian evidence is consistent with this theoretical picture. Okonkwo (2019), studying four interfirm collaborations between foreign and indigenous firms in the Nigerian oil industry, found that in asymmetric alliances where large capability differences exist, partners' motivational characteristics dominate the knowledge transfer process. A technical assessment of 33 indigenous companies in the Niger Delta found measurable capability gaps in tubing inspection, slickline operations, and surface well testing (SPE 184346, 2016).

If capability-related factors are the primary explanation, operators who inherited embedded knowledge, whether through retaining the IOC's operational staff, through bringing equivalent prior experience, or through deliberate capability reconstruction, should perform better than operators who did not. The explanation also predicts selective failure: an operator that inherits equipment and documented knowledge but not experiential knowledge should perform routine operations successfully but fail under conditions that demand experiential judgment.

Failures persisting beyond two to three years of operational experience, during which an operator should have encountered at least one full annual operating cycle and completed initial staffing and familiarization, would suggest structural capability absence rather than transition-period coordination gaps. The threshold is indicative, since some failures only manifest under specific operational conditions. The persistence discriminator gains force when combined with the type of failure: systematic failure in specific knowledge-intensive domains (well control, blowout response) while routine operations succeed is more consistent with absent capability than with governance decay, because governance decay would degrade all functions equally. Operators calling in external specialists for functions the IOC performed internally would further confirm the gap through observable behavior.

The capability test turns on staff retention provisions in deal structures, founding team composition and prior IOC experience, production trajectory versus safety and environmental trajectory, and external contractor engagement for crisis response. It also turns on population-level capability proxies such as flare intensity per barrel, spill frequency per barrel, and lost-time injury rates.

This explanation would be rejected if an operator that inherited no IOC knowledge, had no established organizational architecture for upstream operations at this scale, and nevertheless performed well under crisis conditions. It would also be rejected if no observable correlation between knowledge inheritance and operational outcome appeared across the cases examined. Secondary mechanisms such as absorptive capacity deficit and pre-divestment deterioration

are treated as refinements that specify how the primary mechanism operates. They enter the analysis only after the primary evidence supports capability-related factors, and they are bounded by the same falsification conditions.

Operators who inherited knowledge are also likely to differ from those who did not on other dimensions. They may be better funded (they could afford retention bonuses), better managed (they made a deliberate strategic choice to preserve capability), and more experienced in the industry. Knowledge inheritance may correlate with outcome because it is a proxy for general organizational quality, not because knowledge itself is causal.

The strongest available control is within-operator temporal comparison. If an operator initially lacks inherited knowledge but acquires it later through delayed hiring of experienced personnel, and if performance improves at the point of knowledge acquisition rather than at the point of funding change or management restructuring, the causal attribution is strengthened. Cross-case comparison holding financial resources as constant as possible provides a second control. Where neither control is available and the confound cannot be resolved, the empirical analysis states it as a limitation rather than forcing an attribution.

The five explanations are not symmetric in scope. Capability-related factors, by construction, occupy broader territory than the other four, which are each bounded to a specific domain: financial patterns, fiscal timing, system-level conditions, or commercial survival dynamics respectively. This asymmetry reflects how each theory defines its scope. The analytical discipline it requires is therefore asymmetric: capability-related factors must pass stricter evidentiary tests precisely because the explanation occupies more territory. If the evidence can be explained by a narrower explanation, the narrower explanation is preferred on grounds of parsimony. Capability-related factors earn their verdict only if the evidence pattern is inconsistent with the narrower explanations and consistent with capability's specific predictions, particularly the selective failure signature and the persistence-plus-failure-type pattern.

Table 2.1: Key concepts used in the discrimination analysis. Each concept is defined here once and carried consistently through the case chapters.

Concept	Definition as used in this thesis
Selective failure signature	A pattern in which an operator sustains routine, capital-intensive production while failing in knowledge-intensive domains such as well control, crisis response, and emergency management. The pattern separates capability-related factors, which predict it, from capital constraints, which predict uniform degradation, and from coordination failure, which typically resolves within 12 to 18 months.

Concept	Definition as used in this thesis
Absorptive capacity	An organization's ability to recognize the value of external knowledge, assimilate it, and apply it, which depends on prior related expertise (Cohen and Levinthal, 1990). A buyer's capacity to absorb what a seller transfers therefore rises with its own upstream operating depth.
The five competing explanations	Capital constraints, fiscal regime effects, regulatory inadequacy, political economy, and capability-related factors. Each carries a distinct binding constraint and generates a testable prediction with a falsification condition (Table 2.2).

2.4 Coordination Failure

Coordination failure appears in every unsuccessful divestment: poorly managed transitions, delayed contractor mobilization, inadequate interface management. It is not treated as a sixth root-cause explanation because it is observable in all unsuccessful cases and therefore does not discriminate between explanations on its own. The analytical question is why coordination failed, not whether it did.

If coordination failure were an independent root cause, failures should concentrate in the first six to twelve months after the transaction closes, when coordination gaps are most acute, and should diminish as the operator stabilizes and builds its own routines through operational experience. Outcomes should correlate with identifiable transition design choices rather than with diffuse capability absence.

The empirical analysis tests whether the observed failures extend years beyond the transition period. Persistence on that scale, while routine operations continue, would be inconsistent with coordination failure as an independent cause that self-corrects through operational learning. The test is therefore diagnostic, and a verdict that coordination failure is rejected as an independent cause reports a structural finding: it identifies an observable feature common to the unsuccessful cases, which the deeper conditions explain.

The root-cause explanations account for why coordination failed at a deeper level. Capital constraints imply that the operator cannot afford effective coordination. Regulatory inadequacy means the framework does not require it. Capability-related factors suggest that the knowledge being coordinated does not exist on the buyer's side and the operator does not know what to coordinate because the critical knowledge is tacit and invisible from the data room. Political economy holds that connections substitute for the need to coordinate by ensuring commercial survival through non-operational channels.

The Renaissance transaction provides a design contrast, not an outcome proof (the deal closed recently and its long-term trajectory is not yet established). By design, retaining staff and preserving operational continuity keeps in place the knowledge on which coordination

depends. Whether that design choice produces the predicted outcome is examined in the empirical analysis, and bears on the claim that coordination succeeds when the relevant knowledge is present and fails when it is absent.

Whether coordination failure in each case was independently caused or was produced by one of the deeper conditions is an empirical question that the case evidence must resolve.

2.5 Cross-Cutting Conditions

Two conditions modify how the five competing explanations interact. Neither is a root cause, but both shape how the root causes play out.

2.5.1 Information Asymmetry

The three-party information structure of the divestment transaction creates an adverse selection problem (Akerlof, 1970). The seller knows what operational capability is embedded in the asset: which knowledge is documented and which is not, which operating practices depend on specific personnel who will depart. The buyer cannot assess this from the data room, because the most consequential capability resides in tacit, undocumented forms that do not appear in the documents available for inspection. The regulator has even less information than the buyer, because the regulatory approval process assesses financial capacity, legal compliance, and equity structure, not embedded operational capability. The seller, for its part, has rational incentives not to invest in transferring embedded knowledge: retaining tacit capability protects the seller's remaining portfolio assets. Transferring knowledge creates a potential future competitor, and tacit knowledge is legally non-contractible, meaning the seller cannot charge for it even if willing to transfer. The loss of embedded capability is therefore not an accidental byproduct of the transaction but a structurally likely result of how it is organized.

Transaction cost economics identifies the conditions under which market mechanisms fail and institutional alternatives become efficient: asset specificity, uncertainty, and frequency (Williamson, 1975). The knowledge is specific to the asset and cannot be redeployed; the buyer cannot verify the quality of what it is receiving; and divestments occur repeatedly across the sector.

Post-divestment failure costs fall on host communities (environmental damage, safety incidents, lost employment) and on the Nigerian state (lost revenue, remediation costs, national production decline), not on the transacting parties who negotiated the deal. This externality is the cleanest justification for mandatory regulatory intervention. When the costs of transaction failure are borne by parties not at the table, the transaction market does not reliably self-correct, and institutional governance is warranted.

Information asymmetry applies uniformly across all transactions and therefore cannot discriminate between the five explanations. Its role is foundational: it explains why the capability gap persists without market self-correction and provides the economic justification for the mandatory framework proposed in subsequent chapters.

2.5.2 Absorptive Capacity

Even if capability is available for transfer, the buyer must have the organizational architecture to receive and embed it. Absorptive capacity, the ability of an organization to recognize the value of new external information, assimilate it, and apply it to commercial ends, depends on the organization's prior related knowledge (Cohen and Levinthal, 1990). An indigenous operator that has never managed a complex well intervention, a high-pressure gas compression system, or a suspended-well integrity program lacks the prior related knowledge to absorb the IOC's accumulated expertise in these domains, even if that expertise were made fully available.

The distinction between potential and realized absorptive capacity (Zahra and George, 2002) connects to the selective failure pattern identified in Section 2.3.5. Potential absorptive capacity, the ability to acquire and assimilate documented knowledge, may be sufficient for routine operations. Realized absorptive capacity, the ability to convert that understanding into practiced routines and crisis judgment, requires prior experience that cannot be acquired from documents alone. An operator may possess potential absorptive capacity (and perform routine operations) but lack realized absorptive capacity (and fail under crisis conditions).

Absorptive capacity is invoked only when there is positive evidence that knowledge was available for transfer and that the transition was adequately designed, yet the operator still failed. In the absence of such evidence, capability-related factors (knowledge was not available) or capital constraints (the operator could not afford the transition) are more specific diagnoses. The boundary is important: without it, absorptive capacity could absorb any unexplained failure. The empirical analysis tests absorptive capacity by examining whether operators who received documented knowledge through the data room and had adequate financial resources to act on it nevertheless failed in domains where the documented knowledge existed. If the operating manual covered the procedure and the operator had funding to execute it, yet the operator could not perform the function under operational conditions, the deficit is in realized absorptive capacity: the organization could read the procedure but could not execute it. If the operating manual did not cover the procedure, the deficit is in knowledge availability, and capability-related factors rather than absorptive capacity is the explanation.

Like information asymmetry, absorptive capacity does not directly discriminate between the five competing explanations. Its role is to condition the interpretation of cases where knowledge appears to have been available yet the outcome was poor.

2.6 Synthesis: Discrimination Logic and the Central Empirical Question

The preceding sections have specified five competing root-cause explanations, examined coordination failure as an observable feature of all unsuccessful cases, and identified two cross-cutting conditions. Each explanation carries a causal mechanism and testable predictions. Each also defines the observable evidence patterns it implies and the conditions that would falsify it. What follows consolidates this material into a discrimination apparatus that the empirical investigation applies to the case evidence.

The five explanations differ on what constrains post-divestment operational performance:

financial resources, the external fiscal environment, the institutional framework, political connections, or embedded organizational knowledge. The predictions are distinctive enough that the case evidence can, in principle, discriminate between them, though not always decisively. Table 2.2 sets out, for each explanation, the binding constraint and the distinctive prediction on which that discrimination rests. The decisive falsification condition and the matching process tracing test for each are set out in the methodology (Section 3.4, Table 3.1), and the full discrimination matrix is in Appendix B.

Table 2.2: The binding constraint and key prediction of each competing explanation. The explanations rest on different constraints, which is what lets the evidence separate them.

Explanation	Binding constraint	Key prediction
Capital constraints	Financial resources	Progressive, cumulative degradation through prioritized deferral; failure correlated with funding levels across operators
Fiscal regime effects	External fiscal environment	Temporal break at PIA enactment; uniform effect across operators under same terms
Regulatory inadequacy	Institutional framework	System permits failure at scale; does not predict which operators fail
Political economy	Political connections	Connected operators survive commercially despite lower quality; production rises while quality declines
Capability-related factors	Embedded organizational knowledge	Selective failure (routine success + crisis failure); persistence beyond 2-3 years in knowledge-intensive domains; external contractor engagement

These explanations may not operate in isolation. Regulatory design failure enables the loss of embedded capability by not requiring transfer at the point of transaction. Political economy sustains low-capability operators by ensuring commercial survival through connections that mask operational decline. Capital constraints limit the ability to compensate for capability gaps through hiring, training, or contractor engagement. These interaction pathways mean that the empirical question is which combination of conditions was necessary and which was most proximate to the observed failure in each case.

Three pairwise discriminations are critical. The discrimination between capital constraints and capability-related factors turns on whether an operator that demonstrates financial

capacity (sustained production growth, maintained JV funding) nevertheless fails in a knowledge-intensive domain. Funded routine success coexisting with knowledge-dependent crisis failure in the same operator is inconsistent with capital as the binding constraint, because the same capital pool funds both. The discrimination between political economy and capability-related factors turns on whether operational quality declines after all divestments or only after divestments where capability was not transferred. If quality does not decline when capability is preserved, then quality decline is contingent on capability transfer, not on indigenization itself. The discrimination between regulatory inadequacy and all operator-level explanations turns on cross-operator variation: divergent outcomes among operators under identical regulatory conditions demonstrate that regulation permits failure but does not determine which operators fail.

The empirical analysis assigns each explanation a verdict drawn from a single set, applied consistently in Chapters 4 and 6, and an explanation may receive different verdicts on the two outcome dimensions where it bears on them differently. An explanation is “most consistent with the evidence” if it meets four criteria: it survives all decisive falsification tests applied to it, it has positive observable evidence supporting its predicted patterns. It must also explain the within-case discrimination that the critical experiment tests, and it is not contradicted by cross-case validation. It is “consistent but not decisive” if it meets three of these criteria but the evidence on the fourth is ambiguous. It is “insufficient on its own” where it contributes to the outcome but cannot by itself account for the differentiation, whether because it operates at the system level as an enabling condition that permits failure without determining which operators fail. It is also insufficient where it accounts for only one of the two outcome dimensions. It is “rejected as sufficient” where it fails a decisive test as a primary or sufficient cause, even if it still contributes. It is “rejected for specific cases” where the falsification applies to particular transactions rather than to the whole set. Three further labels record the state of the evidence rather than a disposition: “insufficient evidence to discriminate” where the evidence cannot separate two surviving explanations, “not tested” where a case does not reach an explanation, and “not applicable” where its precondition does not arise.

The critical experiment puts four of the five explanations to a decisive test: capital constraints, regulatory inadequacy, political economy, and capability-related factors. Fiscal regime effects are tested separately through temporal falsification, as the primary transactions predate the PIA. The selective failure pattern would confirm capability-related factors, but it would also confirm capital constraints if the failure turned out to be funding-driven, regulatory inadequacy if the regulator had capability criteria but did not apply them, or political economy if connections explained why quality declined. The anchoring case and the basis for its selection are set out in Chapter 3.

If no explanation emerges as most consistent after all tests are applied, the analysis reports that the evidence supports a multi-causal account without a dominant factor, and specifies what additional evidence or cases would resolve the question. Multiple explanations may be “most consistent” if they operate on different outcome dimensions. Political economy may be most consistent for commercial survival outcomes while capability-related factors may be most consistent for operational quality outcomes.

Falsification and the “most consistent” standard are related but distinct. Surviving falsification is necessary: an explanation that fails a decisive test is eliminated. But survival alone is not sufficient. An explanation may survive all its falsification tests yet generate

predictions that the evidence does not specifically confirm. Regulatory inadequacy, for instance, could survive every test (no one can point to a case where the framework DID include capability criteria) and still not be “most consistent” for individual operator outcomes, because it predicts uniform system-level failure and the cases display differentiated operator-level failure. The discrimination between surviving explanations rests on which generates predictions that the evidence specifically confirms.

Elimination of competing explanations is therefore necessary but not sufficient for a “most consistent” verdict on capability-related factors. Capability must also show positive evidence matching its specific predictions, particularly the selective failure signature and the correlation between knowledge inheritance and outcome across cases. The positive evidence test ensures that any “most consistent” verdict rests on what the record demonstrates, not on what remains after elimination.

These conditions interact as a system. The regulatory framework determines what is assessed before a transfer proceeds. Information asymmetry determines what each party can observe about the asset’s embedded requirements. Absorptive capacity determines whether the acquiring organization can integrate whatever is made available. No single actor has both the information and the incentive to ensure the asset is operated safely after transfer. The seller knows what capability the asset requires but has no obligation to transfer it, the buyer cannot assess what it does not know is missing, and the regulator lacks the mandate and tools to verify operational readiness. The structurally likely result, shown in Figure 2.1, is that operational quality declines after divestment with no single actor bearing the full cost. The corrective feedback that a functioning market would supply is severed: the costs of failure fall on host communities and the Nigerian state, not on the seller or the buyer. No loss registers with the parties to the transaction, and nothing in the transaction forces them to correct it. Breaking this pattern would require coordinated action across all three parties. The central empirical question is whether the evidence confirms this systemic pattern or whether a simpler, single-factor explanation accounts for the observed outcomes.

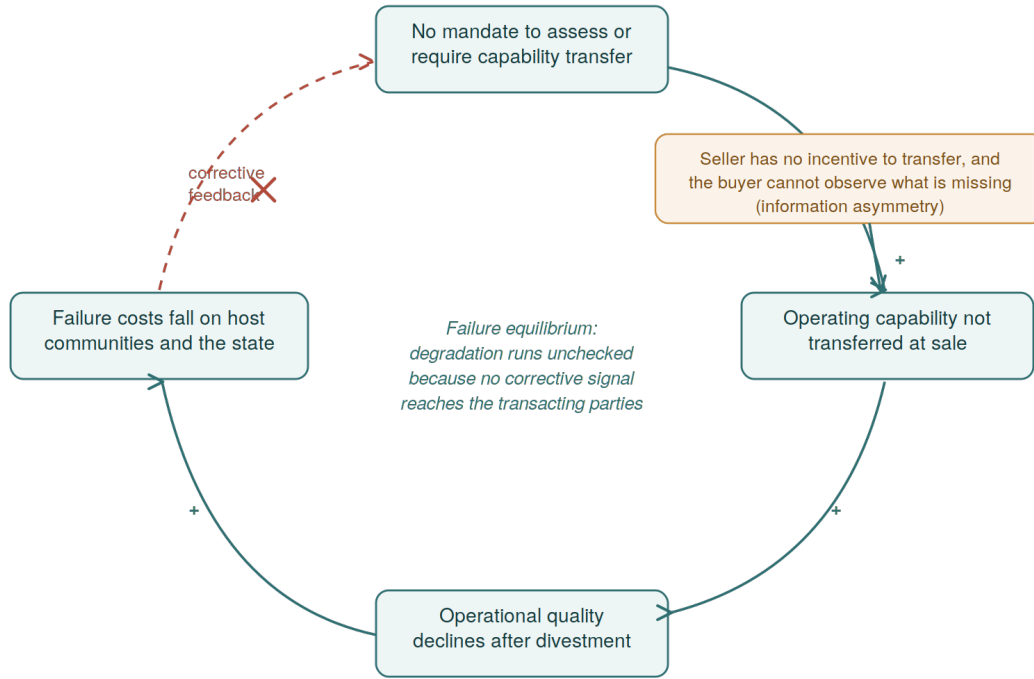


Figure 2.1: Divestment failure equilibrium. No single actor holds both the information (the seller knows the asset’s requirements) and the incentive (failure costs externalize to communities and the state) to ensure operational quality after transfer. With the corrective feedback that would close the loop severed, a low-quality state can persist.

The question reduces to substitutability: can financial resources and better management fully replicate the embedded capability within operationally relevant timeframes, or is some operationally consequential capability non-substitutable? If substitutable, capability-related factors are a temporary problem that resources and time solve, and the case for institutional redesign of the transfer process is substantially weakened. If not substitutable, capability-related factors have unique explanatory content, and institutional intervention to ensure transfer at the point of transaction is warranted. If the evidence further shows that post-divestment failure costs fall on parties not at the transaction table (host communities, the Nigerian state), the economic justification for mandatory intervention is established through the externality and information asymmetry conditions specified in Section 2.5.

This is the question the evidence must answer.

2.7 How This Analysis Advances the Literature

The literatures drawn on here each stop short of the problem this thesis takes up. The knowledge transfer tradition establishes that tacit, embedded knowledge moves poorly across organizational boundaries (Kogut and Zander, 1992; Cohen and Levinthal, 1990; Ferreira et al., 2024). Yet it treats transfer as a process internal to firms that choose to collaborate,

leaving open what becomes of that knowledge when an arms-length sale changes the owner of the asset but not the organization. The enterprise architecture and sociotechnical traditions establish that the organization, and not the plant alone, is what delivers value (Trist and Bamforth, 1951; Nightingale and Rhodes, 2015). Yet they say little about a change of ownership in which the plant transfers and the organization stays behind. The process safety literature specifies the systems that keep a hazardous facility safe (CCPS, 2007) while saying little about how those systems survive, or fail to survive, the point of a transaction. This thesis joins these strands at the divestment boundary and makes three moves the separate literatures do not.

First, it specifies operating capability as something observable at the point of transfer, through staff retention, prior operating depth, standing crisis-response arrangements, and the contractual terms of the deal. Its presence or absence becomes a testable question at the moment of transaction. Second, it identifies the selective failure signature as the diagnostic that separates a capability account from its rivals: an operator short of embedded capability sustains routine production while failing in the knowledge-intensive domains, a pattern the capital, fiscal, and regulatory explanations leave unexplained. Third, it recasts the divestment itself as the transfer of a sociotechnical system. That move turns the observed failures into a control failure in the system's design and places the remedy in the structure of the transaction rather than in the conduct of individual operators. These three moves set up the framework developed in Chapter 5.

Chapter 3

Research Methodology

3.1 Research Design

The discrimination apparatus constructed in the preceding chapter comprises five competing root-cause explanations for differentiated post-divestment outcomes and two cross-cutting conditions, with coordination failure examined as an observable feature of all unsuccessful cases. Each explanation carries testable predictions, observable evidence patterns, and falsification conditions. What follows specifies how the apparatus is applied to evidence: the research design, case selection, evidence architecture, analytical procedures, interview protocol, and safeguards.

This study adopts a theory-testing design. The five competing explanations were derived from existing literature before any evidence was collected. The analytical categories, the decision rules, and the conditions under which each explanation is confirmed, rejected, or left unresolved were all specified before the evidence was examined. This prior specification distinguishes the approach from conventional qualitative case research, where analytical categories often emerge from the data. The discrimination apparatus predetermines what the researcher must look for and how the evidence will be evaluated.

The specific design is a critical experiment with cross-case validation. A critical experiment investigates the specific details on which competing theories generate divergent predictions, so that the evidence can decide between them regardless of which theory turns out to be correct. Field and de Neufville (2021) describe this design as “particularly effective” and “especially desirable” for thesis research because it produces a valid result regardless of which explanation the evidence supports. A case qualifies as a critical experiment when competing explanations generate predictions that point in opposite directions for the same observable outcome, and the case contains evidence that can distinguish between those predictions.

The critical experiment tests the pattern for one case. Cross-case validation tests whether the pattern holds across operators or reflects idiosyncratic circumstances. The validation follows Yin’s (2009) replication logic: each additional case is selected because it tests a specific theoretical proposition, not because it represents a population. Literal replications test whether the pattern recurs in other operators under similar conditions. Theoretical replications test whether operators who differ on the dimension of interest (knowledge inheritance) produce different outcomes under comparable external conditions.

The analytical method is documentary process tracing with cross-case comparison. Process tracing reconstructs the causal chain connecting conditions to outcomes by examining the observable implications of competing explanations at each step in the sequence (Beach and Pedersen, 2013). Documentary process tracing can establish observable event sequences, pattern consistency, temporal ordering, and convergence of independent indicators. It cannot directly observe internal organizational decision-making or subjective motivations.

Applying process tracing with documents alone is a constrained application of the method, defensible when the relevant causal mechanisms leave observable documentary traces, as they do in this context (production data, spill records, regulatory filings, court documents, corporate disclosures). Yin (2009) identifies documents and archival records as core sources of evidence in case study research, and documentary sources constitute the primary evidence base here.

Within this framework, three types of tests apply, illustrated in Table 3.1 below. Hoop tests establish necessary conditions: if the evidence fails the test, the explanation is eliminated. Smoking-gun tests establish sufficient conditions: if the evidence passes, the explanation is strongly confirmed. Straw-in-the-wind tests provide suggestive but not conclusive evidence (Beach and Pedersen, 2013). Each falsification condition from the theoretical framework is mapped to a specific test type in Section 3.4.

The unit of analysis is the divestment transaction, not the individual participant. The transaction is treated as the transfer of a sociotechnical system, in which the licensed hardware and the operating organization that runs it form one working whole, so the question is twofold. It asks whether the assets changed hands, and whether the capability embedded in that operating organization transferred with them (Section 2.2). Data are collected from multiple sources (documentary evidence and, where obtainable, participant testimony through semi-structured interviews or structured surveys), but conclusions are drawn about organizational-level phenomena. These are whether operational capability transferred, whether the acquiring organization could perform specific functions, and whether the institutional design of the transaction permitted or prevented knowledge transfer.

The decision sequence produces, for each explanation on the primary case, one of the five verdicts defined in Section 2.6: most consistent with the evidence, consistent but not decisive, insufficient on its own, rejected as sufficient, or rejected for specific cases. Where the evidence cannot separate two surviving explanations, the result is recorded as insufficient evidence to discriminate. The cross-case validation then tests whether the pattern holds across operators.

3.2 Case Selection and Justification

Cases are selected for their analytical discriminating power between competing explanations. Field and de Neufville (2021) emphasize that case selection should follow the logic of the thesis rather than convenience or familiarity. Each is justified below by the specific theoretical proposition it tests and what it cannot test.

Aiteo (OML 29) is the critical experiment. It acquired the asset from Shell in 2015. Its trajectory, which includes both sustained production growth and a prolonged crisis failure, creates conditions under which the five explanations generate distinguishable predictions. Capital constraints predict progressive degradation across all operational domains. Capability-

related factors predict selective failure in knowledge-intensive domains while routine operations succeed. These predictions point in opposite directions for an operator that funds sustained production growth while also experiencing a crisis-response failure. The critical experiment can potentially confirm or reject four of the five explanations (capital constraints, regulatory inadequacy, political economy, and capability-related factors). Fiscal regime effects are tested separately through temporal falsification, as the Aiteo transaction predates the PIA by six years.

Cross-case validation extends the critical experiment through replication logic. Eroton (OML 18) provides a literal replication: an operator whose JV funding was maintained throughout a period of operational decline, testing whether financial resources alone account for the outcome. Eroton cannot independently test fiscal regime effects because its fiscal and operational timelines overlap. First Hydrocarbon (OML 26, through Afren) adds an additional dimension to the literal replication: an operator with substantial financial resources whose trajectory was interrupted by executive fraud, testing the regulatory screening process. For First Hydrocarbon, the fraud conviction confounds any assessment of post-divestment operational capability.

Three theoretical replications test whether different conditions on the dimension of interest produce different outcomes. Seplat (OMLs 4, 38, 41) provides the primary theoretical replication: an operator whose founding team included former IOC executives with decades of Nigerian upstream experience, testing whether prior knowledge inheritance produces a different outcome under the same external conditions. Pre-divestment deterioration from the seller side is not testable through Seplat because the seller dynamics differed from those in the failure cases. Shoreline and HEOSL (OML 30) constitute a within-OML natural experiment, where the same asset was operated under different operators under the same regulatory and fiscal terms, isolating operator-level factors. In the terms of the sociotechnical frame (Section 2.2), it decomposes the asset: the technical subsystem and the external regime are held fixed while the social subsystem that operates the asset varies. Any difference in outcome is therefore attributable to the operating organization rather than to the hardware or the regime. For Shoreline/HEOSL, initial acquisition conditions fall outside this comparison because operatorship was transferred between indigenous operators, not acquired directly from the IOC. NPDC (multiple OMLs) extends the analysis beyond commercial acquirers to a government entity with political connections but limited operational capability. NPDC cannot test commercial viability dynamics in the same way as commercial operators because its funding structure and institutional mandate differ in kind. Data availability for NPDC is more limited than for commercial operators (NPDC does not produce corporate-style disclosures), but safety incident records, NEITI audit reports, and technical-efficiency studies cover the variables examined here.

Together, these cases cover both operational success and failure, span three operator types (indigenous commercial, international with local partnership, government entity), and include two distinct institutional transfer mechanisms (bilateral IOC sale and government licensing). This coverage is sufficient to address RQ1 and RQ2 because the cases vary on the dimension of interest (knowledge inheritance) while sharing the same regulatory environment, asset type, and broad time period.

The primary cases share a feature that bears on the inference: each acquired its assets from Shell. The Aiteo, Eroton, First Hydrocarbon, Seplat, and Shoreline transactions, along with

the NPDC equity holdings, all derive from the SPDC divestment program. If the patterns identified across these cases reflected Shell’s divestment practices rather than the structural condition under examination, they should not recur when the seller is a different international operator. Three non-Shell transactions test this. Oando’s 2014 acquisition of ConocoPhillips’ Nigerian interest introduces a different seller, but as a non-operating stake in an Eni-operated joint venture it tests governance and seller identity, not operational capability. Oando’s subsequent acquisition of Eni’s Nigerian Agip Oil Company, processed under the seven-pillar divestment framework that the NUPRC introduced in 2024, tests whether the capability assessment criteria established after the primary transactions are applied to a non-Shell transfer. Seplat’s 2024 acquisition of ExxonMobil’s Mobil Producing Nigeria tests absorptive capacity from the buyer side: an operator with demonstrated upstream capability integrating assets from a different international seller, with early indicators available rather than a multi-year operational record. None of the three carries the six-to-fifteen-year post-transfer history of the primary cases, and two close too recently for medium-term outcome assessment. They therefore enter the cross-case synthesis as seller-independence checks, not as cases analyzed through the full decision sequence.

The marginal field program (2003 to 2020) serves as a structural precedent. It allocated petroleum assets to indigenous operators through government licensing rather than bilateral IOC sale, but under the same structural condition: asset transfer without mandatory capability assessment. If a comparable failure pattern appears under a different transfer mechanism, the pattern is more likely institutional than idiosyncratic to any single transaction type.

The Renaissance transaction (2024) serves as a design contrast, not a formal case analyzed through the full decision sequence. It demonstrates what a capability-preserving transaction design looks like in practice, providing a natural contrast with earlier transactions where no such structure existed. At the time of writing, only a short period of post-completion operating data is available, insufficient to assess medium-term operational quality, particularly asset integrity and maintenance outcomes that manifest over three-to-five year cycles. Renaissance is therefore not weighted equally with the primary cases, which have six to fifteen years of post-transfer history. Its early indicators are examined as a real-world instance of the framework design proposed in the following chapters.

3.3 Evidence Architecture

The evidence base consists of two tiers. Tier 1 comprises observable outcomes, documentary records, regulatory framework texts, corporate disclosures, court filings, production data, spill records, and independently compiled datasets. The core evaluative conclusions rest on Tier 1 evidence. Tier 2 comprises participant testimony from industry professionals across three groups (sellers, buyers and regulators), collected through semi-structured interviews or structured surveys where obtainable. Tier 2 evidence deepens the analysis of mechanisms not fully observable from documentary sources.

The research questions are evaluable at the pattern level from documentary evidence alone. Observable outcomes (production, spills, safety incidents) are publicly recorded and sufficient for testing whether the predictions of each explanation match the evidence.

Interviews and surveys add precision where they are available. Specific verdicts can be

upgraded from “consistent with the evidence” to “supported by converging testimony and documentary evidence,” and the mechanisms through which the most consistent explanation operated can be specified more precisely. The specific analytical cost of proceeding without interviews is that the discrimination between coordination failure as independently caused and coordination failure as structurally produced cannot always be closed from documents alone. Documentary evidence can narrow the range of plausible interpretations on the coordination question, but it cannot fully observe the internal organizational conditions that would resolve the distinction in every case.

Where this ambiguity persists, the empirical analysis reports it directly, specifying what additional evidence would resolve the question. This transparency allows readers to assess the strength of the verdict for themselves.

Sources are classified into two quality tiers. Tier A sources anchor all key inferences: peer-reviewed academic publications, government and regulatory primary documents, court and litigation records, audited corporate filings, and official statistical databases. In the context of this thesis, Tier A sources include NOSDRA spill records, NEITI audit reports, the Petroleum Act, the PIA, NUPRC regulatory frameworks, DPR/NUPRC production data, ICC arbitration filings, and UK Serious Fraud Office records. Tier B sources provide context, triangulation, and case reconstruction: professional legal analyses, established NGO reports (SDN, HOMEF), recognized institutional think tanks (Brookings, NRGi), and mainstream industry press (Bloomberg, Energy Voice, World Oil). Case narrative detail sometimes relies mainly on Tier B sources, but the pattern-level variables on which verdicts depend (production volumes, spill frequencies, flare intensity, financing structure, legal framework texts) are anchored in Tier A. When only Tier B sources exist for a factual claim, this is disclosed. Sources that are unattributed opinion pieces, unreviewed student essays, or blog posts without original reporting are excluded.

All documentary evidence carries bias. Corporate filings present favorable interpretations to shareholders, government statements may reflect institutional interests rather than operational reality, and NGO reports may carry advocacy framing. These biases are addressed through triangulation across source types. A claim about Aiteo’s production trajectory, for example, is checked across corporate disclosures (Aiteo statements), regulatory data (DPR/NUPRC production records), and independent reporting (Bloomberg, Energy Voice, World Oil). Convergence across source types increases confidence. Divergence is reported as ambiguity, with weight reflecting each source’s independence from the transacting parties.

Production figures specifically come from three series that carry different measurement bases, serve different purposes, and are reported separately, not spliced into a single trajectory. The Wood Mackenzie modeled series reports sales volumes after estimated crude theft, which it notes can reach 40 percent for some onshore fields, and on that consistent post-theft basis it carries the operator-to-operator trajectory comparison. The NEITI audited metered figure, measured at the flow station before theft and metering losses, anchors annual verification. The NEITI fiscalized figure, the volume reconciled for royalty and tax after deductions and adjustments, supports the revenue-impact estimates. The modeled sales figure and the audited fiscalized figure share a post-theft basis and are the comparable pair, with the metered figure higher by the losses. Where the series diverge, the audited figure governs.

Where interviews or surveys are obtained and the testimony contradicts documentary evidence on quantifiable events, the documentary evidence takes precedence. Interviews may

reveal alternative interpretations of events, contextual factors not captured in documents, or organizational dynamics that documents do not record, but they do not override audited numbers, official production data, or court-filed claims.

3.4 Analytical Framework

The empirical analysis applies the discrimination apparatus through a decision sequence mapped to the three research questions. The sequence describes how the results are organized. In practice, evidence for multiple explanations is examined concurrently; the decision sequence provides the inferential structure, not a procedural checklist.

The sequence proceeds through six operations. Establishing the outcome differentiation comes first: confirming that the dependent variable (differentiated operational outcomes across acquiring operators) exists and is documented precisely enough that the pattern is not disputed. This addresses RQ1's descriptive component. Testing temporal and structural preconditions follows, determining which explanations are even applicable to each case (fiscal regime effects are excluded for pre-PIA transactions on temporal grounds; regulatory design failure applies uniformly). The falsification conditions specified in the theoretical framework are then applied: does the evidence contain any pattern that decisively rejects an explanation for a given case? Rejected explanations are set aside with the evidence stated. These three operations collectively answer RQ1 by identifying which explanations survive initial scrutiny.

For surviving explanations, the weight of evidence is assessed: does the evidence pattern strongly support, suggestively support, or leave the explanation unresolved? Where two or more survive, the discrimination logic is applied to determine which explanation carries the most weight. Where the evidence does not permit discrimination, the analysis reports the ambiguity and specifies what additional evidence would resolve it. These operations collectively answer RQ2. A final operation assesses the substitutability question and the externality condition, providing the empirical and economic foundation for RQ3.

Table 3.1 consolidates the discrimination logic for the competing explanations, mapping each explanation's decisive falsification to a process tracing test type and its evidence base; the binding constraint and key prediction for each are given in Chapter 2 (Table 2.2). Coordination failure appears as a final row, assessed for independent causal status rather than treated as a primary explanation.

Table 3.1: The decisive falsification condition for each competing explanation and the test that applies it, the logic by which the cases tell the explanations apart. The full matrix, with operationalization and source-level detail, is in Appendix B.

Explanation	Decisive Falsification	Test Type	Principal Data Sources
Capital constraints	Funded production growth with crisis failure in the same operator	Smoking-gun for capability	Financing records (ICC, Bloomberg); production data (DPR, NUPRC, NEITI); JV audit (NNPC)
Fiscal regime	Transaction closed before the PIA, so the failure predates the fiscal regime	Hoop (timing); straw-in-the-wind (cross-operator variation)	Legal texts (Petroleum Act, PIA); transaction dates; fiscal analysis (Andersen, Brookings)
Regulatory inadequacy	Pre-2024 framework already contained operational-capability criteria	Hoop (necessary condition); smoking-gun (design versus enforcement)	Statutory texts; NUPRC statements; cross-operator outcome data
Political economy	Operational quality maintained when capability is transferred	Smoking-gun against political economy as sufficient	Corporate registry; Rexer ICPSR dataset; NOSDRA spill records; SDN reports
Capability-related factors	An operator with no inherited knowledge succeeds under crisis	Smoking-gun against capability; hoop on knowledge-outcome correlation	Corporate disclosures; NOSDRA data; production records; flare, spill, and LTI proxies
Coordination failure (symptom)	Failures persist for years beyond any transition period, six or more in the cases examined	Hoop against independence	Post-acquisition timelines; persistence of failures across operators

The full discrimination matrix with source-level detail, including specific document names, database queries, and variable operationalizations, is presented in Appendix B.

Two time thresholds operate here, answering different questions. The capability signal emerges at two to three years, by which an operator should have completed a full annual operating cycle and its initial familiarization, so persistence beyond it points to a structural absence of knowledge rather than a transition gap (Section 2.3.5). The coordination-failure

bar sits higher: a self-correcting coordination gap eases as the operator settles, so only persistence well beyond any plausible transition period, six years and more in the cases examined, rejects coordination failure as an independent cause.

Quantitative capability proxies supplement the process tracing analysis. Flare intensity is calculated as total gas flared (standard cubic feet) divided by total crude oil produced (barrels) for each operator over the same time period. Spill frequency is calculated as the number of spills attributed to equipment failure or operational error per million barrels produced. The empirical analysis normalizes this to per 100 million barrels where international benchmarks (UK, Norway) use that standard, to permit direct comparison. Under NOSDRA’s cause classification, spills attributed to “equipment failure” and “operational error” are included. Spills attributed to “sabotage,” “theft,” and “unknown” are excluded from the operational quality proxy, as these are not directly indicative of operator capability. The distinction is not perfect (sabotage may be facilitated by poor security management, which is an operator responsibility), but it is the best available classification from the data. Data are drawn from the NOSDRA Oil Spill Monitor (queryable at oilspillmonitor.ng, filterable by operator, downloadable as CSV) and NEITI annual audit reports for production volumes.

Where data for a given operator-year are missing, that year is excluded from the operator’s average. Operators with fewer than three years of post-transfer production data are flagged as having insufficient data for reliable comparison. All comparisons are presented as descriptive statistics (means, ranges). The unit of analysis for quantitative comparisons is the operator category (IOC, indigenous commercial, NPDC) rather than individual named operators, unless operator-level data are available and sufficiently complete for direct comparison. Operator-level comparisons are made only for operators with at least three years of post-transfer production data and recorded spill or flaring data covering at least 75 percent of that period. These metrics are used to corroborate or challenge qualitative inferences about capability at the operator-type level.

These proxies are outcome indicators from which capability is inferred, not direct measures of capability. The inference is valid when the outcome pattern matches capability predictions and does not match the predictions of competing explanations.

The epistemological claim is pattern consistency under pre-specified discrimination, not causal identification in the experimental sense. The analysis identifies which explanation is most consistent with the observable evidence after each has been tested against its own falsification conditions. This constitutes the strongest inferential claim available from a case study design based on documentary evidence and is the standard Yin (2009) establishes for analytic generalization.

3.5 Interview Protocol

Semi-structured interviews and, where scheduling constraints prevent interviews, a structured survey instrument with equivalent questions provide Tier 2 evidence from industry participants across three groups. Group A comprises IOC technical staff involved in pre-divestment operations. Group B comprises DOC operators and managers who assumed operational control of the assets. Group C comprises regulatory officials responsible for divestment oversight. The interview protocol received exemption from the MIT Committee on the Use of

Humans as Experimental Subjects (COUHES Protocol E-7658). The full case study protocol, including the participant-group design, informed-consent procedures, and pre-specified coding categories, is provided in Appendix A, and the interview guide for each group is reproduced in Appendix C.

All core tests rely on Tier 1 documentary data. Where participant testimonies are obtained, they are treated as corroborative or clarificatory evidence, upgrading a verdict in the two ways set out in Section 3.3.

Each interview follows a standardized core of questions asked identically to every participant within the same group, supplemented by structured probes tailored to the participant's role and the specific case under discussion. Questions ask about observable operational events and decisions, not about feelings, opinions, or proprietary information. This design choice follows the thesis manual's guidance that action-based questions produce more reliable data than perception-based questions (Field and de Neufville, 2021).

Current employee participants face validity risks that former employees do not. Corporate confidentiality may constrain their responses, they may self-censor to avoid contradicting their employer, and favorable self-presentation or concern about professional retaliation can shape how they describe events. These risks are addressed by focusing questions on observable events rather than proprietary information, triangulating current employee responses against independent sources, and noting divergence between testimony and independent evidence with weight reflecting consistency across the broader evidentiary pattern. The analysis also explicitly acknowledges the validity limitations in the empirical analysis.

The principal investigator is a current employee of Chevron Nigeria. This affiliation is disclosed to every participant at the start of every interview. No current Chevron employees are interviewed about Chevron-specific operations, and none of the cases examined in this thesis involves an asset operated by Chevron Nigeria. The Aiteo, Eroton, First Hydrocarbon, Seplat, Shoreline/HEOSL, and NPDC cases, together with the Renaissance design contrast, involve assets divested through the Shell-operated SPDC joint venture, in which Total and Eni were the other international partners. The investigator has no direct professional involvement with any of them. The research design, case selection, and analytical conclusions are independent of the investigator's employment.

Informed consent is obtained verbally at the start of each interview and through an embedded consent screen at the start of each survey. Participants are de-identified in all transcripts, notes, and thesis text. Names are replaced with alphanumeric codes. Organizational affiliations are generalized to prevent indirect identification. Any quotations used in the thesis are anonymized and lightly edited only for clarity, not substance. Recordings are deleted after transcription is verified. All data are stored on the investigator's password-protected personal computer with encrypted backup.

3.6 Validity, Reliability, and Limitations

Table 3.2 summarizes the design quality tests applied to this study, adapted from Yin (2009). The case study database includes a case-by-case evidence table (Appendix D) and a verdict traceability matrix (Appendix E). Each verdict in the empirical analysis is traceable to specific rows in the evidence table, with all sources listed in the References.

Table 3.2: Design quality tests applied to the study, after Yin (2009), with the tactic used to satisfy each.

Test	Threat	Tactic	Implementation
Construct validity	Study does not measure what it claims	Multiple sources of evidence	Documentary sources (Tier A and B) plus participant testimony where obtained
		Chain of evidence	Case study database: sources to coded categories to verdicts (Appendices D, E, F)
Internal validity	Causal inferences unjustified	Pattern matching	Evidence compared to discrimination apparatus predictions from the theoretical framework
		Rival explanations	All five explanations assessed for every case; none excluded without evidence
		Pre-specified falsification	Falsification conditions and test types specified before evidence examined, reducing post-hoc rationalization
		Logic models	Causal chains with falsification conditions specified in the theoretical framework
External validity	Findings not generalizable	Replication logic	Literal and theoretical replications across six operators plus marginal fields
		Theory-based generalization	Analytic generalization to propositions, not statistical generalization

Test	Threat	Tactic	Implementation
Reliability	Procedures not replicable	Case study protocol	Discrimination apparatus, decision sequence, and process tracing tests documented
		Case study database	Organized evidence repository with traceable analytical chain

Because this is a case-study design, the findings generalize to theoretical propositions through analytic generalization rather than to all IOC divestments through statistical generalization (Yin, 2009). They apply most directly to IOC-to-domestic-operator asset transfers in petroleum-producing developing countries where the institutional design of the divestment process does not include mandatory capability assessment. The sample is not statistically representative. The extent to which the findings apply to other extractive industries, other countries, or transactions with different institutional structures is an empirical question the concluding chapter addresses.

The internal organizational decisions made during and after divestment transactions are not fully observable from outside the organization. What occurred inside acquiring organizations is reconstructed through inference from observable indicators and, where available, from participant testimony. Triangulation across source types reduces but does not eliminate the risk that the reconstruction diverges from what actually occurred. The knowledge-inheritance/organizational-quality confound discussed in Section 2.3.5 remains a limitation; where neither within-operator temporal comparison nor cross-case control resolves it, the analysis states the limitation.

Some regulators and operators may be unwilling to discuss specific incidents due to political sensitivity, ongoing litigation, or professional constraints, which primarily affects the richness of participant testimony. Flaring data may be incomplete for operators without continuous monitoring systems; the analysis uses the best available data from NOSDRA and the World Bank Global Gas Flaring Tracker and notes gaps where they exist. The quantitative capability proxies (flare intensity, spill frequency, lost-time injury rates) are outcome indicators from which capability is inferred, not direct measures. These comparisons do not control for asset age, terrain complexity, or reservoir characteristics.

The discrimination apparatus tests five specified explanations and examines coordination failure as an observable feature of all unsuccessful cases. If the evidence contains patterns that none of these accounts for, the methodology can detect unexplained variance but cannot systematically characterize the mechanism producing it. The Renaissance transaction provides a design contrast with only a short period of post-completion data, insufficient to assess medium-term asset integrity outcomes. Renaissance is not weighted equally with the primary cases.

The apparatus constrains the range of permissible interpretations, but it does not eliminate researcher judgment. Falsification tests and temporal preconditions are governed by pre-

specified rules that leave little room for discretion. Assessments of relative explanatory weight among surviving explanations require guided judgment within the framework's categories. The case study database and documented chain of evidence make these judgments transparent and assessable, but they do not make them mechanical.

Chapter 4

Empirical Analysis

4.1 Introduction

The five competing explanations specified in the preceding chapters (Chapters 2 and 3) are tested here against the documentary evidence: capital constraints, fiscal regime effects, regulatory inadequacy, political economy, and capability-related factors. Coordination failure accompanies them as an observable feature of all unsuccessful cases, a symptom the analysis tracks among the outcomes the five explanations must account for. Support accrues to an explanation only when the evidence matches its own pre-specified predictions; the elimination of its rivals lies outside that standard. The presentation of evidence in each case precedes and grounds the verdict on it, and a single observable pattern can bear on more than one explanation at once. The six operations proceed in order: the outcomes first, then a test of which explanations apply, then each explanation's falsification conditions. Then come a weighing of the survivors, then discrimination among those that jointly survive where the evidence permits, and finally an assessment of substitutability and externality.

The two outcome dimensions, production performance and operational integrity, come from a descriptive account of the differentiated outcomes (Section 4.2). The order of the analytical cases then follows their discriminating power. The critical experiment, Aiteo on OML 29, supplies the primary discrimination: the case where the five explanations generate divergent predictions that the evidence can adjudicate (Section 4.3). Eroton on OML 18 comes next as a test of whether that pattern recurs in another operator, a partial replication complicated by its dependence on a trunk line that Aiteo operated (Section 4.4). First Hydrocarbon on OML 26 follows as a screening case, since executive fraud dominates its trajectory and the case bears on the regulatory screening gap. Three theoretical replications, Seplat on OMLs 4, 38, and 41, Shoreline and HEOSL on OML 30, and NPDC across multiple licenses, test whether operators who differ on knowledge inheritance produce different outcomes under comparable conditions (Section 4.5). The marginal field program then provides a structural precedent, an earlier and independent transfer that used government licensing in place of a bilateral sale (Section 4.6).

Because every primary divestment case originates in the Shell divestment wave, the cross-case synthesis also draws on three non-Shell transactions, Oando and ConocoPhillips, Oando and Eni, and Seplat and ExxonMobil, as seller-independence checks (Section 4.7).

That synthesis renders a verdict on each explanation and discriminates among those that jointly survive. It then weighs the substitutability and externality conditions that bear on the third research question, where the Renaissance transaction offers a design contrast. The unresolved ambiguities come last (Section 4.8).

Documentary evidence grounds the verdicts. Production figures come from audited and regulatory records, principally NEITI annual audits and NUPRC reports. The Wood Mackenzie Nigeria Upstream Summary (2026), a commercial modeled series, provides a consistent panel for the five formal case licenses, where comparable license-level series are available, over 2016 to 2025. The two series appear together where they overlap and stay separate where they diverge, with the reason given. Flare intensity from NEITI audits, spill records via NOSDRA and the SDN environmental index, corporate disclosures, statutory texts, regulatory statements, and court records complete the evidence base, weighted by source quality (Section 3.3). Spill records enter mainly for discrete, undisputed events, since operator-level spill comparison is unreliable across sources (Section 4.2). Where participant testimony was obtained, it appears within the relevant case analysis, with documentary evidence taking precedence.

The resulting claim matches what the design can support. Identifying which explanation is most consistent with the observable evidence, after every explanation has faced its own falsification conditions, is pattern consistency under pre-specified discrimination, a standard weaker than causal proof, and it generalizes to theoretical propositions through analytic generalization (Yin, 2009). The evidence may point to a single explanation, to several that are jointly necessary, or to distinctions that stay open, in which case the ambiguity is the finding. Each explanation receives, for each case and in a consolidated verdict across the cases, the strongest verdict the evidence permits among the categories specified in Chapter 2. These categories are: most consistent with the evidence, consistent but not decisive, an enabling condition insufficient on its own, rejected as sufficient, or rejected for specific cases. Where a case lacks the observable implication needed to test an explanation, that explanation carries the status not tested or not applicable for the case.

4.2 Outcome Differentiation

The operators that acquired the divested assets fared differently; their outcomes diverged across two dimensions, set out here before any explanation is weighed. Production performance covers sustained output, peak, and trajectory; operational integrity covers well-control events, spills, flaring, and safety. The two dimensions stay separate because they move independently: a population study of the divestment wave found output and spills rising together (Rexer, 2025).

Absolute production figures come from the audited and regulatory record, the NEITI annual audits and the NUPRC reports. The Wood Mackenzie Nigeria Upstream Summary (2026), a commercial modeled series, supplies a consistent panel for the five formal case licenses, Aiteo, Eroton, Seplat, HEOSL, and First Hydrocarbon, over 2016 to 2025. NPDC and the marginal fields fall outside this panel; their figures come from NEITI, NUPRC, and Adekunle et al. (2019). These series follow the measurement-basis rule set out in Section 3.3: they are reported separately, and where they diverge the audited figure governs.

Against the long national decline set out in Chapter 1, the 2 million barrels per day target went unmet (NUPRC, 2024). At the population level, the Stakeholder Democracy Network’s 2018 index reported that indigenous operators emitted more per barrel than the majors, subject to the inconsistencies the index itself notes across reporting agencies (SDN, 2020). Rexer (2025), an observational panel of 314 oilfields from 2006 to 2016 with limited ability to isolate operator-level causal effects, associated divestment to indigenous operators with a 33 percent rise in production and 10 percent more spills. Both are population means the worst performers may dominate. Whether the same pairing of rising output with rising integrity problems holds within an individual operator, and which explanation accounts for it, is tested in the case analysis. These figures are population context, and the case-level evidence comes from the operators examined below.

The operator-level trajectories diverge sharply from the mid-2010s onward, from sustained growth to post-acquisition collapse, and Table 4.1 records each operator’s production figures with Figure 4.1 plotting the modeled series against the audited points. Where the modeled and audited series diverge, the audited record governs. Aiteo’s 2024 regulatory figure reflects a June 2024 pipeline leak that shut the field for most of the year, and the modeled 2024 and 2025 recovery for Aiteo and Eroton runs above the audited record as a modeling artifact, marked in Figure 4.1.

Operational-integrity indicators also vary widely across operators, though each carries a caveat. Flaring, the share of an operator’s produced gas that was flared, ran from 49.5 percent for Aiteo and 35.2 percent for Eroton down to 11.4 percent for the Seplat joint venture in 2021. Chevron and NAOC, both with full gas-evacuation infrastructure, stayed under 5 percent (NEITI, 2021). Gas infrastructure confounds the comparison directly, since the OB3 pipeline serving the western fields has been delayed since 2012, so the cross-operator figures are indicative and carry little decisive weight. The recorded spill counts differ across NOSDRA, NNPC, and the regulator for the same year, so any operator-level spill comparison rests on figures that disagree. The integrity evidence therefore rests on the discrete, undisputed events set out in Section 1.2 and recorded in Table 4.1: three negative, the Santa Barbara blowout, the 74 percent metering error in Aiteo’s 2022 audited return against a 6.9 percent norm, and the July 2020 Benin River Valve Station explosion; and one positive, Seplat’s record of more than 11 million man-hours without a lost-time injury.

Setting the explanations aside for a moment, the six formal cases fall into four groups, and the marginal field program forms a fifth. The classification turns on each operator’s later production as a share of its post-acquisition peak. The operators that declined bottomed below 15 percent of peak, and the operators that held stayed above 70 percent. Among the cases classified as sustained growth or deterioration, every one falls either below 30 percent or above 70 percent, so the cutoffs used here, 70 percent for sustained and 30 percent for deterioration, sit inside an empty band. The classification holds regardless of their exact placement. Aiteo fell to 4,900 metered in 2022, 12 percent of its 41,000 peak (700 fiscalized, after a 74 percent metering-error deduction). Eroton’s delivered output fell to near zero, about 600 fiscalized in 2022, below 2 percent of its 32,000 peak (the wellhead crude metered near 9,000 but stranded short of the terminal). Seplat held at 39,000 in 2025, 74 percent of its 53,000 peak, and HEOSL at 35,000, 100 percent. The comparator differs by operator because the usable record does. The two declining operators are measured at their 2022 audited trough: Aiteo on metered output, since its fiscalized figure is depressed by a 74

percent metering-error adjustment, and Eroton on delivered volume, since its metered crude was stranded short of the terminal. The two operators that held are measured at their 2025 modeled level, the only figure available for that year. Because those cases fall outside the 30-to-70 band on any of these bases, the classification is descriptive and insensitive to the choice among them.

Sustained growth covers Seplat and the HEOSL-operated OML 30. Growth then deterioration covers Aiteo and Eroton, the latter a deterioration of delivered output whose attribution is tested in Section 4.4. Low-level survival covers First Hydrocarbon. Underperformance against mandate covers NPDC, which set 500,000 barrels per day for 2020 and recorded the lowest technical efficiency of any operator category at 23 percent, on a basis that differs from the OML-level production series (Adekunle et al., 2019). Non-development covers the marginal field cohort. The classification establishes that the outcomes are heterogeneous, spanning sustained growth, deterioration, low-level survival, underperformance against mandate, and non-development. The reasons behind them come from the case analysis.

Table 4.1: Post-divestment outcomes across the formal cases, operators that acquired assets under broadly similar sectoral conditions but differ in transfer mechanism, operator structure, and asset history.

Operator (asset)	Acquired	Seller	Acquisition cost	Production performance outcome	Operational integrity outcome	Trajectory class
Aiteo (OML 29)	2015	Shell	\$1.7B (OML 29 stake); \$2.562B with the NCTL	Modeled peak ~41 kb/d (Wood Mackenzie, 2026); audited metered 38.2k (NEITI, 2021); 4.9k metered / 0.7k fiscalized (2022); 7.4k annual (NUPRC 2024, mid-year shut-in). Reported corporate peak 90k (Mar 2017), unsupported.	Santa Barbara blowout Nov 2021 (~200,000 bbl, 33 days, 41 communities); 74% metering error in 2022; flared 49.5% of produced gas (2021)	Growth then deterioration
Eroton (OML 18)	2015	Shell, Total, Eni	\$1.2B (45% interest)	Modeled peak ~32 kb/d (Wood Mackenzie, 2026); audited metered 19.3k (NEITI, 2021); delivered volumes far lower, the NCTL trunk line unavailable); 9.0k metered / ~0.6k fiscalized trough (2022); 13.6k gross (NUPRC, 2024). Reported corporate peak 50k.	Output stranded by evacuation failure; flared 35.2% of produced gas (2021)	Growth then delivered-output deterioration

Operator (asset)	Acquired	Seller	Acquisition cost	Production performance outcome	Operational integrity outcome	Trajectory class
First Hydrocarbon (OML 26, via Afren)	2011	Shell, Total, Eni	45% interest (Shell 30%, Total 10%, Eni 5%); ~\$98M reported, likely the Shell portion	~2 to ~15 kb/d modeled (Wood-Mac); audited metered 8.8k (NEITI, 2021), low base	Insufficient data to assess operational integrity. Sponsor Afren entered administration July 2015; executives convicted of fraud; ~\$185M bank losses	Low-level survival
Seplat (OMLs 4, 38, 41)	2010	Shell	\$340M (45%, from a \$610M valuation)	Modeled peak ~53 kb/d for the three OMLs (Wood Mackenzie, 2026); audited metered 42.3k (NEITI, 2021); company-wide peak 85.2k	11M+ man-hours without a lost-time injury; AfriSAFE Energy Company of the Year 2024; flared 11.4% (JV, OB3-confounded)	Sustained growth
Shoreline / HEOSL-operated (OML 30)	2012	Shell, Total, Eni	\$850M (45% interest)	Modeled peak ~35 kb/d (Wood Mackenzie, 2026); audited metered ~24k (NEITI, 2021) and ~30k (2022); grew from ~4k after the 2017 operatorship change to an operator-reported peak near 67k by 2025; an earlier 80k gross claim is set aside.	Rehabilitation of vessels, wellheads, and compression; clear of major safety or environmental incident in the documentary record surveyed; flaring reported only in aggregate for OML 30 in the NEITI 2021 audit	Sustained growth

Operator (asset)	Acquired	Seller	Acquisition cost	Production performance outcome	Operational integrity outcome	Trajectory class
NPDC (multiple OMLs)	2010 on	Shell equity via NNPC transfer	Equity holdings, without a transaction cost	Target 500k by 2020 unmet; 23% technical efficiency, lowest of operator categories; production basis differs from the OML series	Benin River Valve Station explosion July 2020, seven contractor workers killed	Underperformance against mandate

Audited = NEITI metered or fiscalized and NUPRC figures; modeled = Wood Mackenzie. Fiscalized production is production after metering losses and theft are deducted; the Wood Mackenzie series is modeled field liquids on a license-level basis. NUPRC figures are reported by company, so for multi-OML operators they span assets beyond the case licenses; NEITI 2021 metered figures are by license. Acquisition costs: Items in the source list and corporate filings.

Three further transactions inform the analysis at a lower evidentiary weight, their data limited or non-comparable: an earlier transfer mechanism that supplies a structural precedent, and two recent acquisitions whose deliberate structure offers a design contrast. Table 4.2 records them.

Table 4.2: Structural precedent and design contrasts, weighted below the formal cases because the data are limited or non-comparable.

Item	Transaction	Role	Status / data
Marginal field program (2003 round)	Government licensing of fields to indigenous companies	Structural precedent (a different transfer mechanism)	7 of 24 producing at ten years; 17 of 30 onstream by 2022; 11 licenses revoked over the program's life; near 2% of national reserves; regulator judged the benefit to Nigeria minimal
Renaissance (SPDC onshore)	2024 acquisition of Shell's onshore subsidiary; \$2.4B	Design contrast (capability preservation structured into the transaction)	200,000+ bpd by April 2025; limited post-transfer data
Seplat / MPNU	2024 acquisition of ExxonMobil's Mobil Producing Nigeria; \$800M; 40% operated	Design contrast (a capable buyer integrating from a different seller)	Months of post-transfer data; FY2025 group production roughly doubled

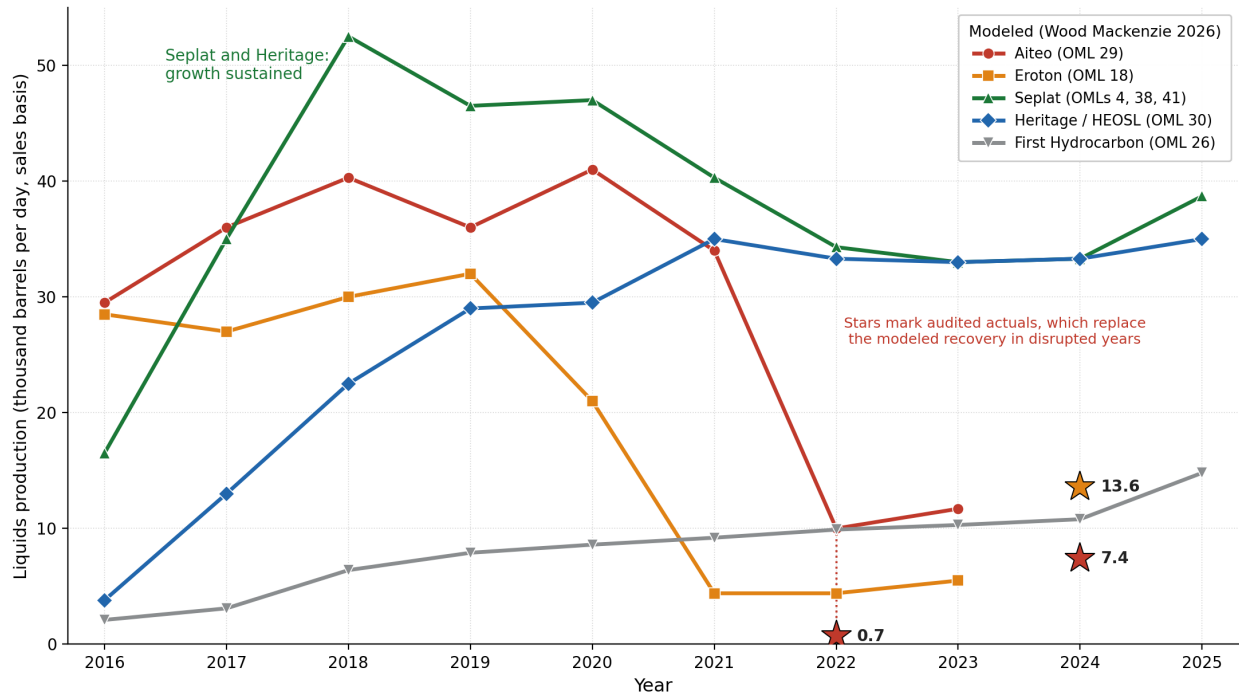


Figure 4.1: Post-divestment production trajectories of the formal case OMLs. Lines show Wood Mackenzie (2026) modeled field liquids by license, on a sales basis after estimated theft; the Aiteo and Eroton lines stop at 2023, before a later modeled recovery the audited record does not support. Stars mark the audited or regulatory figures that govern in the disrupted years and fall well below the modeled lines. Where the two measurement bases diverge, the audited record governs. NPDC, the marginal cohort, and the design contrasts are omitted for want of a comparable license-level series.

4.3 The Critical Experiment: Aiteo on OML 29

Aiteo is the case where the competing explanations generate the most divergent predictions. It is the operator that funded production growth and then failed in a crisis beyond its capacity to contain, on an asset acquired six years before the Petroleum Industry Act. Four of the five explanations are tested here through their divergent predictions about whether and how those two outcomes can co-occur in the same operator; fiscal regime effects are eliminated separately by timing (Section 4.3.2). The documentary record adjudicates several of these differences and bounds the rest. The positive case for capability-related factors is built first, from the shape of the failure, before each competing explanation is tested against its own falsification conditions; the consolidated verdict for the case comes last (Section 4.3.6).

4.3.1 The Selective Failure Pattern

Capability-related factors predict a particular shape of failure: the knowledge-intensive parts of the operation fail while the routine parts succeed. Routine production can often be sustained through established procedures, contractor support, maintenance spending, and

documented operating knowledge, so an operator that inherited assets without the embedded tacit knowledge to run them could still sustain output. The same operator would falter when an event demanded the tacit judgment and practiced routines that crisis response and well control require. Aiteo's record is consistent with that shape.

On the routine side, production grew. The Wood Mackenzie modeled series shows output on OML 29 rising over the post-acquisition years, from about 30,000 barrels per day in 2016 to a peak near 41,000 in 2020. Audited metered production reached 38,200 in 2021, the year before the blowout (Wood Mackenzie, 2026; NEITI, 2021). Aiteo's announcement of a 90,000 barrel-per-day peak in March 2017 is a reported corporate figure that sits well above the audited volumes. Part of the divergence between reported peaks and audited volumes on the Bonny system reflects wellhead potential measured before the pipeline-loss allocation that separates gross from delivered crude. The gap here also exceeds what that mechanism alone explains, and the argument holds without the figure. In the crisis-response domain, the operation failed conspicuously. In November 2021 the Santa Barbara Well 1 blew out and flowed uncontrolled into the surrounding creeks until it was secured on December 8, roughly 33 days later. The joint investigation conducted by NOSDRA, the NUPRC, the Bayelsa State Ministry of Environment, community representatives, and well-control consultants attributed the blowout to third-party interference. The affected community and the Bayelsa State Government insisted that the cause was equipment failure (NOSDRA Joint Investigation Visit, 2021; NOSDRA and NUPRC via Vanguard, 2021). The cause is therefore contested, and the capability finding holds independently of it. What the record establishes is a containment failure. Aiteo assembled its well-control response after the fact, engaging external specialists only once the blowout had occurred, including Halliburton's Boots and Coots working with a local resource. NOSDRA separately engaged Kenyon International for spill response, and the well was secured only after about a month (Upstream Online, 2021; Vanguard, 2021). The director-general of NOSDRA attributed the prolonged failure to a capacity deficit, observing that operators forced to import foreign well-control expertise lose critical time. He contrasted it with the embedded arrangements an international operator carried, noting that Shell had contained the 2011 Bonga spill quickly through a standing worldwide spill-response subscription (NOSDRA via BusinessDay, 2021). The investigation left the quantity spilled unsettled; independent estimates put it near 200,000 barrels across some 41 communities, and Aiteo disputed higher figures (NOSDRA Joint Investigation Visit, 2021; SweetcrudeReports, 2021).

The well was a latent exposure well before it failed. Santa Barbara Well 1 had been non-producing since the 2015 acquisition; both Shell and Aiteo left it undecommissioned, and it lay isolated, unplugged and unmonitored, with further spills recorded in the area in 2018 and 2019. Whether the blowout was triggered by tampering, as the official investigation found, or by equipment failure, as the community alleged, an idle well left in that state for six years was a standing hazard that a fully resourced operator would have retired or kept under watch. The capability reading turns on the containment failure that followed, leaving the trigger aside.

Two inferences follow, and they rest on different evidence. First, the funded production growth weakens capital constraints as the binding explanation, in the sense of a single factor that alone accounts for the outcome. The operator paid about \$2.562 billion for the 45 percent interest in OML 29 and the Nembe Creek Trunk Line (HOMEF, 2021; Nairametrics,

2015), and then funded sustained production growth demonstrably had capital available for production-directed major operational spending. That alone leaves open how Aiteo allocated funds internally, so the failure turns on whether crisis readiness was unavailable through financial scarcity or absent because the organization had yet to put it in place. Second, and separately, the failure points toward capability-related factors. That reading rests on the operational evidence concentrated in the crisis-response domain, with the funded growth bearing only on capital. That evidence is the absence of a contingency plan, the absence of a standing well-control contract, the dependence on contractors engaged after the fact, and the well left undecommissioned and unmonitored. The reviewed record for the OML 29 transaction shows no structured provision for retaining Shell operating staff, well-control routines, or crisis-response systems.

The competing reading treats that operational evidence as a budgeting choice: an operator with capital for production growth might simply have declined to fund crisis readiness. The distinction matters, because a choice to leave a known requirement unfunded is a management decision while the absence of the readiness system altogether is a capability gap, and both stand apart from capital constraints. The observable evidence favors the capability reading over a pure financial-scarcity reading. A standing well-control arrangement is a small commitment relative to the cost of an uncontrolled blowout response, so a funds shortage accounts poorly for its absence. The three weeks of improvised response, in place of an executed plan, point the same way: to an organization that had yet to institutionalize crisis readiness before the event. Capital constraints are rejected as sufficient regardless of which reading holds, because the funds for production growth demonstrably existed.

The leadership's background helps make the capability reading plausible, though on its own it leaves unproven that the gap caused the failure. Aiteo's founder and chief executive, Benedict Peters, holds a degree in geography and regional planning and built his career in downstream petroleum, in trading, storage, and fuel distribution, his upstream exploration and production experience beginning only with the acquisition of OML 29. The downstream-only background carries less than proof on its own; it is a mechanism-consistency indicator, since the failure fell in the upstream well-control domain that lay outside that experience. The operational record is the evidence that the gap operated in practice.

4.3.2 Fiscal Regime Effects

Fiscal regime effects fail on timing. Aiteo acquired OML 29 in 2015, and the Petroleum Industry Act was enacted on August 16, 2021, six years later. The conditions that produced the failure were in place well before that date: the suspended well had been left undecommissioned since 2015, and the unremediated spills date from 2018 and 2019. A cause must precede its effect. The blowout itself came in November 2021, three months after the Act, but the conditions it exposed, the absent well-control capability, the missing contingency plan, and the prior unmanaged incidents, all predated the Act. Fiscal regime effects are therefore rejected for this case as the primary explanation for the observed failure.

4.3.3 Regulatory Inadequacy

The regulatory framework in force at the time of the Aiteo transaction left operational capability outside its screen. Section 29 of the Petroleum Act required ministerial consent to assign an Oil Mining Lease but stopped there, leaving the buyer’s ability to run the asset untested. On the necessary-condition test, the framework indeed omitted a capability requirement.

What the regulator did next points to a gap in the framework’s content. The rules were applied as written; what they required was too little. In 2024 the NUPRC introduced a seven-pillar divestment framework that it described as the first of its kind in the sixty-eight-year history of Nigerian exploration and production. Its first pillar, technical capacity, appeared for the first time, with the Petroleum Act, internal guidelines, and practice all silent on it before. That the NUPRC created a capability criterion is strong evidence that the criterion was new, and the new framework produced operational consequences beyond a statement of intent. A divestment was initially withheld pending assessment, and decommissioning escrow exceeding \$400 million was secured across the 2024 transactions. Post-divestment failures contributed to the timing of the reform, but the content of the reform, a specific technical-capacity test new to the framework, is the mark of a recognized design failure.

The gap’s existence, on its own, sits below proof that it caused the Aiteo failure. Causation would require that a capability gap was identifiable at the point of transfer and non-compensable through the market afterward. The first, identifiability at the point of transfer, is directly supported. NOSDRA’s account that containing the blowout exceeded the operator’s available capacity, the after-the-fact engagement of external specialists, and the absent contingency plan all concern well-control capability, which is assessable before a transfer. An audit at the point of transaction could have tested for well-control protocols, crisis-response arrangements, and blowout contingency planning. The second, non-compensability through the market afterward, is suggested by the persistence of the gap across six years of operation and is tested across the cases that follow (Sections 4.4 and 4.7). And the same pre-2024 regulatory framework, the Petroleum Act and DPR oversight, governed Seplat, HEOSL, Aiteo, and NPDC; as the cases that follow show, their outcomes diverged, so the framework permitted the failure while leaving open which operators would fail. For the single case, then, regulatory inadequacy is consistent with the design-gap interpretation: an enabling condition that permitted the failure while leaving unexplained, on its own, why Aiteo failed where others held. Whether the absence of capability screening is necessary across the population is a question the cross-case synthesis takes up (Section 4.7).

4.3.4 Political Economy at the Case Level

The political-economy explanation is tested here for Aiteo specifically, through observable case-level indicators. The population-level association reported by Rexer (2025) supplies context, and the test here draws on the case record. The relevant indicators are board composition, shareholding, and any change in the theft rate on OML 29 around the transfer. For Aiteo, the available record supports only a weak test of those indicators: corporate registry detail on board composition and shareholding is incomplete, and the search recovered no operator-specific theft-rate series around the transfer. The test here is therefore narrower,

whether political access, even if present, could explain the non-substitutable well-control failure.

The decisive consideration is where political access substitutes for capability and where it reaches its limit. Connections can plausibly reduce certain exposures, oil theft, or community friction managed through elite-mediated access, the substitutable dimensions. Well control and blowout contingency rest on experiential knowledge, and on this non-substitutable dimension the outcome turns on what an operator actually holds. The Aiteo record splits along exactly that line. Whatever connections may have done for theft or community access on OML 29, preventing or containing the Santa Barbara blowout turned on well-control capability, which only operating experience supplies. Political economy is therefore rejected as sufficient for operational quality on the non-substitutable dimension. It may help account for commercial survival, since Aiteo continued operating after the blowout, while the onset of the blowout and the operator's inability to contain it lie outside its reach. The positive form of the test, whether connections sustain commercial viability while operational quality stays poor, is developed on NPDC (Section 4.5.3).

4.3.5 Coordination Failure

Coordination failure, treated throughout as an observable feature of unsuccessful cases and a symptom of deeper causes, is tested here against persistence and failure type. The blowout occurred in November 2021, six years after the 2015 acquisition, well beyond a short transition period. If coordination failure were merely a transition problem, it should appear early and ease as the operator stabilizes. Aiteo's pattern points the other way: specific knowledge-intensive functions failed while routine production grew, and the failure persisted for six years. The unresolved spills of 2018 and 2019 show the exposure persisting years before the blowout, present at year three and still unaddressed at year six.

A failure of this duration and this shape runs counter to a coordination problem that self-corrects through operational learning, so coordination failure is rejected as an independent cause for this case. Its persistence and domain-specificity are consistent instead with a structural absence of knowledge.

4.3.6 The Aiteo Verdict

The five explanations, and the coordination feature that accompanies them, resolve as follows for the Aiteo case. The verdicts use the categories specified in Chapter 2 (Section 2.6), and each rests on the explanation's own evidence, with the elimination of rivals set aside.

Table 4.3: Verdicts for the Aiteo case on OML 29, the critical experiment, where funded production growth alongside an uncontained blowout rejects the capital, fiscal, and political-economy explanations as sufficient for the integrity failure and leaves capability-related factors most consistent with the evidence.

Explanation	Verdict	Evidence anchor	What would change this verdict
Capital constraints	Rejected as sufficient	~\$2.562B acquisition, 45% interest plus the Nembe Creek Trunk Line (HOMEF, 2021; Nairametrics, 2015); audited metered growth to 38.2 kb/d (2021) alongside the crisis failure, modeled peak near 41 kb/d; capital demonstrably available for production-directed major operational spending	Evidence that crisis-readiness funding was cut for financial reasons
Fiscal regime effects	Rejected for this case	Transaction closed 2015; PIA enacted August 2021; the capability gap predates the Act	None; the timing is arithmetic
Regulatory inadequacy	An enabling condition insufficient on its own; system-level role tested in the cross-case synthesis (4.7)	Section 29 lacked capability criteria; the NUPRC created them in 2024; the same pre-2024 framework produced different outcomes across operators, developed in the cases that follow	Evidence that the framework screened for capability and that Aiteo passed
Political economy	Rejected as sufficient for operational quality (non-substitutable dimension)	Preventing or containing the blowout required well-control capability beyond the reach of connections	Evidence that political interference directly caused the operational failure, beyond enabling commercial survival

Explanation	Verdict	Evidence anchor	What would change this verdict
Coordination failure	Rejected as independently caused; an observable feature of the unsuccessful cases	Persistence at six years; domain-specific failure; unremediated prior incidents	Evidence of a specific coordination breakdown independent of knowledge absence
Capability-related factors	Most consistent with the evidence for this case	Selective failure signature; containment failure lasting about a month, which NOSDRA attributed to a capacity deficit; after-the-fact reliance on external well-control specialists against absent embedded crisis-response arrangements; six-year persistence; the documents reviewed silent on any structured knowledge-transfer provision in the transaction	An operator without inherited upstream knowledge handling a comparable crisis successfully

Across the two outcome dimensions, the diagnostic split is clean for the period the critical experiment turns on. In the years before the blowout, commercial production grew while crisis-readiness stayed absent, and the explanation most consistent with that split is one that bears on integrity, with funding set aside, since binding capital constraints would have suppressed production too. Output then fell sharply after the blowout, to about 4,900 barrels per day metered in 2022, 700 fiscalized (NEITI, 2022), so the capability failure spread to both dimensions; that loss came after the blowout, so the capital-constraints explanation stays closed. In the terms of Chapter 2 (Section 2.2), this is a failure of the asset as a sociotechnical system. The transaction moved the wells but not the operating organization that had run them, and when the well reached a crisis the well-control capability, the standing contracts, and the practiced crisis response that the operating organization carried were not present to contain it. This verdict reflects pattern consistency for a single case.

Appendix E reproduces this verdict matrix for every case, recording the evidence, confidence level, and falsification condition behind each verdict; the analyses that follow state their verdicts in prose and refer to that appendix for the full matrices.

4.4 Cross-Case Validation: Failure Replication and Regulatory Screening

The Aiteo verdict rests on a single case. Two further operators test whether it holds. Eroton acquired OML 18 from the same seller in the same divestment round and, like Aiteo, grew production before losing it, which makes it a replication candidate. Its collapse, however, ran partly through a shared export line whose chronic, theft-driven downtime lay largely outside Eroton’s control, so the case is a partial replication, complicated by the shared line (Section 4.4.1). First Hydrocarbon acquired OML 26 through a buyer whose executives were later convicted of fraud, which interrupts any reading of its operational capability but bears directly on whether the approval process screened buyers at all (Section 4.4.2).

4.4.1 Eroton on OML 18: A Partial Replication Complicated by a Shared Trunk Line

Eroton’s early trajectory mirrored Aiteo’s: acquisition from the same 2015 Shell divestment, rapid production growth, and a later collapse. A consortium led by Midwestern Oil and Gas, with the Canadian-listed Mart Resources and Suntrust Oil and Gas, acquired a 45 percent interest in OML 18 from Shell, Total, and Eni in March 2015 for about \$1.2 billion. NNPC held the remaining 55 percent, and Eroton was the designated operator (Mart Resources, 2015; Templars, 2015). Production then grew from a low base, and two independent sources agree on its scale and trajectory once measurement basis is held constant. Wood Mackenzie’s modeled series puts the field near 32,000 barrels per day at its 2019 peak. The operator’s own disclosures, released through its listed financier San Leon, report full-field gross output before pipeline losses of roughly 39,000 to 45,000 barrels per day across 2017 to 2019, about 50,000 on a producing-day basis. Delivered sales after downtime and allocated losses came to roughly 26,000 to 31,000 (Wood Mackenzie, 2026; San Leon Energy, 2017; 2019; 2020). The spread between these figures follows from the measurement bases: the difference between wellhead potential, annual gross, and the volume that survived a pipeline-loss wedge running between 26 and 35 percent in these years. The crude-reconciliation mechanism Wood Mackenzie documents for the entire Bonny system and that Eroton addressed by installing custody-transfer metering in 2018 (San Leon Energy, 2019; Wood Mackenzie, 2026). On production and technical cost, Eroton looked competent for several years, and NNPC recognized it as one of two operators with the lowest technical cost per barrel (Wood Mackenzie, 2026). That initial competence separates Eroton from Aiteo, where the well-integrity exposure later implicated in the blowout existed from acquisition, and it is the first reason to treat Eroton as a partial replication, short of a literal one.

The composition of the acquiring consortium is a secondary, mechanism-consistency indicator, set out here because the verdict that follows relies on it. Financial and business-development principals assembled Eroton, in place of former international-oil-company operators. Its lead figure, Onajite Okoloko of Midwestern Oil and Gas, holds a degree in economics and a Harvard management qualification and built his career in corporate finance and oilfield services. He was managing director of Oando Energy Services, then in a marginal field through Midwestern and in fertilizers through Notore, his career falling outside

well engineering and reservoir management (Africa Oil & Gas Report, 2013; 2015). The contrast with Seplat guards against an easy objection. Okoloko had operated a marginal field, Midwestern's Umusadege at about 13,000 barrels per day, just as Seplat's Austin Avuru had run Platform Petroleum, so marginal-field experience was common to both. What differed was that Avuru carried three decades of hands-on petroleum engineering and reservoir management that scale to a complex asset, where a business-development background scales less readily. Marginal-field success was a necessary condition for a major lease, one step on a longer path. The consortium that took OML 18 held the financial and commercial capability to fund and acquire the asset, while the inherited upstream operating depth to improvise under stress lay beyond it. The background falls below proof that this gap caused the failure; the internal record does that work.

Then delivered output fell to almost nothing. Export volumes collapsed through 2020 and 2021, and by the time NNPC removed Eroton as operator in March 2023 the asset had delivered near-zero export volumes for roughly two years (NNPC via BusinessDay, 2023). The NEITI-audited figure for 2021 had already fallen to about 19,300 barrels per day. Wood Mackenzie's throughput record for the Bonny system shows the OML 18 evacuation fields, Cawthorne Channel and the OML 18 portion of Awoba among them, dropping to zero from 2021, which corroborates the delivery collapse from a source independent of the operator (NEITI, 2021; Wood Mackenzie, 2026). The 2022 audit makes the split explicit: OML 18 metered about 9,000 barrels per day at the flow station but fiscalized only 600 after a terminal adjustment of 1.6 million barrels, the wellhead-to-delivered gap that defines evacuation stranding (NEITI, 2022). The collapse ran through two distinct pathways, and separating them is the work of this case.

The first pathway is external, and it lay largely outside Eroton's control. The Nembe Creek Trunk Line, the export route for OML 18 crude, went into force majeure from the fourth quarter of 2021, and every operator that depended on it went without deliveries (San Leon Energy, 2022; 2023). The line's unavailability extended well beyond that quarter. Wood Mackenzie records the Nembe Creek Trunk Line as frequently targeted by vandals and militant groups, with downtime that can reach 60 percent, and notes that the \$1.1 billion line Shell rebuilt in 2010 specifically to deter theft proved similarly ineffective against it. Terminal force majeure on the Bonny system, the same source observes, follows generically from vandalism or sabotage against the trunk pipelines, a system-wide hazard that falls on every operator on the line alike (Wood Mackenzie, 2026). The vulnerability was therefore structural to the asset and the region, it predated Eroton and Aiteo alike, and it transferred with the hardware from Shell, which had itself repeatedly shut the line to remove theft points. Eroton kept producing at the wellhead through the force majeure, reporting on its own account about 500,000 barrels in November 2021, an interested-source figure of roughly 16,700 barrels per day for the month. The crude meanwhile stayed short of the Bonny terminal (San Leon Energy, 2023). The gap between metered wellhead volume and delivered volume is the signature of evacuation stranding: the crude is produced and metered, then held before it can be delivered.

The line that stranded Eroton's crude was operated by Aiteo on OML 29, a link of shared exposure to the same line. The official joint investigation into the contemporaneous Santa Barbara incident attributed it to third-party interference (Section 4.3), and the trunk-line force majeure's recur on the same third-party-sabotage pattern across operators and

across the periods before and after Aiteo took the line. The available record leaves any Aiteo maintenance or surveillance failure on the line inseparable from the regional theft and sabotage that afflict it. The evacuation shock therefore reads best as the common environment that every Bonny-system operator faced, a feature of the asset more than of either operator's capability. A community suit now before the Federal High Court alleges that Aiteo's trunk-line spills reflected operational negligence and a failure to replace corroded pipelines, which, if established, would reopen that reading. As of this writing the suit is pending, with a finding yet to be made (*Opu Nembe Kingdom v Aiteo*, FHC/YN/CS/284/2024).

The second pathway is internal and concerns whether Eroton could mount an organizational response to that shock. The Federal Inland Revenue Service sealed the company's head office for more than twelve months over unpaid taxes; the EFCC and NUPRC opened investigations; and an NNPC forensic audit identified governance failures (NNPC via *BusinessDay*, 2023; NNPC via *Leadership*, 2023). Most directly, an alternative barge evacuation route won approval, and Eroton left it unexecuted, so the crude beyond the trunk line's reach stayed stranded under the sanctioned workaround as well (*San Leon* via *Energy Voice*, 2019, 2022a, 2022b). These are organizational-performance failures, and they fall in the part of the operation that Chapter 2 treats as organizational capability: managing a regulator, sustaining joint-venture governance, and executing a complex logistics alternative under stress are capability tasks, the organizational work that sits above the wellhead. They are consistent with an organizational-capability gap. The documentary record alone, however, leaves that reading entangled with governance, liquidity, or political dynamics, and participant testimony is the evidence that would distinguish them.

The two pathways resist clean separation, and the case is weaker for it. The external shock alone could have stranded a capable operator; the internal failures alone could have sunk production even with the trunk line intact. In the event both occurred together, and the documentary record can only partly apportion how much of the two-year collapse each contributed. That is the honest limit of the case, and it is why Eroton corroborates the Aiteo pattern while leaving it unconfirmed.

Within that limit, the explanation tests run as follows. Capital constraints are rejected as sufficient. The growth to a modeled 32,000 barrels per day, corroborated by the operator's reported gross output, demonstrates that capital was available for production-directed major operational spending. An operator that funded that growth held the funds to run the asset (*Wood Mackenzie*, 2026; *San Leon Energy*, 2019). NNPC separately stated that the joint-venture partners met their funding obligations through the decline. That statement carries a caveat, because NNPC made it while removing Eroton as operator, a context in which it had an interest in placing the failure on the operator over joint-venture underfunding. The inference against capital constraints therefore rests more securely on the documented production growth than on the statement. The unpaid-tax evidence may indicate liquidity stress or governance failure, though on its own it leaves open whether the joint venture held the capital to sustain production or to execute the evacuation alternatives, so the verdict stands. Eroton's own rebuttal accepted that funding was maintained; it attributed the collapse to the trunk-line force majeure (*Eroton* via *Petroleum Africa*, 2023). Fiscal regime effects are rejected for this case. The production deterioration began before the Petroleum Industry Act, and the acquisition, capability, and evacuation-dependence conditions that produced it were all established under the pre-Act regime. The Act, enacted in August

2021, postdates the conditions that produced the collapse and stays outside the primary explanation, even though the removal of the operator ran into 2023. The regulatory framework in force at the 2015 acquisition left operational and organizational capability outside its screen, the same design gap documented for Aiteo. It permitted Eroton's acquisition while, applying alike to every operator, leaving unexplained why Eroton specifically collapsed, and it functions here as an enabling condition shared across operators, short of a differentiating cause. Coordination failure is rejected as independently caused: the collapse came six to eight years after acquisition, well beyond a short transition period, and a transition problem should appear early and ease as the operator settles. Here, by contrast, the operator demonstrated routine operational competence for several years before the collapse.

The political-economy and capability readings are harder to separate here than at Aiteo, and the evidence is thinner. Corporate registry detail on Eroton's board composition and shareholding is incomplete, so the question is carried by the non-substitutability logic, the registry record being too incomplete for a direct case-level test. Political economy remains live for the case as a whole; it could plausibly bear on the unpaid taxes, the regulatory treatment, or access to approvals. It is rejected only as a sufficient explanation for the operational-capability tasks at issue, in particular executing the approved barge alternative, which lay beyond political access. The capability reading is consistent with the internal pathway but not decisive for the case, because the external trunk-line shock confounds the attribution. The consortium's business-development composition, established above, supports the reading only as a secondary indicator: it is consistent with an organization that ran routine production well while struggling to improvise an operational response under stress. The background indicates the gap without establishing it as the cause; the operational record in each case establishes the link.

Eroton therefore replicates the Aiteo pattern in part. An operator without inherited upstream technical depth grew routine production and then faltered when conditions deteriorated, which matches the Aiteo trajectory. The collapse, however, mixed an external evacuation shock with internal governance failure, and the evidence attributes it to capability with less confidence than the Aiteo case permits. The verdict reflects that ambiguity.

Appendix E records the full verdict matrix for the case, with the evidence and the falsification condition behind each verdict.

The case adds two findings without amounting to a second proof. It rejects capital constraints and fiscal timing on the same logic as Aiteo, and it shows the funded-growth-then-failure pattern recurring in a second operator assembled without inherited upstream depth. The evacuation shock that might have supplied a second, independent line of evidence instead confounds the capability attribution: it ran through the trunk line as the common environment every Bonny-system operator faced, structurally theft-driven and inherited from the seller. Whether the pattern holds where the conditions differ on knowledge inheritance is what the theoretical replications test, against operators facing the same evacuation environment (Section 4.5).

4.4.2 First Hydrocarbon on OML 26: A Test of Regulatory Screening

First Hydrocarbon Nigeria acquired a 45 percent interest in OML 26, in the western Niger Delta, from the Shell joint venture in 2011, completing an agreement struck in October 2010. The interest combined the holdings of Shell at 30 percent, Total at 10 percent, and Eni at 5 percent, with the state company NPDC holding the remaining 55 percent (First Hydrocarbon Nigeria, 2011; Wood Mackenzie, 2026). Two features set the case apart from Aiteo and Eroton. First Hydrocarbon Nigeria was a special-purpose vehicle created by the London-listed Afren plc to hold Nigerian upstream assets, and Afren held 45 percent of it at completion, rising to 70 percent by the time the company failed (Upstream Nigeria, 2018). And First Hydrocarbon Nigeria never operated OML 26: NPDC asserted operatorship under a first-refusal clause in the joint operating agreement and ran the asset throughout, with a joint asset-management team formed only in 2016 (Africa Oil and Gas Report, 2012; Acha et al., 2019). The buyer was a non-operating financial partner whose sponsor collapsed through fraud, a corporate failure that played out entirely away from the asset's operations. For both reasons the case lies outside a test of the buyer's operational capability, and it enters here as a screening case. What it tests is whether the approval framework assessed buyer fitness, and it provides the cleanest rejection of capital constraints in the case set.

The screening question turns on the statute, and the statute is explicit. Section 29 of the Petroleum Act required ministerial consent to assign the lease while leaving the acquirer's fitness, capability, and governance untested, the same statutory void documented for the Aiteo transfer (Section 4.3.3). The buyer commanded ample capital and corporate standing: Afren was a FTSE 250 company that peaked near \$2.6 billion in market value. First Hydrocarbon Nigeria funded the acquisition and its initial development share through \$280 million of facilities (First Hydrocarbon Nigeria, 2011; UK Serious Fraud Office, 2018). The acquiring entity's gap lay in corporate governance integrity, a dimension that the approval framework reviewed here left outside any party's remit, at approval and through any ongoing post-transfer fitness test. Within four years of completion, Afren had collapsed into administration, in July 2015. Its two most senior executives, the chief executive Osman Shahenshah and the chief operating officer Shahid Ullah, were convicted in 2018 of fraud and money laundering for a scheme they ran while leading the company. They diverted roughly \$45 million from a \$300 million transaction with another Nigerian partner into a Caribbean or Bermudan shell company they controlled (UK Serious Fraud Office, 2018). The convictions stood when the Court of Appeal refused permission to appeal in 2019, and a confiscation order of £5.45 million followed in 2020 (Court of Appeal, 2019; Energy Voice, 2020). The case leaves open whether a Nigerian regulator would necessarily have discovered the fraud in 2010 or 2011; it shows that the approval framework lacked any explicit buyer-fitness or governance-integrity screen capable of making that question part of the approval decision. The cost of the collapse reached beyond Afren's shareholders: Nigerian banks that had lent into the structure lost an estimated \$185 million when the company failed (Vanguard, 2015).

Capital constraints are rejected as sufficient. The binding-constraint version of the capital explanation requires that a shortage of funds prevented the acquirer or the venture from running or developing the asset. Here the acquirer commanded the resources of a multi-billion-dollar listed company and dedicated financing facilities, and the venture failed regardless,

through a governance fraud that stood apart from funding scarcity. Capital abundance left both the failure and the asset’s performance untouched.

Appendix E records the case’s verdicts, including the dimensions it leaves not tested, with the evidence and the falsification condition for each.

Because NPDC operated OML 26, the asset’s production reflects the state operator’s performance, the buyer’s role aside. The modeled and audited records agree that OML 26 produced only two to three thousand barrels per day through the mid-2010s and reached roughly eight to ten thousand around 2020 to 2021. The audited record then shows output slipping back, well below the asset’s developed capacity throughout (Wood Mackenzie, 2026; NEITI, 2020, 2022). That trajectory bears on NPDC’s operatorship, taken up in Section 4.5.3, with the buyer’s inherited capability set aside.

The case adds a dimension that neither Aiteo nor Eroton supplies. At Aiteo the screening gap permitted an operator that proved unable to contain a crisis. Here the same statutory gap permitted an acquirer whose leadership was later convicted of serious fraud committed while running the company, on an asset where the buyer never took operational control. The two cases together show that the pre-2024 framework left acquirers unscreened for both operational capability and governance integrity, the void the NUPRC moved to close in 2024 (Section 4.3.3). And the case rejects capital constraints in their starkest form, since here the buyer held every financial advantage and failed anyway. Whether the capability pattern that the critical experiment identified holds where a capable operator inherits and runs the asset is the question the theoretical replications take up next (Section 4.5).

4.5 Theoretical Replication: Operators That Differed on Capability

The critical experiment identified a capability-related failure at Aiteo, and the partial replication at Eroton showed a similar funded-growth-then-collapse pattern, though there the attribution was confounded by an external evacuation failure the documentary record leaves entangled with internal governance (Section 4.4.1). Literal replication predicts the same outcome where the same condition holds; theoretical replication predicts a contrasting outcome where the condition differs, and for a reason the theory names (Yin, 2009). The three cases here differ from Aiteo and Eroton on capability. Seplat acquired its leases with an experienced founding team and a technical partner, and the capability account predicts sustained production and integrity (Section 4.5.1). OML 30 supplies a within-asset comparison in which the asset, owners, regulatory regime, and fiscal terms held constant while operatorship passed from the state company to an indigenous operator with technical capability, which isolates operator capability more sharply than any cross-operator comparison can (Section 4.5.2). NPDC operated its acreage as a state company and underperformed while surviving commercially, which tests political economy in the positive form left open earlier (Section 4.5.3). All three operated under the same pre-2024 regulatory regime as the failure cases, NPDC as a state entity under, if anything, weaker external discipline, so the contrast among them also tests whether regulatory inadequacy, common to every operator, can account for outcomes that diverge.

4.5.1 Seplat on OMLs 4, 38, and 41: A Founding Team’s Technical Depth

Seplat entered upstream operations with the technical depth the failure cases lacked, and it sustained both production and operational integrity, the outcome the capability account predicts for a capable operator. The case is the first theoretical replication.

Seplat Petroleum Development Company acquired a 45 percent interest in OMLs 4, 38, and 41 from the Shell joint venture in 2010 for about \$340 million (Seplat, 2014). The depth originated with the buyer. The reviewed record for the 2010 transaction shows no structured capability-transfer provisions, and the knowledge that ran the assets came from the founding team’s own prior careers. Austin Avuru, the founding chief executive, had spent more than three decades in Nigerian petroleum engineering and reservoir management, beginning at NNPC as a wellsite geologist and reservoir engineer and later running the marginal-field operator Platform Petroleum. Basil Omiyi, who had led Shell’s Nigerian operations, joined the board after the founding (Seplat, 2014). Where Eroton’s principals brought a business-development background, Seplat’s brought operating depth, and the company paired it with a technical partnership with Maurel et Prom, an established international operator (Seplat, 2014). The founding team and the partnership together gave Seplat the absorptive capacity to run the assets, the buyer-side condition specified in Chapter 2; capability was present at the buyer from the start, brought by the founding team itself.

Production grew and held. The Wood Mackenzie modeled series puts the three leases near a combined 53,000 barrels per day at their 2018 peak, and the audited record shows metered production of 42,300 in 2021 and 43,100 in 2020. The modeled series holds the leases near 39,000 in 2025, about 74 percent of peak and inside the sustained band (Wood Mackenzie, 2026; NEITI, 2020, 2021). Where Aiteo and Eroton fell to small fractions of their post-acquisition peaks (Section 4.2), Seplat held above 70 percent of its own.

The available integrity indicators point in the same direction, though they rest on different measurement bases. Seplat reported more than 11 million man-hours without a lost-time injury and was named AfriSAFE Energy Company of the Year in 2024, and its joint-venture flaring ran below the indigenous-operator figures in the 2021 audit (NEITI, 2021; Seplat Energy, 2024). The flaring comparison is confounded by gas-evacuation infrastructure and carries little decisive weight, and the award is a form of recognition, carrying less evidentiary weight than the audited regulatory data, so the indicators support the reading jointly, as a set. Set against Aiteo’s uncontrolled blowout and the weaker integrity indicators in the failure cases, Seplat’s safety record points in the expected direction and is consistent with the capability account.

A later seller-independence check points the same way and is developed in the cross-case synthesis (Section 4.7). In 2024 Seplat acquired Mobil Producing Nigeria Unlimited from ExxonMobil and integrated a producing asset from a different seller (Seplat Energy, 2024, 2026). Because that transaction is only months old, it stands outside the historical evidence here; the absorptive-capacity reading rests on the fifteen-year record of the 2010 leases, and the MPNU integration enters as a design contrast, short of a second case.

The case establishes that a buyer with inherited technical depth, under the same regulatory and fiscal regime as the failure cases, sustained the outcomes that eluded the failure cases. The case leaves open which capability element, the founding team or the Maurel et Prom

partnership, was decisive, since both were present throughout. The three leases also differ in asset characteristics from OML 29, so the comparison is one of operator type under common institutional conditions, a contrast short of a matched-asset experiment.

The full verdicts for this case are in Appendix E. The question for the alternatives is whether any of them accounts for why Seplat succeeded where capability-poor operators failed.

Seplat's outcome matches the capability prediction. An operator that brought the technical depth Aiteo and Eroton lacked produced the sustained production and operational integrity that eluded them, under the same regime that governed them all.

4.5.2 OML 30: A Within-Asset Operator Change

OML 30 offers the strongest control available in this study. The asset, its owners, the regulatory regime, and the fiscal terms all held constant while operatorship changed, first the state company and then an indigenous operator with technical capability, and the outcome changed with it. The comparison isolates operator capability more sharply than any comparison across different leases can, and it is the second theoretical replication. Within the asset's sociotechnical system, the change holds the technical subsystem constant and varies the social subsystem that operated it, so the recovery isolates the contribution of the operating organization (Section 2.2).

Shoreline Natural Resources, a consortium of the Nigerian company Shoreline Power and Heritage Oil, acquired the 45 percent interest held by Shell, Total, and Eni in OML 30 in 2012 for about \$850 million. NNPC's subsidiary, the Nigerian Petroleum Development Company (NPDC), held the remaining 55 percent and operated the asset (Africa Oil and Gas Report, 2014; Heritage Oil, 2020). Under NPDC the lease underperformed: the Department of Petroleum Resources judged the operator's performance unsatisfactory in 2015 and recommended replacing it. The modeled series shows production falling to a trough near 3,800 barrels per day in 2016 amid a Trans Forcados Pipeline shutdown (ThisDay, 2015; Wood Mackenzie, 2026). In 2017 the government approved transfer of operatorship to Heritage Energy Operational Services Limited, hereafter HEOSL, the indigenous Nigerian operating company, whose parent Heritage Oil Limited is UK-listed. Salvic Petroleum Resources served as third-party technical operator under a technical services agreement (Heritage Oil, 2020; Orient Energy Review, 2018).

The outcome reversed. NEITI audited the field's metered production at about 24,000 barrels per day in 2021 and 30,000 in 2022, and the operator reports production reached a peak near 67,000 by 2025, a level last reached in the 1980s (NEITI, 2021; The Energy Year, 2026). The recovery was achieved without new drilling: independent reporting records production held above 45,000 barrels per day across three years without new wells. The operator's first exploration drilling on the lease in more than three decades began only in 2025 (Africa Oil and Gas Report, 2025; The Energy Year, 2026). The operator's earlier claim of about 80,000 barrels per day gross runs ahead of the audited and modeled record and is set aside here. The documentary record surveyed stays silent on any major safety or environmental incident, though that silence counts as weak evidence, well below proof of integrity performance.

The within-asset design carries most of the analytical weight. The lease, the owners,

the regulatory regime, and the fiscal terms were identical before and after; the principal change in the record was operatorship, although governance, incentives, and the technical-service arrangement changed with it. The recovery without new drilling removes new-drilling capital as the differentiator and weakens a pure capital explanation; it leaves in place the rehabilitation, pipeline-restoration, and well-optimization spending the turnaround required, so the remaining contrast turns on how the existing asset and that spending were managed. The capability arrived through specific people. The Salvic team included an engineer with twenty-seven years at Shell's Nigerian operating company and a second with thirty-nine years spanning NNPC's upstream division and project work across the major joint ventures, who had managed earlier Shell-to-NPDC operatorship transfers (Salvic Petroleum Resources, n.d.). The contrast with the prior operator therefore runs along capability, with nationality common to both. NPDC and the HEOSL and Salvic team were both Nigerian; what separated them was upstream technical capability embedded in the operating organization. The case shows that an indigenous operator succeeds when that capability is present and the same asset underperforms when that capability is absent.

The comparison tests the capability prediction in its sharpest form and rejects regulatory inadequacy as the differentiating cause, since the regime held constant across the operator transition. The comparison controls for much while leaving governance and funding differences between a state company and a commercial operator, which changed alongside operatorship, so the within-asset isolation is strong while incomplete. With the hardware and the regulatory regime unchanged across the transition, the case shows that the operating organization's embedded capability drove the outcome.

OML 30 demonstrates the capability account on a single asset. The same lease, under the same regime and owners, underperformed under a state operator and recovered under an operator carrying embedded upstream capability, with new-drilling capital held constant by the absence of new wells.

The verdict matrix for the case, with the evidence and the condition that would overturn each verdict, is in Appendix E.

4.5.3 NPDC: A State Operator and the Limits of Political Access

NPDC underperformed operationally while surviving commercially, and the case tests political economy in its positive form. State backing sustains commercial viability without supplying operational capability, and the case separates the dimension the political-economy account speaks to from the dimension the capability account speaks to. It is the third theoretical replication, and it supplies the positive evidence on political economy that the critical experiment left open (Section 4.3.4).

NPDC operates as the upstream production subsidiary of the national oil company and holds its acreage through the state company's participating interests, the acreage coming to it as state equity. Its operational record is the weakest of the operator categories. It set a target of 500,000 barrels per day for 2020, from about 180,000 in 2017, and fell short of it, on a company-wide basis that differs from the lease-level series (Vanguard, 2017). Its technical efficiency stood at 23 percent on the Adekunle and colleagues measure, against about 75 percent for joint-venture operators and 67 percent for marginal-field operators measured the same way (Adekunle et al., 2019). And the within-asset comparison gives the deficit a

case-level anchor: on OML 30, NPDC's operatorship ran the lease into a formal finding of unsatisfactory performance and a trough near 3,800 barrels per day before the recovery under HEOSL (Section 4.5.2; ThisDay, 2015). A second NPDC-operated lease points the same way on the audited record. On OML 26, the lease where the First Hydrocarbon acquisition failed the governance screen and which NPDC operated thereafter (Section 4.4.2), NEITI audited metered production at about 8,800 barrels per day in 2021 and 3,700 in 2022, with 2022 fiscalized volume about 3,200. Because both the field-metered and the terminal-fiscalized figures are low, the 2022 shortfall reflects production itself, whereas Eroton's fiscalized volume was depressed by evacuation stranding. On the same evacuation route, in the same year, the HEOSL-operated OML 30 rose, which points the contrast back to operatorship (NEITI, 2021, 2022).

NPDC nonetheless continues to operate across multiple licenses. A private operator with this record would, by inference, on a point the documentary record leaves open, face stronger market, lender, partner, or regulatory discipline; state ownership sustains NPDC where such discipline would otherwise bite. This is the positive form of the test the critical experiment left open. At Aiteo, preventing or containing the blowout turned on well-control capability beyond political access, so political economy was rejected as sufficient for operational quality while left open as an account of commercial survival. The NPDC case supplies the positive side: state backing sustains commercial viability without producing operational quality. The two accounts operate on different dimensions, with some overlap between them. Political and state factors can sustain commercial survival and may also weaken the discipline, incentives, and accountability that bear on performance, while the technical and organizational routines that determine operational quality answer to capability instead. That distinction is the substitutability question the cross-case synthesis develops (Section 4.7), and NPDC supplies its clearest single-case illustration.

NPDC tests political economy in its positive form and tests capability as a state-entity deficit. The closest the evidence comes to a matched comparison is the within-asset OML 30 test; NPDC's multi-license, state-equity structure and company-level reporting differ too far from the lease-level acquisitions for a full match. The case establishes the direction of the two effects while leaving their precise magnitude open.

The verdicts for this case are recorded in Appendix E. The case carries two findings on different dimensions, which the matrix records separately.

In NPDC the two dimensions appear separately, where the other cases show them jointly. A state operator survives commercially on backing beyond a private firm's reach, while underperforming operationally in the dimension capability governs.

The three theoretical replications strengthen the pattern the critical experiment identified, working through contrast. Seplat carried capability through its founding team and sustained its assets. OML 30 recovered when an operator with embedded capability replaced the state company on the same lease, with new-drilling capital held constant. NPDC, a state operator without that capability, underperformed operationally even as state backing kept it in business. Capability tracks operational quality across the cases; capital, fiscal timing, and the regulatory regime were common to operators whose outcomes diverged, and political economy bears on commercial survival, while operational quality tracks capability. The marginal field program tests the same pattern through an earlier and independent transfer mechanism (Section 4.6), and the cross-case synthesis then renders the consolidated verdict

and weighs the substitutability and externality conditions that bear on the third research question (Section 4.7).

4.6 The Marginal Field Program: A Structural Precedent

Every primary case originated from the same seller. The Aiteo, Eroton, First Hydrocarbon, Seplat, and Shoreline transactions, and the NPDC equity holdings, all came out of the Shell-operated joint venture in which Total and Eni were the other international partners. The pattern those cases show could in principle be specific to how that venture prepared and handed over its assets, leaving open whether the pattern is a general feature of indigenous-operator capability. The cross-case synthesis tests this directly through three non-Shell transactions, examined as seller-independence checks (Section 4.7). The marginal field program adds a structural precedent on a different axis. It moved assets to indigenous operators through a government licensing round in place of a bilateral sale, under different fiscal and regulatory rules, a decade before the divestment wave. The fields were carved from the acreage of several majors, Shell, Chevron, and Total, with Shell only one among them (Wood Mackenzie, 2026). If a similar pattern of capability-unassessed transfer followed by underperformance appears there as well, it is less likely to be an artifact of any single seller's divestment practice. The program carries corroborating, secondary weight: individual marginal-field operator trajectories are less documented than the divestment cases, so it supports the chapter's pattern at the cohort level, short of supplying another decisive case.

In 2003 the Department of Petroleum Resources awarded 24 marginal fields, small accumulations the majors had left undeveloped, to 31 indigenous companies, with the stated objective of building indigenous technical capacity (Sweet Crude Reports, 2020). The allocation turned on signature bonuses, work-program commitments, and farm-out arrangements with the majors, while leaving the recipients' capability unscreened, short of the technical-capacity criteria the NUPRC introduced in 2024, as the development record and the regulator's later criticism both indicate. The round was also litigation-prone, with disputes between awardees and their partners stalling development for years (Sweet Crude Reports, 2020).

The development record was poor and slow. Ten years after the awards, only about seven of the 24 fields were producing, together contributing roughly 30,000 barrels per day (Africa Oil and Gas Report, 2013). The regulator revoked eleven operators' licenses for non-performance over the program's life, in several cases after more than sixteen years of holding an asset without submitting a field-development plan or reaching production, and reassigned some of the fields to new operators (ThisDay, 2021). Development came slowly and unevenly. Wood Mackenzie's field-level record for the cohort shows 17 fields onstream, 8 idle, and 3 revoked, a split consistent with the regulator's own tally of 17 of 30 producing by 2022 (Wood Mackenzie, 2026; NUPRC, 2022). The denominator varies across sources, since the 2003 round awarded 24 fields, the regulator's count of 30 spans all awards from 1999 to 2010, and Wood Mackenzie tracks 28 with a current status. The figures are used directionally, with the source-specific denominator stated in each instance. Many fields therefore reached production eventually, but only after fifteen years or more and repeated regulatory intervention, and the cohort's realized contribution stayed near 2 percent of national reserves. The regulator's

own director stated that the 2003 round brought Nigeria little benefit (Sweet Crude Reports, 2020). The 2020 round of 57 fields then repeated the pattern: many awardees, described by the regulator as newcomers with limited operational and financial capabilities, struggled to move to development, and a number of the assets lapsed by 2025 (NUPRC, 2022).

Marginal fields are smaller, less proven, and economically more marginal than the producing OMLs, so a high failure rate among them could reflect asset quality as much as any absence of capability. Two features bound that confound. The fields were undepleted. Wood Mackenzie’s reserves accounting shows the cohort’s undeveloped fields holding their initial reserves in full, 78 million barrels of oil and 584 billion cubic feet of gas still in the ground. The 2020 round carries roughly 1.6 billion barrels of largely undeveloped oil across its 57 fields (Wood Mackenzie, 2026). The reserves evidence weakens a depletion explanation; it leaves open that field size, economics, infrastructure access, or technical complexity contributed to non-development. And the difficulty concentrated at the development stage, in operators that took many years to reach production, or stalled short of it, which is a capability-and-commitment outcome more than an asset-grade one. The confound stays in place to some degree, and the claim is correspondingly bounded: the marginal-field record is consistent with the institutional-design interpretation, short of proof on the part asset quality played.

The two mechanisms also pose different tests. A divestment hands over a producing asset with an embedded operating system, where the question is whether capability transfers with the hardware; a marginal-field award hands over an undeveloped accumulation, where the question is whether the recipient can build capability from the start. The marginal field program therefore corroborates the broader proposition, that transferring assets to operators whose capability goes unassessed produces failure, short of replicating the specific inheritance mechanism of the divestment cases. On that broader proposition the two mechanisms converge, as Table 4.4 sets out.

Table 4.4: Two transfer mechanisms, one shared gap. Although the IOC divestment wave and the 2003 marginal field round differ in almost every term, both handed assets to indigenous operators whose capability went unscreened, and both produced widespread underperformance; across two programs otherwise unlike, what they share is the absence of a capability assessment.

Feature	IOC divestment (2010-2024)	Marginal field allocation (2003)
Transfer mechanism	Bilateral negotiated sale	Government licensing round
Asset originator	Shell, Total, Eni	Federal government, from fields the majors relinquished
Asset condition at transfer	Producing OMLs with an embedded operating system	Undeveloped marginal accumulations
Capability assessment of recipient	None	None

Feature	IOC divestment (2010-2024)	Marginal field allocation (2003)
Documented outcome	Differentiated; most operators examined here underperformed on at least one dimension	Slow, intervention-heavy development: 7 of 24 producing at ten years, 11 licenses revoked, 17 onstream by 2022 against near 2 percent of reserves; regulator judged the benefit to Nigeria minimal
Shared institutional feature	Absence of capability assessment	Absence of capability assessment

The two mechanisms differ on nearly everything the alternative explanations turn on. The seller differs, a private major in one and the government in the other; the fiscal terms differ; the regulatory rules differ; and the asset condition differs. What they share is the absence of any assessment of the recipient’s capability to operate the asset, and both produced widespread underperformance among the operators who received assets under that gap. The shared institutional feature across two otherwise-dissimilar mechanisms is the absence of a capability assessment, the regulatory design gap identified here, the one feature common to both programs whatever else divides them.

The marginal field cohort is also the one context in this chapter where capital constraints have genuine purchase. The 2003 awardees were small companies, and the regulator attributed the 2020 round’s stalled development in part to the awardees’ limited financial capacity (NUPRC, 2022). For these operators, underfunding and a capability gap are confounded: a company too small to fund a development program is typically also a company without the technical organization to run one, so the cohort leaves capital and capability entangled. What it does do is locate where capital constraints bite. Where operators were genuinely underfunded, as here, capital contributed; where operators were well funded, as Aiteo and Eroton were, their failures fall outside a funding story, and capability is what separated them from Seplat. The marginal-field record places capital’s effect at the small, underfunded end, away from the well-funded divestment failures.

The cohort contains a within-population signal as well. Two of the figures central to the divestment cases came through the marginal field program first. Austin Avuru’s Platform Petroleum operated a marginal field at about 2,100 barrels per day before he founded Seplat, and Onajite Okoloko’s Midwestern Oil and Gas operated one at about 13,000 barrels per day before he co-founded Eroton (Africa Oil and Gas Report, 2013). Both cleared the marginal-field bar, and at the major-OML scale their outcomes diverged sharply, Seplat sustaining its assets and Eroton declining. Okoloko’s marginal field was the larger of the two, yet his consortium’s OML 18 collapsed while Seplat’s assets held. The divergence is consistent with the argument that marginal-field success alone fell short at scale: domain-specific petroleum-engineering depth mattered once the operating challenge moved to major-OML complexity, a reach Avuru’s background had and business development lacked.

The program bears on the explanations as structural corroboration. The marginal-field

outcomes all predate the PIA, so fiscal regime effects are rejected for this cohort as for the others. The allocation framework omitted a capability test, the regulatory design gap in its most direct form, so regulatory inadequacy stands here too as an enabling condition that permitted the failures while leaving open which operators would fail. Capital constraints contributed for this underfunded cohort, the one place they do, without explaining the well-funded divestment failures. And the cohort is consistent with the capability account. Operators who received assets without an assessed capability to run them underperformed at the cohort level under a licensing mechanism wholly unlike IOC divestment. Development that took fifteen years or more for many of the fields, with eleven licenses revoked along the way, is consistent with capability resisting reconstitution by time alone. The contemporaneous record agrees, tracing the high failure rate among marginal-field licensees to inadequate technical and commercial due diligence at the point of award (Idigbe et al., 2013; Danmadami et al., 2021). At the cohort level, the marginal-field record is consistent but not decisive for the capability account, and it shows capital constraints contributing for genuinely underfunded operators.

The marginal field program shows that the pattern the divestment cases display, asset transfer to capability-unassessed indigenous operators followed by widespread underperformance, appears under a second mechanism that stands apart from the Shell seller. The cross-case synthesis now consolidates the verdicts across the critical experiment, the replications, and this precedent, and turns to the substitutability and externality conditions that bear on the third research question (Section 4.7).

4.7 Cross-Case Synthesis and Discrimination

Across the critical experiment (Aiteo), the partial replication (Eroton), the regulatory-screening case (First Hydrocarbon), the three theoretical replications (Seplat, the OML 30 operatorship, and NPDC), and the structural precedent (the marginal field program), the five explanations have been tested case by case. Together, the cases support a single verdict on each explanation, and three of the five survive in a form that requires discrimination. The core divestment cases share broadly comparable pre-PIA institutional conditions and asset type, so comparison across those operators isolates knowledge inheritance, while the marginal-field and non-Shell cases test whether the pattern travels across different transfer mechanisms and sellers.

4.7.1 Consolidated Verdicts

Capital availability and outcome move independently: every failed operator commanded substantial capital and funded production growth, while Seplat sustained its assets on the most modest outlay of the group, and the sector-wide cost premium falls on every operator alike (Table 4.1; Sections 4.3 to 4.5). Capital constraints are therefore rejected as sufficient for the differentiated outcomes among the adequately funded operators, and consistent but not decisive for the underfunded marginal-field cohort, where they confound with capability (Section 4.6). At Aiteo, capital was adequate but allocated away from crisis readiness when the funds were available (Table 4.5).

Every primary transaction and the marginal-field awards closed before the Petroleum Industry Act of August 2021, so fiscal regime effects, which cannot postdate the outcomes they are said to cause, are rejected for the cases examined, reinforced by the sharp divergence among operators under identical pre-PIA terms; whether the Act shapes future transactions is left open (Table 4.1; Table 4.5).

The identical permissive pre-2024 framework governed every operator and so cannot explain the divergence among them, leaving regulatory inadequacy as the common system-level enabling condition that permitted capability-deficient transfers at scale but is insufficient on its own to differentiate operators (Section 4.3.3). The gap is one of content rather than enforcement, which bears on the form any intervention should take; how it nests with the operator-level capability finding is developed in Section 4.7.2 (Table 4.5).

Political connection tracks commercial survival and leaves operational integrity untouched: NPDC, connected by definition, sustained survival while posting the lowest technical efficiency of any category and a fatal integrity failure, and Rexer's population study shows the same split of connected operators raising output even as quality declines (Section 4.5.3; Rexer, 2025). Political economy is therefore most consistent for commercial survival and insufficient on its own for operational integrity; its boundary with capability, the line between the substitutable and the non-substitutable, is drawn in Section 4.7.2 (Table 4.5).

The positive signature for capability appears in the critical experiment and recurs across the replications: Aiteo's selective failure in a single knowledge-intensive domain while funded production grew (the pattern capability predicts and capital leaves unexplained), the OML 30 within-asset recovery when an operator carrying inherited upstream expertise replaced the state entity, and the correlation across cases between inherited upstream depth and sustained integrity, with Eroton's attribution bounded by the inherited evacuation constraint (Sections 4.3, 4.5.2). Every operator that handled a comparable knowledge-intensive challenge successfully carried inherited upstream capability, so capability-related factors are most consistent with the evidence for operational-integrity outcomes, on positive evidence matching capability's own predictions and identified before any alternative was eliminated; the claim is pattern consistency under pre-specified discrimination, below experimental proof of cause, and the domain-specific character of that capability is developed in Section 4.7.2 (Table 4.5).

The practitioner survey (Appendix F), small and drawn only from sellers, is consistent with the capability emphasis but carries little independent weight and departs from the documentary ranking. All three respondents judged that stronger capability transfer would have improved the outcome they observed and that it should condition divestment approval, two calling it mandatory, yet each ranked a different factor first, a qualification Section 4.8 takes up with the survey's other limits.

Coordination problems are visible in the failed cases but fall short of an independent cause: failure persisted for years, far beyond the transition gap that attends any handover, and concentrated in knowledge-intensive functions while routine ones continued, which points to coordination failing because the knowledge to coordinate was absent (Section 4.3.5). Coordination failure is thus an observable feature of the unsuccessful cases, accounted for at a deeper level by the capability gap; the persistence test does not reach the successful cases (Table 4.5).

Table 4.5: Consolidated verdicts across the cases. After synthesis across the critical experiment, the replications, and the structural precedent, only capability-related factors are most consistent with the evidence on the operational-integrity dimension; regulatory inadequacy is the common system-level enabling condition, political economy explains the separate commercial-survival dimension, and capital, fiscal, and coordination explanations fall away as differentiating causes. Each verdict traces to its supporting evidence in the verdict traceability matrix (Appendix E).

Explanation	Cross-case verdict	Principal basis
Capital constraints	Rejected as sufficient for the funded operators; contributes for the underfunded marginal-field cohort	Funded growth then failure (Aiteo, Eroton); Afren collapse by fraud; Seplat success without a funding advantage; system-level cost premium
Fiscal regime effects	Rejected for the cases examined	All transactions predate the PIA; sharp divergence under identical pre-PIA terms
Regulatory inadequacy	The common system-level enabling condition; insufficient on its own to differentiate operators	Capability unscreened before 2024; identical framework for all operators; the 2024 criterion was new
Political economy	Most consistent for commercial survival and continued participation; insufficient on its own for operational-integrity outcomes	NPDC connection without quality; Rexer population pattern; the non-substitutability boundary; complementary to capability on operational integrity
Capability-related factors	Most consistent with the evidence for operational-integrity outcomes	Selective failure (Aiteo); the OML 30 within-asset comparison; knowledge-inheritance correlation; positive evidence before elimination
Coordination failure	Rejected as independently caused; an observable feature of the unsuccessful cases	Persistence of six to fifteen years with domain-specific failures; the successful cases fall outside the test

4.7.2 Discrimination Among the Surviving Explanations

Three explanations survive the falsification sequence: regulatory inadequacy, political economy, and capability-related factors. Across the formal cases, sustained production and integrity outcomes align more closely with demonstrated upstream operating capability than with

capital, fiscal regime, or regulatory exposure alone. Each surviving explanation addresses a different outcome, and discriminating among them is a matter of identifying which dimension each governs.

Regulatory inadequacy and capability-related factors are nested, the one inside the other. The regulatory gap operates at the system level: by leaving capability unassessed, the framework permitted capability-deficient transfers to proceed, which is the condition under which an operator-level capability gap can produce a failure. Capability operates at the operator level: it explains which operators failed and which succeeded under that permissive condition. One explanation accounts for why the pattern was possible at scale; the other accounts for its distribution across operators. The system-level gap enables the operator-level outcome.

Political economy and capability-related factors survive on different outcome dimensions. The two dimensions move independently, commercial and production viability on one side and operational and sociotechnical integrity on the other. Political connection explains why an operator with access stays commercially viable despite poor quality; capability explains why an operator without upstream knowledge fails in the crisis domains. The boundary between the two explanations is the boundary between the substitutable and the non-substitutable. In the potentially substitutable domains, such as exposure to organized theft, community relations, and commercial standing, connection has real purchase. In the non-substitutable domains, well control, crisis response, and equipment-specific operational practice, only capability explains the outcome. The verdict that capability is most consistent is therefore specific to the operational-integrity dimension, narrower than a single-winner claim across every dimension of performance.

The capability that correlates with outcome is domain-specific upstream operational knowledge, narrower than general organizational competence. The operators that held integrity were led by figures who had spent careers in upstream exploration and production. They included Austin Avuru at Seplat, a wellsite geologist and reservoir engineer with three decades in the field, and the Salvic team that operated OML 30, built around a former Shell production engineer and an NNPC upstream veteran. The operators that failed on integrity were led by figures whose careers were in downstream distribution, trading, or finance. They included Benedict Peters at Aiteo, whose background was in fuel storage and petroleum trading, and Onajite Okoloko at Eroton, whose background was in business development and energy services. The pattern holds across the two dimensions, as Table 4.6 sets out. The background is read here as a mechanism-consistency indicator: it explains why a knowledge gap existed. The causal link in each case is operational and documented. At Aiteo, a well was left undecommissioned and unmonitored, the blowout contingency plan was absent, and prior spills went unremediated, against the well-integrity readiness expected for a suspended well.

Table 4.6: Buyer background and the two outcome dimensions. Operators led by figures with upstream technical backgrounds held operational integrity, while operators led by figures with downstream or finance backgrounds maintained commercial production but failed on integrity, which locates the relevant capability as domain-specific, narrower than general.

Operator (lead figure)	Pre-acquisition background	Commercial viability	Operational integrity
Seplat (Avuru)	Upstream: wellsite geologist and reservoir engineer, three decades in E&P	Sustained	Held; strong safety record
Shoreline / HEOSL, OML 30 (Salvic team)	Upstream: former Shell production engineer, NNPC upstream veteran	Recovered from a trough	Clear of major incidents in the record surveyed (weak positive evidence)
Aiteo (Peters)	Downstream: fuel storage and petroleum trading	Grew, then declined	Failed; blowout and unremediated spills
Eroton (Okoloko)	Finance and business development; energy services	Grew, then collapsed	Weak; governance and delivery failure
NPDC (state entity)	Government E&P; political access; institutional upstream history, short of an embedded capability-development pathway	Underperformed against target	Weak; valve-station fatalities, lowest efficiency

The pattern holds beyond Shell-divested assets. Three transactions with different sellers test whether it is an artifact of Shell’s divestment practice. Seplat’s 2024 acquisition of Mobil Producing Nigeria from ExxonMobil placed a producing asset from a different international seller in the hands of an operator with demonstrated upstream capability. The integration reported full-year 2025 output near 131,500 barrels of oil equivalent per day (Seplat Energy, 2026), consistent with the proposition that an upstream-capable buyer succeeds regardless of who sold the asset. Oando, led by a principal whose background lies in law and trading, acquired ConocoPhillips’ Nigerian interests in 2014 and Eni’s Nigerian Agip Oil Company a decade later, and encountered governance difficulties across that period, including a Nigerian securities investigation and a Deloitte forensic audit. The Oando interests were largely non-operating, with the joint-venture assets operated by others, so the case bears on seller identity and governance, and it enters as a seller-independence check, one the analysis does not carry through the full decision sequence. Seplat’s integration points the same way as the primary cases on the operational-capability dimension. The Oando experience is weaker on that dimension, because the interests were largely non-operating, but it is consistent with

the broader point that post-transfer governance and performance turn on more than seller identity alone.

4.7.3 Substitutability and the Distribution of Failure Costs

The question the theoretical foundation left open is whether financial resources and better management can replicate embedded capability within the timeframes that matter operationally. The cases weigh against it. In the terms of Chapter 2 (Section 2.2), each transaction moved the technical subsystem, the wells and facilities, but not the social subsystem that ran it, and the cases show that the operating organization is not quickly reconstituted through capital and time alone. Aiteo had six years and demonstrable funding and still failed in the crisis domain. NPDC has operated with institutional backing for decades and posts the lowest technical efficiency of any operator category. Much of the marginal-field cohort held assets for fifteen years or more without developing them. Across all three, time and money proved unable to reconstitute operational capability. The successful cases leave the substitution claim untouched. Seplat and the HEOSL operatorship carried capability from the start, through an experienced founding team and an indigenous technical operator. The Renaissance transaction, which structured capability preservation into the deal, shows that such a design is commercially feasible and shows early signs of operational continuity, though with limited post-transfer history it is a design contrast, short of outcome evidence. The evidence across Aiteo, NPDC, and the marginal fields indicates that embedded capability remained an incomplete substitute within operationally relevant timeframes.

Table 4.7: Where political access and capital can substitute for operating capability, and where the cases show they cannot. Connections and funding reduced theft exposure, eased community friction, and sustained commercial survival in the record, while well control, crisis response, and equipment-specific expertise were the domains in which the most serious failures fell.

Domain	Substitutable by political access or capital?	Evidence in the cases
Oil theft exposure	Yes	Rexer (2025): political connections reduce theft losses
Community relations	Partly	Elite mediation eased friction in several cases
Commercial survival	Yes	NPDC operated for decades at the lowest technical efficiency of any operator category
Well control	No	Santa Barbara blowout on OML 29, uncontrolled for 33 days
Crisis response	No	Specialist intervention engaged only after the blowout

Domain	Substitutable by political access or capital?	Evidence in the cases
Equipment-specific expertise	No	After-the-fact engagement of well-control specialists

The costs of post-divestment failure fell largely outside the parties to the transaction. The seller takes some reputational cost and the buyer some financial loss, while the heaviest and least reversible costs fell on those outside the transaction. The Santa Barbara blowout on OML 29 affected dozens of communities across the surrounding area. Seven workers died in the valve-station explosion on OML 40. Other integrity failures in Nigeria’s post-transfer operating environment also imposed costs beyond the transacting parties, though they stay outside the primary case evidence here. Production shortfalls on the transferred assets contributed to revenue losses borne well beyond the transacting parties. Spill sites went unremediated and flaring rose. Because the transacting parties bear only a fraction of these costs, the transaction market leaves much of the cost of failure external: the seller and buyer bear too little of it to change their conduct. That distribution of costs, together with the information asymmetry between seller, buyer, and regulator established earlier, is the structural reason a transaction left to the parties alone will under-provide capability transfer.

Two empirical conditions follow from the cross-case evidence: embedded capability proved an unreliable substitute within operationally relevant timeframes, and the costs of its absence fall disproportionately on parties outside the transaction. These are the conditions the third research question takes up. What intervention they warrant, and what the seller, the buyer, and the regulator must each be required to do, is the work of Chapter 5.

4.8 Limitations and Unresolved Ambiguities

The empirical claims established here are bounded by the limits of the documentary evidence and the discrimination apparatus.

The first boundary is the confound between knowledge inheritance and general organizational quality (Section 2.3.5). Operators that inherited upstream capability tend also to be better funded and better managed, so inheritance may correlate with outcome partly as a proxy for organizational quality. The within-asset comparison on OML 30 is the strongest available control, though it leaves the confound in place, since the change of operator carried management, governance, and contractor arrangements along with the inherited expertise. For cases without a comparable within-operator or cross-case control, the attribution to capability is correspondingly weaker.

The second boundary is documentary. Whether coordination failure was independently caused or followed from absent capability lies, in some cases, beyond what documents alone settle. The persistence test narrows the range, since a coordination gap enduring for years while routine functions continue is hard to read as transitional, though documents reach the internal conditions that would close the question only partly, which participant testimony would be needed to do.

The quantitative evidence has limits of its own. Flare intensity and spill frequency are

indicators from which capability is inferred, indirect measures, and asset age, terrain, and reservoir characteristics stay uncontrolled here, though the OML 30 within-asset comparison holds these constant by construction. The production evidence is similarly measurement-bounded: Wood Mackenzie modeled series, NEITI metered and fiscalized figures, and NUPRC annual production serve different purposes, so audited and regulatory figures are treated as decisive where available and each basis is labeled. The classification holds because the cases separate into wide bands, though the percentages stay approximate.

A further boundary concerns security and external operating conditions. Theft, sabotage, vandalism, and community conflict affect several of the cases and can produce failures that originate outside any operator's control. Such disruptions are treated as external where the record supports it, as with the pipeline-theft evacuation stranding at Eroton, and capability is implicated only where the evidence concerns preparedness, response, governance, or recoverability within the operator's control. This bounds the capability claim and leaves in place the external pressures the operators faced.

The Renaissance transaction is bounded differently. It is too recent to speak to the integrity outcomes that emerge over three-to-five-year cycles, so it enters as a design contrast, carrying less evidentiary weight than the historical cases.

The central claim is calibrated to what the design supports. The result is pattern consistency, without causal identification, and it generalizes through analytic generalization to theoretical propositions, with statistical generalization to all IOC divestments set aside (Yin, 2009). A single set of cases stays below proof of a general theory; it can corroborate one and show it plausible under the conditions examined. The structural conditions that generate the pattern extend beyond the idiosyncrasies of these cases, though whether the findings reach other countries, industries, or institutional structures is left open. The apparatus tests five explanations and treats coordination failure as an observable feature; it can detect variance beyond what the five explanations account for while leaving its mechanism uncharacterized.

The case selection also concentrates on a single seller. Every primary case involves Shell or SPDC, raising the possibility that the pattern reflects Shell's divestment practice. The first and strongest answer is internal to the case set: the outcomes diverged sharply among operators that all acquired from the same Shell-operated joint venture. Whatever produced the divergence therefore operated at the level of the acquiring operator, which a single seller's uniform practice cannot explain. Three further checks bound the concern from outside the set. The marginal field program transferred assets through a government licensing round, from acreage relinquished by several majors, with Shell only one among them, and showed a comparable failure pattern, with an asset-quality caveat bounded in Section 4.6 and treated as structural corroboration. Seplat's integration of the ExxonMobil MPNU assets is an early seller-independence check consistent with the pattern, though too recent to carry the weight of the older cases. Oando's non-Shell transactions bear on governance and seller identity, and because the interests were largely non-operating they make a weaker check than a direct test of operational capability would. The domain-specificity finding holds across sellers: a downstream-trading background would produce the same knowledge gap whichever company sold the asset. Seller concentration remains a structural feature, which the finding survives while acknowledging.

Finally, the practitioner evidence is a small, seller-only, perception-tier sample. The three completed responses (Appendix F) all came from IOC sellers, two of them describing

the Renaissance transaction, where they rated the capability transfer adequate. Each ranked a different factor as the single most important driver of the outcome they observed, ranking untransferred capability lower. Because two of the responses concern Renaissance, a transaction structured to preserve capability, they bear on a case of deliberate capability retention, so the perception evidence from one side of the transaction qualifies the documentary reading while leaving it standing. The survey is reported for transparency and treated as contextual corroboration, with the case-level verdicts resting on the documentary record.

These boundaries define the scope within which the findings hold, and they are why the conclusions in Chapter 6 are calibrated to the cases examined, holding to those cases.

Chapter 5

A Capability Assurance Framework for IOC Divestments

5.1 From Explanation to Design

The empirical analysis answered a question of explanation: why broadly comparable upstream assets, transferred under the same sectoral regime, produced operators that held operational integrity and operators that did not. The explanation most consistent with the differentiated outcomes was capability-related: the operating capability each acquirer brought, retained, or contracted (Section 4.7). The operators that failed had not secured the domain-specific upstream operating knowledge their assets required, and the transaction that transferred the hardware did not transfer that knowledge with it. The question now becomes one of design: what transaction and institutional arrangement would make the capability an asset needs observable before the asset changes hands, testable against the particular asset, and enforceable afterward? Such a design would also protect the value placed at risk when operating capability does not survive the transfer.

The governing idea is a re-specification of an existing rule, not a new regime. NUPRC's 2024 reforms already require an acquirer to demonstrate technical capacity, but they assess it primarily at the corporate level, and corporate-level capacity can pass without any demonstration that the buyer can run and protect the specific asset it is acquiring. That variable is re-specified here as asset-specific operating capability, with the seller disclosure, the structured transfer, and the post-transfer verification the existing rules leave out added to it, and each obligation grounded in a hook the law already contains. The aim is to extend the direction NUPRC has already set, not to work against it.

Why the intervention must be mandatory rather than left to the transacting parties is a matter of where the value sits. Most of what a producing asset is worth to the people around it is borne by parties outside the sale, which is why a price agreed between a willing seller and a willing buyer underprotects it. Those dimensions, in order of how far each falls outside the transaction, are set out in Table 5.1.

Table 5.1: What operating-capability gaps put at risk: the value dimensions of a divested asset. The two dimensions measured in the empirical analysis begin inside the operator’s commercial case and are priced into the sale; operational integrity is the hinge, since its catastrophic failure externalizes costs onto the third parties and the state listed below. The remaining dimensions are borne mostly outside the transaction, which is why a voluntary sale underprotects them and why mandatory verification is warranted. The status column marks which dimensions were demonstrated in the empirical analysis and which are warranted under the framework.

	Value dimension	Borne by (incidence)	Status
	<i>Internalized by the operator, and priced by the market</i>		
V1	Production performance	The operator, with a fiscal and national shadow	Demonstrated (DV1)
V2	Operational integrity and reliability	The operator while running; third parties on catastrophic failure	Demonstrated (DV2)
	<i>Externalized, borne by others</i>		
V3	Safety and environmental harm to third parties	Workers, contractors, communities, ecosystems, the state’s remediation liability	Warranted
V4	Host-community welfare	Host communities	Warranted
V5	National capability development	The Nigerian workforce and economy	Warranted
V6	Fiscal and public-revenue protection	The state	Warranted
V7	Energy security and production resilience	The nation	Warranted, with the theft, sabotage, and price confound
V8	Decommissioning and abandonment responsibility	The state and future generations	Warranted, with a financial-security mandate

The hinge is in the second row. The upkeep of operational integrity is internalized, so an operator captures the savings from deferring it, but the catastrophic failure that deferral invites is externalized onto the dimensions below. The transaction market therefore does not reliably correct the problem on its own, and verification cannot be left to the parties. Production capability is protected alongside safety, since the same operating knowledge

carries both; what is verified is readiness, not a guaranteed level of output.

The dimensions below the hinge fall outside the transaction in distinct ways, and each names a reason the private bargain underprotects the public interest. Safety, environmental, and community harm (V3, V4) are the catastrophic-failure costs the hinge externalizes directly. The remaining four fall outside the sale for different reasons. National capability development (V5) is the workforce capacity that the local-content regime exists to build and that a hollowed-out transfer sets back. Fiscal protection (V6) is the royalty and tax base that erodes as production and integrity decline. Energy security (V7) is the national production resilience that weak operations put at risk, with theft and price as confounds. Decommissioning (V8) is the abandonment liability that outlives the operator and falls to the state and future generations. Each sits further from the sale than the last, and that distance is the measure of how little an agreed price protects it.

The claim is bounded in two ways. The empirical regularities and the design logic warrant the framework, short of proof from a tested intervention, and what it assures is demonstrated capability and readiness, not a guaranteed outcome. It is also calibrated to a particular setting. That setting has aged Niger Delta assets carrying integrity backlogs and suspended wells, a security and theft environment that raises the cost of weak operations, and a community and public-revenue structure that bears the failures the transacting parties do not. It works within an existing architecture of NUPRC regulations, the Nigerian Oil and Gas Industry Content Development Act (NOGICD), and the Petroleum Industry Act (PIA) that the framework works through rather than replaces. A design built for these conditions would need re-fitting for others.

What follows is a warranted design, not a costed implementation plan and not a claim that any single design guarantees success. What operating capability is, and why it does not reconstitute on its own, comes first (Section 5.2), followed by why the market and the existing rules do not deliver it (Section 5.3). The framework itself, three mutually constraining roles across the transaction lifecycle, is set out next (Section 5.4), then calibrated against Nigerian and comparator international practice (Section 5.5) and scaled to the risk a given transaction carries (Section 5.6). Finally, the jurisdictional and operational limits of the warrant are established (Section 5.7).

5.2 The Embedded Nature of Operating Capability

The relevant capability is not generic managerial competence. It is domain-specific upstream operating knowledge embedded in people, routines, integrity and emergency systems, decision rights, and data, and when its carriers are not retained, the gap can surface as a safety failure. A divestment moves the license, the wells, the platforms, and the pipelines. It does not, on its own, move the knowledge of how that particular asset is run safely and productively, because most of that knowledge lives in the people who operate it and in the routines, records, and relationships they have built around it over years.

That capability, defined in Section 2.2, can be described as a bundle of six families. The first is the operating workforce itself, the engineers, controllers, and supervisors who hold the tacit feel for how the asset behaves and what it tolerates (Kogut and Zander, 1992). The second is the body of operating routines, the procedures and practices that translate general

engineering standards into the specific handling of this reservoir, this terminal, this set of aging wells. The third is the integrity and emergency systems, the inspection histories, the maintenance backlogs, the contingency plans, and the standing arrangements to respond when something fails. The fourth is the structure of decision rights and accountability, the question of who is authorized to act, and how fast, when a well begins to lose control. The fifth is the asset's data and history, the record of what has been done, deferred, and discovered across decades of operation. The sixth is the contractor and technical-support architecture, the web of specialist firms a competent operator keeps on call before a crisis rather than after one. Taken together, these six families are the architecture of the operating enterprise, the organization, routines, knowledge, decision rights, data, and external relationships through which the physical assets are made to produce safely. Enterprise architecting treats this organizational architecture, not the physical plant alone, as the enterprise (Nightingale and Rhodes, 2015), so a divestment that conveys the plant without it transfers the hardware without the system that runs it.

These families map the same ground that the process-safety literature organizes, at finer grain, into an established set of elements. The Center for Chemical Process Safety organizes operating capability into twenty elements across four pillars, under the heading of Risk-Based Process Safety, or RBPS (CCPS, 2007). The four pillars are a commitment to safety, an understanding of hazards and risk, the management of that risk in daily operation, and the discipline of learning from experience. That four-pillar structure is the backbone for operating integrity in the analysis that follows, chosen because it is independent of any single operator, widely used across the industry, and auditable. It was also chosen because mapping each obligation onto a recognized element keeps the design traceable rather than ad hoc. The condensed bundle appears as Table 5.3 (Section 5.4); the full element-by-element trace, from each RBPS element to Ferreira's transfer factors, the empirical evidence, the legal hooks, and the value dimensions, is in Appendix G.

Risk-Based Process Safety carries the integrity side of performance. It does not, by itself, capture production. Production performance and operational integrity are the two dimensions of value at risk in the cases examined (Sections 4.5 to 4.7). The capability that sustains output, the subsurface understanding and the production-operations discipline that keep a field flowing, is as embedded and as perishable as the capability that keeps it safe. A production and subsurface operating-capability element therefore sits alongside the four process-safety pillars and on the same footing, not beneath them. Production performance is monitored as a value dimension and linked to that subsurface and production-operations capability; what is verified is readiness, not guaranteed output.

If the four pillars and the production element describe what capability is, a separate body of work describes what becomes of it during a handover. Ferreira et al. (2024) study the transfer of knowledge between operators during the sale of an aging offshore asset and identify twenty-two elements that determine whether knowledge actually moves from seller to buyer. The elements range from the structure of the contract and the availability of the seller's professionals to the access the buyer is granted to databases and the time allowed for the transfer to happen. They are not the capability itself but the conditions under which it transfers or is lost, and they depend as much on the buyer's capacity to receive and absorb knowledge as on the seller's willingness to release it (Cohen and Levinthal, 1990).

The link between the two, between a failed transfer condition and a degraded operating

element, has been traced directly. In a content analysis of the same transfer, Ferreira et al. (2026) map these transfer conditions onto the process-safety elements each degrades when it fails, and report that process-safety incidents rose, in both frequency and severity, after an aging asset changed hands without retention of its operating team. That content analysis directly codes ten of the twenty process-safety elements; in Appendix G the mapping is extended to the remaining ten from the cases and the legal hooks, with the directly coded links flagged as coded and the inferred ones as inferred. The path is concrete. When the contract is silent on operational requirements, the elements that suffer are emergency management, asset integrity, and hazard analysis. When the transfer period is compressed and records arrive piecemeal, the management of process knowledge erodes first, and it is the element on which the largest number of these transfer failures converge. When the people and records that hold operating knowledge are not retained, the gap can stay latent for a time, but the evidence shows it tends to surface as higher incident frequency and severity.

The six cases divide on exactly this point. The operators that sustained both output and integrity had absorbed the capability rather than merely acquired the assets. Seplat brought an experienced upstream organization under Austin Avuru and operated in partnership with Maurel et Prom; HEOSL operated OML 30 through Salvic as its technical operator, retaining the operating competence the license assumed (Sections 4.5.1 and 4.5.2). Neither rebuilt that capability after the fact; each had it in place at transfer or held it through the operator it brought. The operators whose performance diverged had taken on the hardware without the embedded system that made it work. Aiteo, whose founder and chief executive came from a downstream trading background, inherited the wells of OML 29 without the well-control capability they demanded. The Santa Barbara blowout of November 2021 is where that absence became visible: no contingency plan, no standing well-control contract, and a specialist response contractor engaged only once the well was already flowing uncontrolled (Sections 4.3.1 and 4.3.6). The license transfer moved the asset; what it did not do was make the missing well-control capability visible, testable, or enforceable before close.

Operating capability, then, is embedded, perishable, and specific to the asset, and in the cases examined it did not reconstitute itself in the hands of a new owner within operationally relevant timeframes. Whether it survives a sale depends on conditions that are observable at the point of transfer, what the seller discloses, what the buyer can absorb, and what the regulator verifies, and on whether the buyer is built to receive what is offered. Nigeria's 2024 divestment framework now creates a technical-capacity route at consent. Whether that route reaches what this asset needed, asset-specific crisis readiness, a structured transfer, demonstrated buyer absorption, and verification after close, is the open question.

5.3 The Market and Regulatory Failure to Transfer Capability

In the case analysis, the embedded operating capability a buyer inherited, brought, or contracted before performance depended on it, rather than capital constraints or fiscal terms alone, best separated the operators that sustained performance from those that did not (Sections 4.5 and 4.7). If that capability is what a divested asset needs, why does a buyer not

reliably acquire it, and why does the consent regime that governs the transfer not require it? The transaction structure makes that loss structurally likely rather than accidental, through four gaps that compound across the seller, the buyer, and the parties outside the deal. And the rules that gate the transfer measure corporate and compliance attributes instead of demonstrated operating capability, and confirm nothing once the asset has changed hands.

5.3.1 The Pre-Transfer Condition and Disclosure Gap

Operating capability is not a static inventory. Its tacit component erodes when it is not exercised, and the record of an asset's embedded requirements, its integrity history, and its near-misses lives partly in documents and partly in the people who ran it (Kogut and Zander, 1992; Nonaka, 1994). A seller preparing to exit holds that record, and has incentives to limit transition exposure, protect proprietary operating knowledge, and avoid new spending whose benefit would accrue mainly after close. What was never written down cannot be recovered from a data room.

The evidence is the condition of what Aiteo inherited: a well at Santa Barbara that was suspended and undecommissioned at transfer. The reviewed record does not show that the well's condition became an approval condition, a readiness test, or a verified handover obligation (Section 4.3.1). The corpus contains no admission by a seller that it cut spending before a sale, and none is needed. The provable claim is narrower: the information needed to assess what is being inherited sits with the seller, and the reviewed rules do not yet compel its disclosure in asset-specific, verifiable form. Nigeria's content law requires a Nigerian Content Plan and Certificate of Authorization before acquisition (NOGICD Sections 7-9), and the NUPRC divestment guidance calls for a facility catalog and asset register. A catalog lists what exists, not its condition or the operating knowledge the asset demands. Neither obliges the seller to disclose asset condition, integrity and incident history, and embedded operating requirements in a form a buyer and a regulator could use to judge the inheritance.

5.3.2 Information Asymmetry and Adverse Selection

The condition gap becomes an economic problem once the seller's incentives are added. The seller knows the embedded difficulty of the asset, the buyer does not, and disclosure of that difficulty lowers the price. The better-informed party therefore gains from limiting what becomes price-relevant. This is the adverse-selection structure developed earlier from Akerlof (1970): where quality is hidden and the informed side has reason not to reveal it, the market does not price it correctly, and the uninformed side overpays for what it cannot verify. Tacit operating knowledge is the hardest case, because it is not fully contractible even when the seller is willing. It cannot be reduced to a schedule and warranted.

The asymmetry is visible directly in the cases. The seller knew the asset's operating requirements, the buyer could not assess them from the data room, and the regulator lacked the tools to close the gap on either side (Section 4.3.3). The existing consent gate turns on ministerial approval and a technical-capacity test conducted mainly at the corporate level (PIA Section 95; NUPRC Reg. 7), with asset-specific demonstration and site inspection not mandatory in every case; neither obtains standardized disclosure from the seller nor

independently verifies demonstrated operating capability in the buyer. The asymmetry is therefore left to voluntary seller disclosure, by the party with the least reason to volunteer.

5.3.3 The Buyer's Absorptive Capacity

Even full disclosure and a willing seller do not transfer capability if the buyer cannot receive it. Absorptive capacity, a firm's ability to recognize the value of external knowledge, assimilate it, and apply it, is itself a function of prior related expertise (Cohen and Levinthal, 1990). A buyer without upstream operating depth lacks the foundation needed to receive and integrate what is transferred. This is a buyer-side constraint rather than a seller incentive, and the market cannot supply it on demand.

The contrast across the cases is sharp. Seplat, built on a founding team of upstream operators and reinforced by Maurel et Prom, and the OML 30 operator, staffed through Salvic, retained or brought equivalent operating people and absorbed what they inherited. Aiteo, led from a downstream trading background, did not (Sections 4.5.1 and 4.5.2). The transfer-process evidence points the same way: incidents rose in frequency and severity after handover where the operating team was not retained and the transfer terms did not secure continued access to the people and data that held the asset's knowledge (Ferreira et al., 2026). Nigeria's content law holds the nearest statutory analogue, a four-year understudy succession and technology-transfer provisions (NOGICD Section 31; Sections 43-47). It was written to localize a single operator's workforce over time, not to guarantee that a new operator can absorb an asset at the point of handover.

5.3.4 The Externality

The first three gaps explain why a buyer may end up without the capability an asset needs. The fourth explains why the public, and not the contracting parties, must be the one to act. When operating capability fails, the largest costs do not fall on the buyer or the seller. They fall on the communities exposed to a spill or a blowout, on the state that loses production and revenue, and on a future public that will bear the abandonment liability. Welfare economics has named this divergence since Pigou: where private actions impose costs on third parties that market prices do not reflect, the private optimum sits below the social optimum, and correction calls for public intervention rather than private goodwill (Pigou, 1920).

The obvious objection is Coase. If the affected parties could bargain with the transacting parties, they would reach an efficient outcome without a regulator, whatever the initial allocation of rights (Coase, 1960). That route is closed here. The parties who bear the cost of a capability failure run to tens of communities and extend to a general and future public that cannot be assembled at the table before consent is granted. Transaction costs are prohibitive, the Coasean solution is unavailable, and the Pigouvian case for mandatory public verification stands. The empirical analysis documented the more direct externalities: the 41 communities affected by the Santa Barbara blowout and the seven deaths in the separate Benin River explosion. The national production shortfall, near 1.5 against a target of roughly 2 million barrels per day, provides sector-level context rather than a case-specific causal estimate, since theft, sabotage, investment cycles, and price conditions also bear on it (Section 4.7.3). Nigeria already addresses parts of this externality after the fact, through Host Community

Development Trusts and a decommissioning fund, and the 2024 framework secured more than US\$400 million in decommissioning escrow (PIA Chapter 3; Sections 232-233; NUPRC pillars 4 and 5). Those mechanisms compensate for harm or fund eventual abandonment. None of them verifies the operating capability that would prevent the harm in the first place.

The four gaps are set out together in Table 5.2, with the evidence for each and the requirement each implies.

Table 5.2: The four structural gaps that make an operating-capability gap the likely outcome of an unassisted divestment. The gaps compound, from the asset condition and capability the seller does not disclose, through a buyer that may be unable to absorb what is transferred, to the failure costs that fall on parties outside the transaction, which is why correction cannot be left to the contracting parties. In each row, the existing consent rules verify a corporate or compliance attribute, short of the demonstrated operating capability the gap concerns.

The gap	Evidence (Chapter 4)	What the current rules verify	What the framework requires
1. Condition and disclosure. Tacit knowledge erodes when unexercised; the seller holds the asset’s condition and operating record (Kogut and Zander, 1992; Nonaka, 1994).	Aiteo inherited a suspended, unde-commissioned well at Santa Barbara (Section 4.3.1).	A Nigerian Content Plan and a facility catalog: what exists, not its condition (NOGICD Sections 7-9; NUPRC guidance).	The seller discloses asset condition, integrity and incident history, and embedded operating requirements.
2. Adverse selection. The seller knows the embedded difficulty, the buyer cannot, and disclosure lowers the price (Akerlof, 1970).	The buyer could not assess operating requirements from the data room, and the regulator lacked the tools (Section 4.3.3).	Ministerial consent and a technical-capacity test conducted mainly at the corporate level (PIA Section 95; NUPRC Reg. 7).	Standardized disclosure and independent verification of demonstrated operating capability.
3. Absorptive capacity. A buyer without upstream depth cannot receive and integrate transferred knowledge (Cohen and Levinthal, 1990).	Seplat and the OML 30 operator absorbed; Aiteo, downstream-led, did not; incidents rose where teams were not retained (Sections 4.5.1-4.5.2; Ferreira et al., 2026).	A four-year under-study written for one operator over time, not for a handover (NOGICD Section 31; Sections 43-47).	The buyer demonstrates absorptive capacity; a technical operator or under-study and a team-retention plan for de novo buyers.

The gap	Evidence (Chapter 4)	What the current rules verify	What the framework requires
4. Externality. The cost of a capability failure falls on third parties who cannot bargain before consent (Pigou, 1920; Coase, 1960).	41 communities affected at Santa Barbara; seven deaths at Benin River; sector-level production shortfall (Section 4.7.3).	Host-community trusts and a decommissioning fund: compensation and abandonment cover after the fact (PIA Chapter 3; Sections 232-233).	Public, enforceable verification; financial security tied to decommissioning; community standing.

5.3.5 The Wrong-Variable Diagnosis

The four gaps make an operating-capability gap structurally likely in an unassisted transfer. The natural question is whether the existing consent regime catches it, and the regime is more thorough than it is often given credit for. NUPRC’s upstream asset divestment framework assesses seven pillars at consent through the Nigeria Upstream Petroleum (Assignment of Interest) Regulations 2024, and the content law adds pre-acquisition participation and succession requirements. The difficulty is not that the machinery is thin. It is that it measures the wrong variable.

A single test makes the point. Walk the consent requirements, and after each one ask whether it obliges the operator to demonstrate that it can contain a well-control event of the kind that occurred at Santa Barbara. The seven-pillar technical-capacity assessment evaluates capacity mainly at the corporate level, with a site visit left to the regulator’s discretion (NUPRC Reg. 7); it does not. The Nigerian Content Plan and Certificate of Authorization establish local-content participation (NOGICD Sections 7-9); they do not. The understudy and technology-transfer provisions localize a workforce over years (NOGICD Section 31; Sections 43-47); they do not. Ministerial consent is a legal and fiscal gate (PIA Section 95); it does not. The Host Community Development Trusts provide for welfare once the asset is operating (PIA Chapter 3); they do not. The answer is the same down the list. None of these requirements compels a mandatory, asset-specific demonstration of crisis readiness before close, or verification after it. The diagnosis is therefore one of specification and closure, not absence: the technical-capacity route exists, but it reaches none of the asset-specific disclosure, transfer, absorption, and verification the four gaps require.

This is a re-specification, not a condemnation. The rules are well-intentioned and have grown more demanding, and the 2024 reform, including the decommissioning escrow, narrows the gap without closing it. The First Hydrocarbon transfer of OML 26 illustrates the older gap rather than the current one. It passed under the prior regime to a buyer whose backers were later convicted of fraud (Section 4.4.2), and the 2024 Regulations have partly closed that particular opening by screening directors for prior convictions. The point is not that the later fraud could have been detected at the time of the transfer; it is that buyer fitness and governance integrity were not structured approval questions under the earlier framework. What the regime still does not do is require the operating-capability demonstration that the four gaps make necessary.

5.3.6 The Same Gap Across Acquisition Pathways

The gap is not peculiar to the bilateral IOC-to-indigenous sale, although that transaction is the textbook adverse-selection setup. It recurs wherever Nigeria has allocated operatorship without verifying operating capability. The marginal field program awarded acreage on signature bonuses, work-program commitments, and farm-out arrangements, with no effective operating-capability screen, a structural precedent for transferring operations to parties whose ability to run them was never tested (Section 4.6). The state route shows the same pattern. At NPDC, political standing sustained operatorship and commercial participation without the operational demonstration a capability test would require (Section 4.5.3). An NPDC-authored assessment of indigenous service companies found that table-top scoring passed firms that failed on facility inspection, with fewer than half meeting the benchmark once equipment and personnel were examined on site (Oriaku et al., 2016). Paper capability and demonstrated capability are not the same thing, and only a site-level test separates them.

Voluntary standards cannot close the gap either. The industry’s process-safety and management-system guidance describes the right capability content. The guidance is voluntary, internal to a single operator, blind to the moment of transfer, and dependent on voluntary adoption by the buyer, including buyers whose capability is the issue in question, a point developed in Section 5.5.3. An independent civil-society proposal, the National Principles for Responsible Petroleum Industry Divestment, is consistent with much of the same logic. It calls for seller disclosure of liabilities, demonstrated technical and financial capacity assessed by an independent body, and an asset-integrity review with deficiencies remedied before approval (Steiner, 2023). The design set out in the next section is grounded in a capability theory and the critical-experiment evidence. It goes beyond those proposals by incorporating the buyer’s absorptive capacity and the structure of the transfer itself, extending assurance past close into verification, and working through existing statutory hooks in place of a new voluntary instrument.

The case for intervention follows from these findings together. On its own, the transaction underprovides capability transfer: the structure leaves the gap unpriced, the existing rules verify corporate and compliance attributes and stop at consent, well short of demonstrated capability, and the cost of failure falls on parties never at the table (Table 5.2). The institutional design that resolves this, set out in Section 5.4, makes operating capability something a seller must disclose, a buyer must demonstrate, and a regulator must verify, before and after the asset changes hands.

5.4 The Capability Assurance Framework

A specific institutional design follows from that diagnosis. Corporate-level technical capacity can clear consent without asset-specific operating capability being disclosed, absorbed, or verified. The remedy is to re-specify the existing hooks so that those three things happen, and to connect them, so that none is satisfied on paper while another is left undone. Three roles carry the design. The seller discloses, the buyer absorbs, and the regulator verifies, across the life of the transaction and within the consent, content, and labor provisions Nigeria already has. Together the three roles form a control structure, in the systems-theoretic sense of the

control actions and feedback that hold a hazardous process within safe bounds (Leveson, 2011), organized around the asset as a sociotechnical system. The divestment moves the technical subsystem, the wells and the facilities, but not the social subsystem that operated it (Section 2.2). The design exists to make the disclosure, transfer, and verification of that social subsystem happen across the transaction instead of being left to the discretion of the transacting parties. It thereby installs the corrective function the market does not supply, the one severed because the costs of operational failure fall outside the sale (Section 2.6). The framework can be built primarily through existing consent, due-diligence, content, labor, and decommissioning hooks, although some elements would require regulations, consent conditions, or license terms.

5.4.1 Three Roles in a Closed Loop

The three roles are not a sequence of independent checks. Each depends on what the others produce. Disclosure gives the buyer and the regulator something concrete to work with, the asset's condition and the operating-capability bundle defined in Section 5.2. Absorption is meaningful only when it is tested against what was disclosed, so that a buyer's claim to competence is matched to the specific demands of this asset instead of asserted in general. Verification constrains both and rests on both, because the regulator can confirm only what the seller has surfaced and the buyer has shown. Each role enables and constrains the others, which is why the design closes into a loop rather than running down a checklist.

The loop is built around a feature of the documented failures: they were not isolated to one party. At Aiteo, an inherited undecommissioned well, the buyer's missing well-control capability, and the regulator's lack of any tool to test for it were present at the same time (Sections 4.3.1 and 4.3.6). A checklist that cleared each party in isolation would have passed the transaction, because each gap was not visible, within the consent process, to the party that might have caught it. A loop checks each role against what the others produce, so that a gap left open in one place has somewhere to be caught.

The regulator's role is a verification overlay rather than a party to the transfer. The transfer-process literature describes a bilateral exchange between seller and buyer (Ferreira et al., 2024); the third role rests on a different basis. It draws on the consent and fit-and-proper powers the regulator already holds (PIA Section 95; NUPRC Reg. 7), on the externality that places the cost of failure on parties outside the transaction (Section 5.3.4). It also rests on the practice of mature petroleum regulators that assess capability by demonstration rather than declaration, Norway through prequalification before entry and Alberta through assessment of both transferor and transferee at transfer (both developed in Section 5.5). The loop is shown in Figure 5.1.

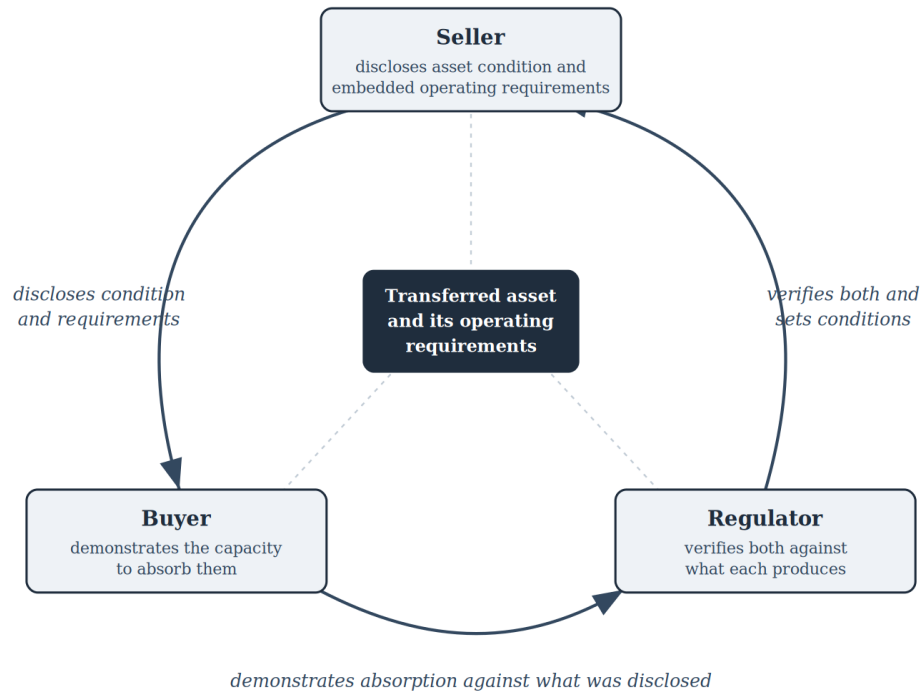


Figure 5.1: The capability assurance loop. Seller disclosure, buyer absorption, and regulator verification act as three mutually constraining roles around the transferred asset. Weakness in any one role breaks the assurance, because the next role is left with nothing to act on.

That is the structure of the loop, which role acts on what. Its dynamics are the complement: the same three roles are three balancing interventions on the degrading cycle of Figure 2.1. Mandatory seller disclosure closes the information gap at the point of sale; buyer absorption verification confirms the capability before handover; and the regulator’s verification mandate, with rule-based triggers, installs the requirement and restores the corrective signal the market does not send. Each intervention cuts a different link, and the degrading cycle is broken only when all three act together: a single intervention is unlikely to close it, which is why disclosure without verification, or verification without a buyer able to absorb, does not hold. Figure 5.2 places the three interventions on the same loop. Across the thesis the three figures form one sequence: Figure 2.1 sets out the failure equilibrium, Figure 5.1 the capability assurance loop that would close it, and Figure 5.2 the points at which the framework cuts that loop.

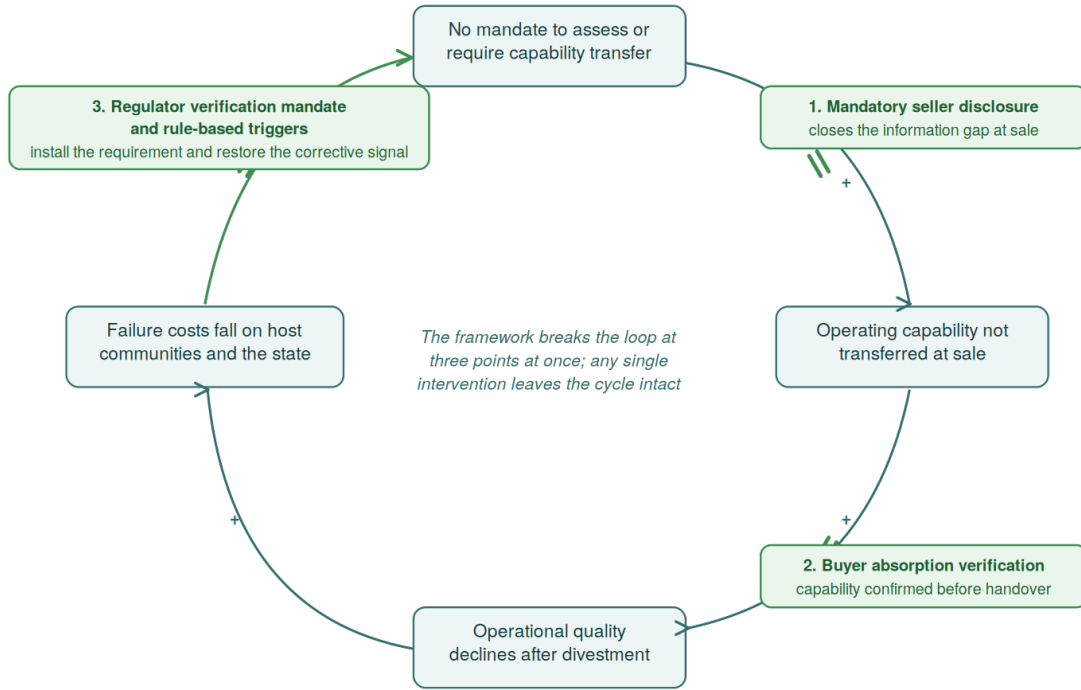


Figure 5.2: Breaking the failure equilibrium. The degrading cycle of Figure 2.1, with the framework’s three roles acting as balancing interventions on different links. The loop is broken only when all three act together; any single intervention leaves the cycle intact.

5.4.2 Where Each Obligation Attaches Across the Transaction

The three roles act at different points, and the points follow the order in which knowledge actually moves between operators: before the transfer, during it, and after close (Ferreira et al., 2024). Five phases organize the obligations, two before the transfer commits, two through the transition, and one after close.

Diagnosis comes first, before a buyer is committed. The seller discloses the asset’s condition and its embedded operating requirements, the integrity and incident history, the status of wells including any that are suspended or undecommissioned, and the catalogue of facilities and their state. This extends the due diligence the Assignment of Interest Regulations already require (Reg. 7) and the asset register that should accompany it, and it reaches the information a buyer cannot recover from a data room on its own. The case for it is direct: Aiteo inherited a suspended well at Santa Barbara, and the reviewed record does not show that its condition became an approval condition, a readiness test, or a verified handover obligation (Section 4.3.1).

Transfer design follows, and it carries the most weight. The assignment and transfer agreement sets out what moves with the asset and what support the seller provides through the handover. These cover the retention or secondment of operating staff, the transfer of data and process knowledge, and a defined understudy period in which the buyer’s people work alongside those who hold the asset’s operating knowledge. These terms are mandatory in the

design and beyond negotiation, because they are the elements a buyer cannot supply for itself after close, and the transfer-process evidence places them in its most-critical tier (Appendix G, criticality tiers). Nigeria's content law already contemplates this support in its succession and technology-transfer provisions (NOGICD Section 31; Sections 43-47), written for a single operator's workforce; the design binds them to the operatorship handover itself. The contract is the instrument for two reasons. The transfer-process evidence places contractual terms at the center of whether knowledge moves, reaching more operating elements than any other single factor (Ferreira et al., 2026). The operators that sustained performance had retained, brought, or contracted equivalent operating capability before performance depended on it (Sections 4.5.1 and 4.5.2).

Readiness is tested during the transition, before operatorship passes. The buyer demonstrates that it can run this asset, not that it is competent in the abstract: a working well-control response capability, a safety-critical integrity backlog brought to a defined state, and an understudy handover completed rather than promised. The competency assessment can include the site visit the Regulations already permit (Reg. 7). What justifies it is the gap between paper and practice: a study of indigenous operating capacity found that firms passing a table-top assessment failed when their facilities were inspected (Oriaku et al., 2016). Aiteo's unreadiness was visible in the suspended well it had not addressed.

Verification closes the transaction. The regulator confirms the disclosed condition against the demonstrated capability and attaches conditions to consent where a gap remains, including financial security for decommissioning and for suspended wells (PIA Section 95; Sections 232-233). The contrast between two consents shows why: the regulator withheld approval of the SPDC divestment to Renaissance until it was satisfied on capability. The earlier transfer of OML 26 to a buyer whose backers were later convicted of fraud passed under the prior regime (Section 4.4.2).

Enforcement continues after close. The regulator monitors the indicators that show whether capability held, for a defined initial period and then on a trigger basis. It combines leading measures that signal readiness before harm, drill completion, integrity-backlog closure, suspended-well monitoring, and audit close-out, with the lagging process-safety event rates that confirm it only once an incident has occurred. Where readiness is incomplete at close, operatorship can be staged instead of transferred whole, for instance through a phased transfer of operational responsibilities across the transition, with the detailed options left to Section 5.6. The financial security that backs decommissioning liability follows the principle Alberta applies, held within the existing Nigerian powers (PIA Sections 232-233; Section 95) rather than imported as foreign law. Monitoring earns its place because differentiated outcomes are observable only if they are measured, and the cost of an unmeasured failure falls on the communities and the state (Section 5.3.4).

The seller's duty in this design is bounded. It is to disclose fully and to support the transition for a defined period, typically measured in months, bounded well inside the life of the asset. Beyond that window, the seller is not held liable for the buyer's operation.

The obligations are not uniform in weight. The disclosures and transfer terms that only the seller can provide, and that the buyer cannot remedy after close, sit in the most-critical tier of the transfer-process evidence (Appendix G, criticality tiers), and these the design makes mandatory. Obligations the buyer can manage are scaled to the risk of the asset and the buyer, as developed in Section 5.6.

5.4.3 The Bundle, by Capability Domain

The bundle is organized in Table 5.3 by the capability domains defined in Section 5.2: the production and subsurface capability that sustains output, and the four process-safety pillars that sustain integrity. Each domain is paired with the corresponding seller, buyer, and regulator obligations, with the empirical finding and the legal hook that warrant it.

The cases populate the columns. On the seller side, the inherited undecommissioned well, the absent blowout contingency plan, and the facility catalogue the divestment did not produce are what a disclosure obligation would have surfaced. On the buyer side, the operators that absorbed capability show what the column asks for. They are Seplat under Austin Avuru with Maurel et Prom, OML 30 under HEOSL with Salvic as technical operator, and the SPDC divestment to Renaissance, designed to preserve the operating capabilities, management systems, and staff of the outgoing operator. Renaissance is evidence that capability-preserving design is feasible and already emerging in practice; it is not yet evidence of outcome, because the transfer is recent. On the regulator side, the contrast between the decision to withhold the Renaissance consent until capability was satisfied and the earlier consent to a buyer it had no tool to test is what the verification column is built from. The column also draws on the finding that the response to the Aiteo blowout exceeded the operator's internal capacity.

Table 5.3: The capability bundle: what the seller discloses, the buyer absorbs, and the regulator verifies, by capability domain. Each row carries the empirical finding and the legal hook that warrant it; the full element-by-element trace is in Appendix G.

Capability domain	Seller discloses	Buyer absorbs	Regulator verifies	Warrant: empirical finding and legal hook
Production and subsurface operating capability	Reservoir and production history, subsurface models, intervention and decline records	Subsurface and production-operations competence, through a retained or equivalent team	Demonstrated production-operations readiness, and post-transfer production stability against the pre-transfer baseline, with theft and sabotage distinguished from operator performance	Absorbed operators sustained output where de novo operators did not (Section 4.5); Reg. 7(a) and 7(g); pillar 7
Commit to Process Safety (Pillar 1)	Safety management system, competency records, workforce-retention plan	Safety culture and competency, through retained staff and an understudy period	Corporate competency and director fitness, and the team-retention plan	Aiteo’s downstream-only leadership against Avuru and Salvic’s upstream pedigree (Sections 4.5.1-4.5.2); NOGICD Sections 29-31; Reg. 7(5)
Understand Hazards and Risks (Pillar 2)	Process knowledge, hazard analyses, asset data and databases	Process-knowledge management, through retained data and the people who held it	Data repatriation complete, hazard analyses current, and the asset’s crisis scenarios analyzed	The buyer could not assess embedded requirements from the data room (Section 4.3.3); Reg. 7(g); NOGICD Sections 43-47
Manage Risk (Pillar 3)	Operating procedures, integrity and maintenance records and backlogs, emergency plans, well status, contractor arrangements	Operating procedures, integrity management, emergency-response capability, contractor architecture	Asset-specific crisis readiness, a standing well-control arrangement, the integrity backlog, and security for suspended wells	Aiteo well-control readiness evidence: no contingency plan, no standing well-control contract, after-the-fact specialist engagement, and an unde-commissioned well (Section 4.3.1, Section 4.3.6); Reg. 7(d); PIA Sections 232-233; pillar 4

Capability domain	Seller discloses	Buyer absorbs	Regulator verifies	Warrant: empirical finding and legal hook
Learn from Experience (Pillar 4)	Incident history, prior investigations, lessons learned, metrics	Incident-investigation capability against the inherited history, and the metrics	Post-close verification, leading indicators (drills, backlog closure, suspended-well monitoring, audit close-out) and lagging process-safety event rates	Capability is observable only if measured; table-top assessment passed firms that site inspection failed (Oriaku et al., 2016); Reg. 7(4); IOGP Report 456

Footnote (below the table): In each row, the seller-disclosure and transfer-agreement obligations are mandatory: they are the elements the buyer cannot remedy after close, and they fall in the most-critical tier of the transfer-process evidence (Appendix G, criticality tiers). The buyer-absorption obligations are scaled to the risk of the asset and the buyer (Section 5.6).

5.5 Calibration: Existing Nigerian Rules and International Models

The design has precedent on both sides. NUPRC's 2024 reforms already move in its direction, and the comparator regimes reviewed below require a buyer to show capability, fitness, or organizational capacity before they permit entry or a transfer. Where it goes beyond them is in seller-to-buyer continuity of operating knowledge, and in verification after close. Both points are established below, and from them follows a specification of the buyer's own technical organization, which the next section develops alongside the proportionality that scales it.

5.5.1 The Domestic Foundation

NUPRC's 2024 reforms moved in the right direction, and the design here extends them, building on those reforms. The Nigeria Upstream Petroleum (Assignment of Interest) Regulations 2024 already require due diligence across technical capacity, financial capacity, decommissioning, host-community and environmental funds, labor relations, and data repatriation, and they permit site visits for competency evaluation (NUPRC Reg. 7). The reviewed pre-2024 consent regime did not require the operating-capability demonstrations the cases turn on: no blowout contingency review, no well-control demonstration, no HSE management-system assessment, no crisis-response or knowledge-transfer requirement. The 2024 pillars reach adjacent concerns directly and some operating-capability concerns indirectly, and the decommissioning escrow secured under them, more than US\$400 million in 2024, is real progress (NUPRC, 2024).

Specification and closure are achieved at four points. Reg. 7's technical capacity, assessed largely at the corporate level, becomes a demonstration of asset-specific operating capability tied to the particular asset transferred. The discretionary competency visit becomes mandatory and asset-specific for higher-risk transfers. Data repatriation extends from a transfer of records into a structured disclosure of asset condition and operating history. And a seller-to-buyer transfer obligation and post-transfer verification, absent from the pillars, are added, applying the logic of NOGICD's succession and technology-transfer provisions, written for a single operator's workforce, to the operatorship handover itself (NOGICD Sections 31, 43-47).

5.5.2 Demonstrated Capability in International Practice

Three comparator petroleum regulators assess operating capability before they permit a transfer, and all three require evidence beyond self-declaration, though they differ in how

directly they test operational capability. The claim is narrow and comparative: requiring a buyer to show it can run an asset is not a novel demand in petroleum regulation.

The United Kingdom's North Sea Transition Authority requires the buyer and seller to prepare a capability pack, setting out the buyer's financial and technical capability, before it consents to a license assignment. It also assesses the fitness of a licensee and of those who control it, including the incoming directors, whenever a licensee takes on a new commitment. It holds a standing power to require a change of control, or to revoke a license, where control passes to an unfit holder (NSTA, 2023). Fitness, in the UK system, is a continuing criterion rather than a one-time check.

Norway is more demanding on organization. A company cannot hold a license or operate on the Norwegian continental shelf until it is prequalified. Prequalification requires it to document technical petroleum competence, health and safety competence, and financial strength, assessed jointly by the Norwegian Offshore Directorate and Havtil through meetings and site visits (Havtil. Norwegian Offshore Directorate). The Petroleum Act requires a competent organization inside Norway: licensees must keep technical staff in the country able to follow the activity. Operators must maintain an in-country organization capable of discharging the operator's duties, a materially higher bar than the one set for a licensee (Norwegian Petroleum Act). Capability is assessed at entry and supervised continuously after it.

Alberta assesses both parties to a transfer before it approves one. Under the Licensee Life-Cycle Management framework introduced in 2021, the regulator runs a multifactor licensee capability assessment that weighs financial health, liability, and closure performance, and uses it to decide eligibility to hold a license and the security required on a transfer (AER, Directive 088). The framework ties the security a transferee must post to that assessment and sets mandatory closure-spend targets, keeping decommissioning capacity in step with the liability a licensee carries.

These three regimes differ in emphasis, and each is built for its own setting. None, as reviewed here, combines asset-specific crisis-readiness verification for the particular asset, an obligation to provide for continuity of operating knowledge through retention, secondment, transition services, or equivalent technical support, and post-transfer verification, in the form the Nigerian cases require. They assess a buyer's corporate and financial capability, and Norway and Alberta supervise it over time, but the combination Nigerian conditions call for is not found in any one of them. The comparison, and what the framework draws from each regime and adds beyond them, is set out in Table 5.4.

Table 5.4: Capability assessment as a condition of transfer across four petroleum regimes, and what the proposed framework draws from them and adds. Every regime assesses a buyer’s capability before transfer; the framework extends that practice by requiring asset-specific operating and crisis readiness, seller-to-buyer continuity of operating knowledge, and verification after transfer, none of which the comparators require.

Regime	Capability assessed before transfer	Form of the assessment	Continuing after transfer
NUPRC, Nigeria (2024)	Seven-pillar due diligence: corporate technical capacity, financials, decommissioning, host-community and environmental funds, labor, data repatriation (Reg. 7)	Documentary; competency site visit permitted but discretionary	Decommissioning escrow; no mandatory post-transfer asset-specific operating-capability verification
United Kingdom (NSTA)	Buyer’s financial and technical capability via a capability pack; fitness of directors and controllers	Documentary capability pack plus a fitness assessment	Fitness a standing criterion; power to require a change of control or revoke
Norway (NOD and Havtil)	Prequalification for technical, HSE, and financial competence; a competent in-country organization	Joint meetings and site visits	Continuous supervision; in-country organization maintained
Alberta (AER)	Multifactor licensee capability assessment of financial health, liability, and closure performance (Directive 088)	Assessment of the transferee’s record and finances	Lifecycle monitoring; security tied to liability; mandatory closure-spend targets
Proposed framework (Nigeria)	Asset-specific operating and crisis readiness for the particular asset; seller-to-buyer continuity of operating knowledge; buyer absorptive capacity	Mandatory asset-specific demonstration and site verification for higher-risk transfers	Leading and lagging integrity verification after transfer

5.5.3 The Limits of Voluntary Standards

The technical content of operating capability is already codified. The International Association of Oil and Gas Producers and IPIECA publish an operating-management-system framework (IOGP Reports 510 and 511), asset-integrity guidance (IOGP Report 415), and process-safety performance indicators (IOGP Report 456) that describe, in detail, what a competent operator does. The gap is not the content; it is the delivery. These standards are voluntary, internal to a single operator, and silent on the moment of transfer, and their adoption depends on the buyer, including the buyers whose capability is the very thing in question. That gap is closed by mandating through existing hooks what the standards recommend, by naming them as the buyer's integrity-management benchmark, and by using their process-safety event rates, the Tier 1 and Tier 2 indicators, as lagging post-transfer metrics. These are paired with leading indicators that signal readiness before harm: readiness-drill completion, well-control-contract status, integrity-backlog closure, suspended-well monitoring, management-of-change compliance, near-miss reporting quality, and audit close-out. A voluntary standard becomes an enforceable condition only when a regulator attaches it to consent.

The calibration is therefore two-sided. The gate installed here has precedent in Nigerian law and in the comparator regimes, and where it reaches further, into seller-to-buyer transfer and post-transfer verification, it does so to close a gap those regimes leave open. The comparator models also specify what the buyer's own organization must look like, an accountable operating entity, an inspectable in-country organization, fit controllers, and a qualified technical operator where in-house depth is absent. Those requirements are developed in Section 5.6, together with the proportionality that scales them to the asset and the buyer.

5.6 Proportionality and Structural Levers

A uniform requirement would be the wrong design. What is asked of a buyer scales with two variables: the hazard the asset presents and the operating capability the buyer brings to it. A buyer with demonstrated comparable capability acquiring a low-hazard asset needs little beyond confirmation of what it already holds; a *de novo* buyer acquiring an asset with suspended wells, an evacuation dependency, and dense community exposure needs the full apparatus. The cross-case evidence defines the variables, and the obligations fall most heavily where the process-safety criticality is greatest (Ferreira et al., 2026). The requirements enforced at the gate, the risk tiers that scale them, and the structural levers that carry them into a transaction are set out below.

5.6.1 Requirements for the Buyer's Organization

The comparator regimes do more than assess capability elsewhere; they provide a template for an acquirer's operating organization, and the case evidence populates it. Four requirements follow, and their intensity scales with the asset's hazard and the buyer's profile in the way the next subsection sets out.

The first is demonstrated upstream operating competence. The accountable operating entity, whether through its own leadership or an approved technical operator, must hold

subsurface, operations, and well-engineering experience on comparable assets, with named operational accountability and the authority to command a well-control crisis. This re-specifies Reg. 7(5)'s requirement of sufficient technical knowledge and experience as asset-relevant operating depth in the entity that runs the asset, not general managerial or commercial standing. The failure it targets is the Aiteo containment failure, where the leadership-background evidence pointed to downstream trading in place of comparable upstream operating depth for directing a well-control response (Sections 4.3.1, 4.5.1, and 4.5.2; Table 4.6). Leadership background is a mechanism-consistency indicator rather than a proof of cause, but the requirement follows from the absorptive-capacity argument in any case: a firm cannot absorb operating knowledge it has no prior expertise to receive (Cohen and Levinthal, 1990).

The second is an inspectable, accountable operating organization. The buyer must run the asset through a real operating organization a regulator can inspect, rather than a thin holding structure whose operational depth cannot be independently demonstrated. The depth this requires scales with asset risk, on the Norwegian in-country-organization model at the high end, as the next subsection sets out. The failure it targets is First Hydrocarbon Nigeria, a holding vehicle that took an interest it never operated, leaving no operating organization a regulator could have inspected (Section 4.4.2).

The third is controller fitness beyond a clean record. Reg. 7(5) screens directors for prior convictions, which closes the narrow opening a single prior conviction would present but not the wider one. The fitness test should reach whether those who control the licensee can provide, govern, and resource the technical and managerial capability the asset requires, on the North Sea Transition Authority's model, drawing on the fit-and-proper and disqualification provisions the PIA already contains. The failure it targets is the governance collapse behind First Hydrocarbon, whose principals were convicted of fraud committed while leading the company. The point is not that the fraud would necessarily have been discovered at approval, but that the earlier framework made neither buyer fitness nor governance integrity a structured approval question (Section 4.4.2).

The fourth is a technical operator or partner where in-house depth is absent. A de novo buyer without operating depth of its own must supply it through a qualified technical operator or a partnering arrangement, the route the OML 30 operator took through Salvic and Seplat reinforced through Maurel et Prom (Sections 4.5.1, 4.5.2). The failure it targets is the de novo operator that runs a high-risk asset with neither in-house depth nor a technical operator engaged, as at Aiteo. NOGICD's technology-transfer and partnering provisions already contemplate the arrangement (NOGICD Sections 43-47); whether it is mandatory or optional is the proportionality question the next subsection answers.

5.6.2 Scaling Requirements to Risk

The four requirements describe a ceiling, not a uniform floor, and three variables set where a given transaction falls beneath it. The first is the buyer's profile, read as demonstrated capability over corporate age or size: a buyer that has run comparable assets carries capability into the transaction, while a de novo buyer carries none and must build or contract it. The second is the asset's hazard. Suspended wells that require active integrity management, a single evacuation route whose failure strands production, and proximity to populated communities each raise the cost of an operating lapse, as the records of OML 29 and OML 18

show (Sections 4.3, 4.4). The third is the seller-transition structure: a handover that retains the operating team and transfers the management system presents a lower absorption risk than an abrupt exit that strips both.

One part of the apparatus does not scale. The seller’s disclosure, the transfer of data and records, and a time-bounded mechanism for continuity of operating knowledge are mandatory at every tier, because they address failures a buyer cannot remedy on its own once the seller has gone. The criticality finding identifies them as the most safety-consequential of the transfer elements (Ferreira et al., 2026). The transfer agreement is the instrument that carries them, and it must specify the data and records to be transferred, the continuity mechanism, and the transition period. What scales is the buyer-facing burden: the depth of capability demonstration, whether a technical operator is mandatory, the form the continuity mechanism takes, and the intensity of post-transfer verification. The three tiers, and what each requires, are set out in Table 5.5; the mandatory floor in the final row applies in addition to every tier above it.

Table 5.5: Risk-tiered divestment assurance. The apparatus required of a buyer scales with the capability it demonstrably brings, read independently of corporate age, and with the hazard the asset presents; the mandatory floor in the final row applies at every tier, since it addresses failures a buyer cannot remedy after transfer. In the retrospective examples, the cases examined are classified under the framework, which did not exist at the time of those transactions.

Tier	Retrospective example	When it applies	Required of the buyer	Transfer and verification apparatus
Tier 1 (full)	Aiteo (OML 29); OML 30	A de novo buyer, or a buyer without demonstrated comparable capability acquiring a high-hazard asset	All four requirements; a qualified technical operator or partner is mandatory	Full-form continuity mechanism; staged operatorship on demonstrated readiness; financial security tied to decommissioning; leading and lagging integrity triggers; community-readiness measures where exposure is dense
Tier 2 (intermediate)	Seplat (OMLs 4, 38, 41)	A buyer with demonstrated comparable capability acquiring a moderate-to-high-hazard asset	Operating competence, inspectable organization, and controller fitness demonstrated; technical operator at the regulator's discretion	Continuity mechanism scaled to need; integrity triggers; staged operatorship at discretion
Tier 3 (baseline)	None in the present set	A buyer with demonstrated comparable capability acquiring a low-hazard asset	Operating competence and controller fitness confirmed	Documented handover; standard post-transfer reporting
Mandatory floor (all tiers)	All transactions	Every transfer, in addition to the requirements above	Receipt and integration of the disclosed knowledge	Seller disclosure; data and records transfer; a time-bounded continuity-of-knowledge mechanism, all specified in the transfer agreement

The tiers track the cases in retrospect. Aiteo on OML 29 would classify as Tier 1, a de novo buyer on a high-hazard asset, and the apparatus it would have required, a mandatory technical operator and a demonstrated containment capability, is what the record shows absent (Section 4.3). The OML 30 transaction would classify as Tier 1 on the same reading, an indigenous buyer without its own operating depth. It cleared the bar the framework would set by supplying that depth through Salvic, which is why its outcome diverged from Aiteo's under otherwise similar conditions (Section 4.5.2). Seplat illustrates why the tiering turns on demonstrated capability over corporate age: newly formed in 2010, it brought upstream operating capability through its founding team and its technical partnership with Maurel et Prom. It would fall within Tier 2 instead of carrying the full Tier 1 burden (Section 4.5.1).

5.6.3 The Structural Levers

Five levers carry the tiered requirements into the transaction, and each rests on a case in the record and an existing legal hook.

The first is a mandated technical-operator or partnering arrangement for de novo buyers at Tier 1. Where the buyer's own organization cannot demonstrate operating depth, an approved technical operator must supply it, the arrangement OML 30 used and Aiteo did not (Sections 4.3.1, 4.5.2). NOGICD's technology-transfer and partnering provisions supply the statutory basis (NOGICD Sections 43-47).

The second is a time-bounded mechanism for continuity of operating knowledge. Because people cannot be transferred like equipment, the transfer agreement must provide a defined route by which the knowledge that ran the asset reaches the buyer, through secondment, retention incentives, transition-service arrangements, or guaranteed access to named subject-matter experts. The arrangement must run for a period long enough to transfer working knowledge and short enough not to leave the seller indefinitely responsible for an asset it has sold. The process-safety record supplies the warrant: incidents rose after handovers that retained no operating team (Ferreira et al., 2026). The Renaissance transaction is a design contrast that shows the arrangement is structurable. Shell stated that the sale was designed to preserve SPDC's operating capabilities, including its management systems, and that its staff would continue to be employed through the change of ownership (Shell, 2024). That transaction completed only in 2025, so it offers feasibility evidence rather than validation of outcomes. NOGICD's succession and understudy provisions, written for a single operator's workforce, supply the closest statutory analogue (NOGICD Section 31). The handover runs against a shared knowledge repository that both parties review before transfer, the transfer instrument the inter-organizational knowledge-transfer literature places at the center of whether operating knowledge moves (Ferreira et al., 2024).

The third is staged operatorship, a phased assumption of operational responsibility in which critical functions pass only after readiness is demonstrated, instead of at a single administrative moment. This gives the absorptive-capacity requirement a point of verification before full responsibility passes (Cohen and Levinthal, 1990).

The fourth is financial security tied to decommissioning liability. The buyer must post security scaled to the abandonment obligation it assumes, the mechanism Alberta operates through its licensee assessment, and the one the PIA and the 2024 escrow already begin to build in Nigeria (AER, Directive 088; PIA Sections 232-233; NUPRC, 2024).

The fifth is a set of post-transfer integrity triggers, leading and lagging. Lagging indicators, the IOGP Tier 1 and Tier 2 process-safety event rates, register harm after it occurs. Leading indicators, readiness-drill completion, integrity-backlog closure, suspended-well monitoring, management-of-change compliance, near-miss reporting quality, and audit close-out, signal a developing capability gap before it produces an incident, which gives the closed loop a point of intervention ahead of failure rather than only after it.

Indicators are only as strong as the consequences attached to them. The triggers should connect to graduated, rule-based consequences applied on defined thresholds, escalating conditions on operatorship and financial consequences scaled to the breach, rather than to consequences that depend solely on discretionary regulatory action. The NPDC case shows why the distinction matters: where enforcement turns on discretion, political access can deflect it (Section 4.5.3). Rule-based triggers reduce, though they do not eliminate, that vulnerability, and they tie the verification loop to the financial security the fourth lever establishes. Where the asset carries dense community exposure, the readiness test extends beyond the technical to a set of community-readiness measures: host-community notification protocols, spill-response interface arrangements, and linkage to the Host Community Development Trust the PIA requires (PIA Chapter 3).

The obligations placed on the seller are those only the seller can initiate: disclosure of asset knowledge, access to records and subject-matter experts, and a time-bounded continuity mechanism. They attach to the transaction and expire at the end of a defined transition, short of extending the seller's liability for the asset indefinitely.

No framework can guarantee that a buyer which clears every tier will succeed. What it offers instead is assurance in the proper sense: it makes the capability an asset needs observable, testable, and enforceable before and after the asset changes hands. The boundaries of that assurance are the subject of Section 5.7.

5.7 Design Risks and the Limits of the Warrant

The framework is warranted, not proven. It rests on the regularities established across the cases and on the logical completeness of the design that follows from them, not on a tested intervention; no implementation data yet exists. What the design assures is capability that can be observed, demonstrated, and verified; it does not reach the non-contractible tacit knowledge no instrument can compel a seller to surrender, and it does not guarantee outcomes. Its contribution is to make the capability an asset needs observable and testable at the point of transfer, not to ensure on its own that the capability is there. The seller's obligations are deliberately time-bounded, a defined transition in place of indefinite liability; after that window, responsibility for operating the asset rests with the buyer, though the post-transfer verification and enforcement remain in force.

The central confound is one the design tolerates rather than resolves. The evidence did not cleanly separate whether an operator's difficulty traced to the specific capability it inherited or to its general organizational quality, and the two run together in the cases, as noted in Section 4.8. The design does not hinge on the distinction. The same assurance screen detects both, though the response it calls for differs: a structured transfer of the seller's knowledge answers an asset-specific gap, while technical-operator support or, in the limit, refusal of

consent answers a broader organizational one. The confound limits the precision of the causal account in the case analysis; it does not weaken the design, which detects a capability gap without needing to attribute it to one source over the other.

Two further limits follow from the evidence base. The internal coordination through which a firm absorbs operating capability was inferred from outcomes and documents rather than observed directly. The absorptive-capacity mechanism the design relies on is warranted by theory and by the outcome pattern, not by direct observation of how each organization worked inside. And the primary cases concentrate on a single seller. The seller-independence checks bound the concern, since an acquirer's capability gap can arise under any seller where buyer absorption and regulatory verification are weak. The empirical base, however, remains weighted toward one divesting company, and the generalization to other sellers rests on the logic of the capability argument more than on a balanced sample.

The framework can also misfire in implementation, and the most likely way is the substitution of paper for practice. Its premise is demonstrated capability, not a certificate attesting to it, and a regulator that accepted documentary compliance in place of a site-verified demonstration would reproduce the failure the design exists to prevent. Norway's reliance on meetings and site visits, and the finding that firms which passed a table-top assessment failed a subsequent facility inspection, both warn that the demonstration must be of the operating reality and not of the paperwork (Oriaku et al., 2016). A deeper implementation risk is institutional capacity. The design asks the regulator to verify capability on site, sustain post-transfer monitoring, set and enforce conditions, and share evidence across agencies, and it will not work where the regulator lacks the staff, the data access, the independence, or the political backing to do so. Regulatory inadequacy was an enabling condition of the failures (Chapter 4), so the framework presupposes a capacity whose weakness it must also withstand. A second misfire runs the other way: a bar set too high deters capable acquirers and narrows an already thin pool of credible operators. The tiering is the instrument meant to hold the burden in proportion to risk, but its boundaries are qualitative and left to regulatory judgment, and if set too conservatively, the framework would raise entry barriers without a commensurate gain in assurance. The design is calibrated against under-protection and over-deterrence both, and the balance between them falls to implementation rather than to design alone.

The enforcement and verification mechanisms are specified in principle rather than as worked schemes. The integrity triggers connect to rule-based consequences, but who sets the thresholds, who administers them, and at what levels are questions of institutional design left open here. The rule-based direction addresses the political-economy vulnerability documented in the case analysis, but it is not a finished enforcement architecture. The same holds for the community-readiness measure, whose force depends on the linkage a regulator builds between it and the Trust structure the PIA already requires.

These limits are bounded by jurisdiction, and the boundary is not uniform across the design, which operates at two levels. The general level, the capability-assurance logic, the seller-disclosure obligation, the absorptive-capacity requirement, and the regulator-verification role, is calibrated to the institutional conditions that characterize petroleum-producing developing countries. Those conditions are constrained regulatory capacity, high externality exposure, and the active transfer of producing assets from international to domestic operators. The framework would carry to any divestment regime that shares them. The specific level, the

particular legal hooks and the indigenous-operator framing, is tied to Nigeria, to the PIA, the NUPRC regulations, and NOGICD, and to the operating environment of the Niger Delta. The demonstration here is on upstream producing-asset transfers, where integrity failures and their externalities are most acute. The same logic extends to midstream and downstream transfers, which proceed under Section 117 and the Authority instead of Section 95 and the Commission, though that extension is not demonstrated by the cases examined. Transfer to another setting or subsector would keep the logic and re-fit the hooks, subject to local validation. The broader implications and the directions for further research are taken up in the conclusions.

Chapter 6

Conclusions

The question was why the transfer of these assets from international to indigenous operators produced such different results, and what should follow from it. The answer is that capability-related factors best account for the differences, which do not track the capital each acquirer raised, the fiscal terms it faced, or the regulatory regime it worked under. The gap that produces this divergence is built into the structure of the transfer rather than into any single operator. What the evidence warrants is a mandatory, asset-specific assurance mechanism for high-risk producing-asset transfers.

Behind that answer is a particular way of seeing the asset. A producing field is a sociotechnical system, the physical facilities and the operating organization that runs them working as one, and a divestment that conveys the facilities while leaving the organization behind separates the system from the capability that kept it safe and productive. In those terms, the cases reveal a control failure in how the transfer is built, one whose cost falls largely outside the sale (Figure 2.1). The framework developed in Chapter 5 is a control structure designed to close that loop.

The answers to the research questions come first (Section 6.1), followed by why the gap calls for a mandatory rather than a voluntary response (Section 6.2). The recommendations that follow from it are set out next (Section 6.3), then the limits of the findings (Section 6.4) and the boundaries of their generalizability (Section 6.5). The contribution is stated last (Section 6.6).

6.1 Answering the Research Questions

This thesis opened with three questions, and the case analysis answers each (Chapter 4). First, the differentiated outcomes are most consistent with capability-related factors: the acquirers separated according to the operating capability each inherited, brought, contracted, or embedded before performance depended on it. Second, the evidence contradicts fiscal regime effects for these cases and rejects capital constraints as sufficient for the funded operators, finds regulatory inadequacy and political economy each enabling but insufficient on its own. It leaves capability-related factors as the explanation most consistent with the integrity outcomes. Third, because the explanation is capability-related, the remedy is a three-actor Capability Assurance Framework that makes operating capability observable,

testable, and enforceable at the point of transfer, with the seller disclosing requirements, the buyer demonstrating absorption, and the regulator verifying both.

Across the two outcome dimensions, production performance and operational integrity, the acquirers did not converge: they separated into distinct trajectory groups, and the separation lines up most closely with the operating capability each acquirer inherited, brought, contracted, or embedded before performance depended on it. The separation does not track the capital each raised, the fiscal terms each faced, or the regulatory regime each worked under.

Capital constraints, the common explanation, are rejected as sufficient. On the OML 29 critical experiment, the operator that suffered the most serious integrity failure had funded and delivered production growth beforehand, and an operator able to raise and deploy capital for production cannot have its later integrity failure explained by capital scarcity alone. A separate body of evidence points to capability: no standing well-control contract or matching contingency plan when the Santa Barbara well blew out, specialist intervention engaged only after the fact, and a suspended, undecommissioned well the operator was unready to control. The pattern points to capability, with the national spill agency recording that the event exceeded its internal capacity (Section 4.3). The two readings rest on different evidence.

Regulatory inadequacy is an enabling condition, not the primary driver. The test is differentiation: were a weak regime the cause, acquirers under it would perform alike, yet under the same pre-2024 regime that governed Aiteo, Seplat and the HEOSL operatorship sustained production without an equivalent integrity failure (Section 4.5). A regime common to all cannot explain outcomes that varied sharply among them; what it does is let the consequences of a capability shortfall go uncaught: it is the setting in which a capability shortfall produces harm, while the shortfall itself originates in the transfer.

Fiscal regime effects are rejected on timing: every primary transaction, and the capability gaps that produced the failures, predate the Petroleum Industry Act of August 2021, and the divergence under identical pre-Act terms reinforces the point; whether the Act shapes future transactions is left open. Political economy explains which firms acquired and kept operatorship, and the production gains that connections secure by suppressing theft, but it does not reach the case-level capability to sustain integrity or contain a well-control crisis. Why two firms on comparable terms then diverged in whether they could run their assets safely is a question it leaves open.

The within-asset comparison on OML 30 is the strongest evidence against the obvious competing reading, that outcomes track asset quality. Operatorship passed from NPDC to HEOSL on the same wells, reservoirs, and facilities, and performance improved under the new operator, which is hard to reconcile with an asset-quality account. The population-level evidence, and the marginal-field program of 2003 to 2020 in which a majority of licensed operators also failed to sustain their assets, are consistent corroborating context that does not, by itself, carry the verdicts.

The five explanations resolve as follows. Capital constraints are rejected as sufficient; fiscal regime effects are rejected for the cases examined, the transactions and capability gaps predating the PIA. Regulatory inadequacy is an enabling condition, insufficient on its own. Political economy accounts for commercial survival and continued participation but is insufficient on its own for operational integrity. Capability-related factors are most consistent with the combined pattern, with the OML 30 comparison strengthening that reading against an asset-quality explanation. That answers the first two research questions and grounds the

design argument that follows.

6.2 The Structural Case for Mandatory Intervention

The differentiated outcomes are not reducible to individual mismanagement, to be corrected one operator at a time. They follow from how the transaction itself is built. At the point of divestment the information about an asset's embedded operating capability sits with the seller, the capacity to absorb that capability sits with the buyer, and the full social cost of a capability shortfall falls on neither. The gap is structural, a control failure in the design of the transfer, not a defect peculiar to any one acquirer.

Three features of the transfer make it so, each set out in the economics of the transaction (Section 2.5). Information asymmetry comes first. The seller knows what its operating organization does to hold an aging asset within safe bounds; a buyer cannot read that from a data room, and the regulatory screen at consent does not reach the asset-specific capability at issue. The party that knows the most about the embedded capability has little incentive to surface it, since fuller disclosure can lower the price, slow the close, or create transition obligations, and the party that carries the operational risk cannot verify what it is buying. Absorptive capacity is the second, and it sits on the buyer's side. A seller willing to hand over everything it knows still cannot make that knowledge land in an organization without the prior expertise to receive and embed it. The capability has to be present before performance depends on it, which is the line the cases divided along. The third is the externality. When an operator cannot contain a well-control event or sustain integrity, the cost falls on host communities and on the Nigerian state, parties outside the sale and absent from its terms.

These three features have one consequence in common. Voluntary transfer cannot solve a problem whose critical information is held by the seller, whose absorption depends on the buyer, and whose failure costs are borne by communities and the state. Voluntary instruments do not escape it. The industry's operating and process-safety standards and the civil-society National Principles for Responsible Petroleum Industry Divestment describe the right content, but they attach no consequence to ignoring it and leave the underlying incentives untouched (Section 5.5.3). Each describes what a responsible transfer should look like. None compels it.

The instruments that do carry legal force have a different weakness: the ones meant to protect the exposed parties are funded out of the operations that fail. The Host Communities Development Trust is funded by a contribution equal to three percent of the operator's actual annual operating expenditure (PIA Section 240). When an operator falters and operations contract, that contribution contracts with them, leaving the fund thinnest when the hazard is greatest. The decommissioning obligation runs the same way from the other direction. Where the Decommissioning and Abandonment Fund is left to accrue across the producing life of the asset rather than secured at transfer (Sections 232 and 233), an operator that fails early has set aside only a fraction of what closing its wells will cost. Both mechanisms are necessary, and each delivers least when it is needed most.

The structure points to a single remedy, security taken at the front of the transaction rather than left to accrue. The Act already provides an analogous structure: its environmental remediation provision requires a financial contribution the Commission can reach (Section 103), fixed at the point of operation rather than dependent on the operator's later solvency.

Extending that logic to operating capability and to community exposure is the work of the recommendation developed below. The deeper reason the transaction cannot correct itself is that the parties who bear the residual cost are not at the table. Host communities and the state cannot bargain *ex ante* over terms they never see, so the externality cannot be priced into a private deal and must be placed there by the body that licenses the transfer.

The third research question asked what institutional design follows from these findings, and how responsibility should be allocated across the seller, the buyer, and the regulator. A voluntary framework would suffice only if market forces or the existing rules already secured capability transfer. They do not. Outcomes diverged sharply under a common regulatory regime, the transacting parties hold neither the information nor the incentive to close the gap on their own, and the full social cost of failure falls outside the sale. The rules already in force do not reach the asset-specific operating-capability question at consent. The Commission's 2024 reforms created a technical-capacity route, but it is assessed at the corporate level, and a corporate screen can pass an operator that has not demonstrated the asset-specific crisis readiness the cases show to be decisive. The provisions that reach past consent are triggered by failure rather than preventing it. What the evidence warrants is a mandatory, asset-specific assurance mechanism for high-risk producing-asset transfers, the control structure designed to close the severed loop (Figures 5.1 and 5.2), allocating disclosure, demonstrated absorption, and verification across the seller, the buyer, and the regulator.

6.3 Prioritized Recommendations

The full framework, with its three roles, five lifecycle phases, and risk tiers, is developed in Chapter 5. For implementation, six recommendations follow from it, ordered from before consent to after close, each re-specifying an instrument the Nigerian regime already contains rather than adding a new one. They carry the framework's disclosure floor and its five structural levers (Section 5.6.3) into instrument form. Together they put the framework's three roles into operation: what the seller must disclose, what the buyer must demonstrate, and what the regulator must verify and enforce. They are risk-tiered: a disclosure, data, and continuity floor applies to every transfer, while staged operatorship, technical-operator coverage, and intensified post-transfer verification apply to the highest-risk assets and lowest-capability buyers. Disclosure and continuity come first, because the recommendations that follow depend on them.

1. A regulator approving the transfer of a high-risk producing asset should require, as a condition of consent, that the seller disclose the asset's condition and the operating requirements it imposes, and that the buyer demonstrate the operating capability and crisis-response readiness to meet them. This follows from the documented condition that the information determining whether an aging asset can be held within safe bounds sits with the seller and cannot be read from a data room. In Nigeria the requirement attaches to the ministerial consent already mandatory for assignment of an upstream license or lease (PIA Section 95) and to the due-diligence requirement under the Nigeria Upstream Petroleum (Assignment of Interest) Regulations 2024 (Reg. 7). It makes asset-specific disclosure and a demonstrated well-control and crisis-response capability

explicit conditions of consent. Neither is left to the buyer's discretion after close. Aiteo's transfer closed with no documented well-control contingency for OML 29 and no standing well-control contract, the gap that surfaced in the Santa Barbara blowout (Section 4.3). A disclosure-and-readiness condition at consent is the point at which that gap is still visible and correctable.

2. The transfer should carry binding terms that keep an aging asset's operating knowledge available to the buyer through and beyond the handover. That knowledge holds the asset within safe bounds, and it resides in the people who hold it, the procedures that codify it, and the working practices through which it is applied. Hardware transfers at completion; the embedded operating knowledge does not, and tacit knowledge in particular leaves with its holders. Nigeria already requires a succession plan and a multi-year understudy of incumbent expertise under the Nigerian Oil and Gas Industry Content Development Act (NOGICD, Section 31). That mechanism was built for expatriate-to-Nigerian succession. Re-specified toward seller-to-buyer transfer at divestment, and written into the assignment agreement under Section 95 as binding terms for retention, secondment, data, and understudy rather than negotiated ones, it would function as an obligation to carry the operating knowledge across the change of hands. The HEOSL recovery on OML 30, where a technical operator held that knowledge through the transition, shows what such terms are meant to secure.
3. Consent should depend on the buyer demonstrating the organizational capacity to receive and embed the disclosed capability, not on financial and legal standing alone. Disclosed knowledge cannot land in an organization that lacks the prior expertise to absorb it, which is the line the cases divided along. This re-specifies the technical-capacity criterion already within the consent assessment (Section 95; Reg. 7), raising it from a corporate-level screen to an asset-specific demonstration. For Tier 1 assets, the highest-risk tier, the demonstration should normally include a qualified technical operator with a record on comparable operations, drawing on the technology-transfer and partnering route the content law already contemplates (NOGICD Sections 43-47). In Aiteo, the leadership-background evidence pointed to downstream trading rather than comparable upstream operating depth. Seplat brought that depth, through its founding team and its technical partnership with Maurel et Prom. That contrast is the case basis for testing absorptive capacity before consent, not discovering its absence after.
4. For high-risk transfers, operatorship should pass in stages rather than at a single completion date, with critical operating functions passing only after the buyer has demonstrated readiness for them. Time-bounded transition support, technical-operator coverage, or secondment holds the asset within safe bounds during the transition, without leaving the seller indefinitely responsible for an asset it has sold. An abrupt handover concentrates the capability discontinuity at the moment of greatest exposure, before the buyer has shown it can run the asset. In Nigeria staged operatorship is a condition the Commission can attach to consent under Section 95, supervised under its standing authority to monitor and enforce the terms of leases and licenses (Section 7(d)). It requires no new instrument. The phased technical-operator arrangement

through which OML 30 was returned to stable operation is the design pattern. The single-date handover at Aiteo, into an organization without demonstrated upstream operating depth, is the failure that staged operatorship is built to prevent.

5. The regulator should verify operating capability on a continuing basis after transfer, prompted by rule-based triggers, production or integrity deviations, missed maintenance intervals, the loss of key operating personnel, rather than by discretion alone. A capability shortfall invisible at consent becomes visible only in operation, and an unmonitored latent exposure is the most dangerous kind. Nigeria’s regulator already holds the powers this requires: to monitor and enforce compliance with the terms of leases and licenses (Section 7(d)), and to carry out audits and investigations of licensees (Section 7(g)), with administrative penalties available for contravention (Section 231). Binding those powers to defined triggers connects an integrity finding to a graduated, pre-defined consequence rather than to discretion, subject to the operator’s opportunity to respond. The powers already exist; the triggers are what is missing. The well left suspended and undecommissioned on OML 29 from 2015, unmonitored until it failed in 2021 (Section 4.3), is the exposure continuing verification is built to catch.
6. The financial security for decommissioning and for community exposure should be taken at the point of transfer rather than allowed to accrue out of the operations that may fail. An instrument funded from ongoing operations delivers least when the operator falters, which is the moment it is most needed. The remedy is to secure it at consent. The amount should scale to the asset’s risk and inherited liabilities. This removes the dependency the existing instruments carry, the Host Communities Development Trust’s three-percent-of-operating-expenditure base (Section 240) and the decommissioning fund’s accrual across the producing life of the asset (Sections 232 and 233). It extends the upfront structure the Act already uses for the environmental remediation contribution (Section 103) to decommissioning security and community-readiness provisioning fixed at consent. The undecommissioned well on OML 29 and the 41 communities exposed by the blowout are the costs this provisioning is meant to cover before, not after, an operator’s capacity to pay has eroded.

The six divide by where they attach. Recommendations 1 through 4 attach to the consent gate, the one-time approval at transfer. They therefore apply only to transfers not yet completed, the divestments still to come in a wave of IOC onshore exits that is not finished, and cannot be re-run on a transaction already closed. The consent-stage screening they require is not hypothetical: the Commission applied a version of it when it initially withheld consent for the Shell-to-Renaissance transfer in 2024 on the ground that the consortium had not demonstrated the requisite technical capacity. The Commission granted consent later that year (The Energy Year, 2025; BusinessDay, 2025). Recommendations 5 and 6 attach to powers and obligations that bind every current operator continuously: the Commission’s standing supervisory powers (Sections 7(d) and 7(g)) and the standing fund obligations under Sections 232-233 and 103. They therefore reach the assets already transferred, Aiteo and Eroton included. Applying them to current operators enforces standing obligations rather than reopening closed consent decisions. It would not re-run the original transfer; it would

apply continuing powers and standing obligations to operating risks that are still live. That is where the exposure now sits.

None of this builds a second regulatory track alongside the one the Commission introduced in 2024. That route assesses capacity at the corporate level; the recommendations re-specify the same consent gate and the same standing powers to the asset-specific operating capability and crisis readiness the cases show to be decisive. Nor do they depend on settling whether a given operator's capability came from what it inherited at the transfer or from its own organizational quality. Each tests for demonstrated operating capability directly, whatever its source, a limit taken up in the next section (Section 6.4). These measures work as a set, not a menu. For a new transfer, requiring disclosure at consent without the matching post-transfer verification would let a shortfall that clears the gate go uncaught, so the consent-stage and post-transfer measures belong together rather than sequenced years apart. What the six do, operating together, is make the operating capability an asset requires observable before consent, testable at the gate, and enforceable after transfer. They are warranted by the conditions the cases document, not by evidence of their effect, which no implementation has yet produced; the claim is that they make capability assurable, not that they will prevent the next incident or raise national production.

6.4 Limitations

The findings carry weight within stated bounds, and naming those bounds is part of stating the findings accurately. Each limitation below constrains the scope of a particular claim, and none overturns the central finding.

The deepest of these concerns the regulator. Verification rests with the same Commission whose limited reach was an enabling condition of the failures (Section 6.1), which invites the objection that an institution unable to catch the old shortfall cannot be relied on to operate the new safeguard. The design depends less on regulatory judgment than the status quo does. It re-specifies the technical-capacity screen from corporate-level standing to asset-specific readiness, shifts the burden of proof to the buyer to demonstrate capability, and ties consequences to rule-based triggers rather than to discretion. The parts most exposed to weak capacity, site visits and post-transfer audits, can be risk-triggered rather than applied to every transfer, while disclosure and transfer-agreement content can be enforced as documentary requirements. The limit is real even so. If the Commission lacks the capacity, independence, data access, or political backing to apply the design, the framework cannot function as intended. The framework and the building of regulatory capacity are jointly necessary, not substitutes, and a phased adoption that begins with disclosure and transfer terms and builds toward full verification is the realistic path in a constrained environment.

A second limit concerns the source of the capability the cases turn on. The evidence cannot fully separate whether a given operator's capability came from what it inherited at the transfer or from organizational quality it already held, so the causal attribution to the inheritance itself is bounded. The prescription does not depend on settling that attribution. Each recommendation tests for demonstrated operating capability at the point of transfer, whatever its origin (Section 6.3), so the confound limits what can be claimed about cause without weakening what the framework requires.

A related reading attributes the divergence to the assets rather than to the operators, setting Aiteo's failure on a large and difficult asset against acquirers that performed on easier ones. The within-asset comparison on OML 30 is the direct answer (Section 6.1): the operator changed while the asset was held fixed, and the outcome changed with it. Where that holds, the asset-quality explanation is substantially weakened, and the operator-capability finding stands.

The framework has not been implemented, and no implementation data yet show that it improves performance, prevents incidents, or raises production. The claim made for it is therefore the bounded one: it is warranted by the regularities documented across the cases and by the design logic, not proven by results it has yet to produce. What the absence of outcome data constrains is any claim about effect. It does not weaken the warrant, which rests on harms that are documented, that fall on parties outside the sale, and that are difficult to reverse once they occur (Section 6.2). Action is warranted under those conditions even though the framework remains untested as an intervention. What the framework changes, independently of any later results, is that the capability the cases tie to failure must be shown at the point of transfer rather than assumed.

Verification reaches the capability that is observable, demonstrable, and verifiable. The non-contractible tacit knowledge that no instrument can compel a seller to surrender stays with the individuals and teams who hold it. For that reason the recommendations secure the conditions under which knowledge transfers, retention, secondment, and a documented handover, rather than the knowledge itself. The boundary limits what verification can guarantee. It does not remove the value of verifying what is observable, since several of the capabilities the cases turned on, crisis-readiness arrangements, technical-operator depth, and continuity of operating knowledge, were demonstrable before close.

The evidence is a set of six cases built around a single critical experiment with cross-case validation (Chapter 3). A design of this kind can corroborate an explanation and show it to be plausible across the cases examined; it cannot establish that no counter-example exists, and it does not prove a universal law. The finding is therefore stated in calibrated form: capability-related factors are the explanation most consistent with the observed pattern, not a demonstrated universal regularity. What these cases support beyond their own setting is taken up next (Section 6.5).

The reach of the consent-stage recommendations into transactions already closed is partial by design. Because they cannot be applied to a divestment that has already completed (Section 6.3), their effect is bounded to pending and future transfers, which limits what they can change without bearing on the diagnosis they follow from. The standing-power recommendations do reach the assets already transferred, so the framework is not inert for the operators now holding them.

These limits together fix where the thesis's claims stop. They bound the causal attribution, the outcome claim, the reach of verification, and the generality of the result, but they do not unsettle the diagnostic finding or the design warrant. What stands is the case that mandatory, asset-specific assurance is warranted for high-risk producing-asset transfers under the conditions documented here.

6.5 Boundaries of Generalizability and Future Research

How far these findings reach is a question of analytic transferability, not demonstrated generalizability. The framework was developed on Nigerian divestment cases between 2010 and 2024. It is calibrated first to Nigeria and, analytically, to the wider institutional context set out in Objective 3: petroleum-producing developing countries in which a wave of divestment meets constrained regulatory capacity and an externality placing the cost of failure on host communities and the state. To other settings that share those conditions the framework may transfer, subject to local validation. The basis is analytic rather than statistical, because the cases were chosen to test competing explanations rather than to represent a population (Chapter 3).

The comparator regimes do not validate the framework, but they show that its core move, requiring capability to be shown before and after a transfer rather than merely declared, is established practice in mature petroleum jurisdictions, not an invention of this thesis. The United Kingdom requires the buyer and seller to file a capability pack documenting the buyer's financial and technical capability before it consents to a license assignment. It also treats the fitness of those who control the licensee as a continuing criterion, backed by a standing power to require a change of control or to revoke the license. Norway prequalifies operators before entry and requires a competent in-country organization. Alberta assesses both parties to a transfer and ties the security a buyer must post to that assessment (Section 5.5). What differs is the target. Those regimes run the logic inside strong institutions, whereas the framework adapts it to a setting where regulatory capacity is thinner and the consequences of failure fall on parties outside the sale, the harder case for which it is designed.

Two features of the evidence bound how far the finding extends. Some of the Nigerian setting is specific to it: the theft and sabotage that confound production outcomes at the sector level (Chapter 4) limit what the production evidence can show, though they touch neither the existence of the externality nor the need to verify capability. The cases are also weighted toward divestments originating with Shell, which narrows the seller-independence of the finding. That weighting does not unsettle the core result, because the outcomes diverged sharply among buyers who all acquired from the same seller, so the divergence cannot be an artifact of seller identity. What it leaves open is whether the same pattern recurs across a broader base of sellers.

The research agenda follows from these limits, not from a general call for more study. The first and most important open question follows from the absence of implementation data (Section 6.4): the framework is warranted by the documented conditions, not tested by results. The major post-2024 divestments, the Shell-to-Renaissance transfer and the recent ExxonMobil/MPNU and Eni/NAOC transfers to Seplat and Oando, are closing under the 2024 reforms and will produce the outcome record the cases here could not. As that record accumulates, it moves the design from warranted toward tested. A comparison between transfers screened for asset-specific capability and those screened only at the corporate level would then test, refine, or qualify it. These transactions are future tests, not present evidence; none has yet run long enough to establish medium-term production, integrity, and governance outcomes.

Four narrower questions follow from limits already stated. The organizational-quality

confound (Section 6.4) would be settled by operator-level data that separates capability inherited at transfer from capability a buyer already held, the attribution the cases could only bound. The seller-independence question would be answered by cases drawn from a wider set of divesting parties than the Shell-weighted record here. A third question concerns the boundary of non-contractible tacit knowledge (Section 6.4): how far retention, secondment, and a documented handover can carry what no instrument can compel directly. The last is one of scope, whether the framework holds in other petroleum-producing developing countries where divestment, constrained regulatory capacity, and externalized failure recur.

Each of these is a specific open question left by a stated limit, not a gap to be filled for its own sake. With the evidence already in hand, they mark the path by which a framework warranted on the Nigerian record could be tested, extended, and, where the conditions recur, carried beyond it.

6.6 Contribution and Conclusion

The primary contribution of this thesis is a systems-design one, and it reduces to a single reframing. What fails to transfer in a divestment is the operating capability embedded in the organization that ran the asset: the wells, the facilities, and the licenses change hands intact, while the capability that kept them safe and productive stays behind. Closing the gap that opens is a problem of institutional design, met by a control loop that holds that capability observable, demonstrable, and verifiable across the transfer. An upstream divestment is then the transfer of a sociotechnical system, and the failures observed across the cases sit in the control structure of that system. The empirical finding, that capability-related factors best account for the differentiated outcomes, underwrites the reframing; the reframing and the design that follows from it are the contribution.

The diagnosis is grounded in a tradition that treats a producing asset as a coupling of a technical subsystem and the social subsystem that operates it (Trist and Bamforth, 1951; Section 2.2). Safe and productive operation is an emergent property of the two together rather than of the equipment alone. Unless the operating capability is deliberately carried across, a divestment divides that system along the interface between them, conveying the wells, the facilities, and the data while leaving the operating organization behind. What that organization holds, from the operating workforce to the contractor architecture, is the operating enterprise itself, which enterprise-architecting treats as an object of design in its own right (Nightingale and Rhodes, 2015; Section 5.2). Through that lens, the failure has a precise location. The party that knows what is being lost has no incentive to transfer it, the corrective feedback that would catch a shortfall before harm is severed, and much of the cost falls on parties outside the sale (Figure 2.1). That is a control failure in the systems-theoretic sense, the absence of the control actions and feedback that hold a hazardous process within safe bounds (Leveson, 2011; Section 5.4).

The design follows from the diagnosis. The Capability Assurance Framework is a control structure for the transfer: three mutually constraining roles, seller disclosure, buyer absorption, and regulator verification, that reconstitute the operating system across the transaction by closing the loop the divestment severs (Figures 5.1 and 5.2). Each role supplies what the next must act on, so the structure holds only when all three operate together, and it installs the

corrective function the market does not supply. The interlock sits at the level of capability. A divestment re-architects the operating enterprise only in part: it transfers the physical infrastructure intact while leaving the knowledge that runs it to be reconstituted by the buyer or lost. Because that gap is not captured by ordinary transaction documents, it can remain unverified until it surfaces in operation, the Aiteo pattern (Section 4.3). The buyer's absorptive capacity (Cohen and Levinthal, 1990) is, in these terms, the receiving enterprise's capacity to reconstitute that knowledge through the transition, and the framework's verification step makes the difference observable before handover rather than after failure.

The systems-design contribution is supported by three others, and none displaces it. The first is theoretical, combining the sociotechnical-systems tradition, enterprise-architecting, systems-theoretic control, and the absorptive-capacity and knowledge-transfer literatures into one account of a problem usually treated in financial, fiscal, or regulatory terms. The second is methodological: a discriminating-case design, built around a critical experiment with cross-case validation and the explicit adjudication of competing explanations (Chapter 3), adaptable to other divestment and asset-transfer settings where the same alternatives compete. The third is practical: the recommendations are attached to specific instruments a regulator already holds (Section 6.3), which makes the design deployable rather than only diagnostic.

The scope of this contribution is bounded by what the evidence supports. The reframing and the design are warranted by the documented regularities and the design logic, not by implementation results the framework has yet to produce. The claim that holds is narrower and, within the cases examined, secure. A divestment leaves the control loop open when capability is not assured across it, and a framework of this kind makes that capability observable, testable, and enforceable at the point of transfer.

Yet the stakes reach beyond the transaction itself. A divestment is not only a private sale; it is the transfer of responsibility for a public hazard. When the control loop is left open, the cost of a capability that does not survive falls not on the seller that exits or the buyer that underperforms, but on the host communities and the Nigerian state that never sat at the table. Closing the loop places the burden of demonstrating capability on the parties to the transfer, where the cost of failure says it belongs. Divestment, seen as a systems-design problem, comes down to a single requirement: that the transfer of the hardware not become the transfer of a hazard no one has agreed to carry.

Appendix A

Case Study Protocol (COUHES Protocol E-7658)

The full case study protocol is set out below. It received exemption from the MIT Committee on the Use of Humans as Experimental Subjects under COUHES Protocol E-7658.

A.1 Study Overview

Title: Inheriting Hardware Without the System: A Sociotechnical Knowledge and Capability Transfer Framework for IOC Asset Divestments

Principal Investigator: Ikenna Ekekwe (ike1@mit.edu), MIT System Design and Management Program

Faculty Sponsor: Dr. Donna Rhodes (rhodes@mit.edu)

Thesis Supervisor: Dr. George Lordos (glordos@mit.edu)

A.2 Participant Groups and Sample Design

The target sample is 20 participants distributed across three groups:

Group A (IOC Technical Staff and Management): 6 participants (3 technical/operational, 3 management/strategic). Former or current employees of international oil companies who were involved in the operation or divestment of upstream assets in Nigeria. Group A provides evidence on what operational capability existed before divestment, how it was organized, what was included in or excluded from the transfer, and how the transition was coordinated from the seller's side.

Group B (Domestic Oil Company Operators and Managers): 8 participants (4 technical/operational, 4 management/strategic). Current or former employees of domestic oil companies that acquired assets from IOCs. Group B is the primary source on post-divestment capability gaps, staffing and organizational decisions, financial resources, fiscal regime effects, security and community dynamics, and coordination from the buyer's side.

Group C (Regulatory Officials and Industry Observers): 6 participants. Current or former officials of the Department of Petroleum Resources, the Nigerian Upstream Petroleum

Regulatory Commission, the NNPC, or independent industry analysts and consultants with knowledge of the divestment process. Group C contributes evidence on what the regulator assessed at the point of divestment, the regulatory mandate, and cross-transaction patterns.

The minimum sample for analytical viability is 16 (4 Group A, 6 Group B, 4 Group C, with at least 2 of each role subclass in Groups A and B). The maximum is 25 if theoretical saturation has not been reached.

A.3 Informed Consent Procedures

Informed consent is obtained verbally at the start of each interview and documented in the investigator's notes. Before each interview, the investigator provides the participant with a description of the study's purpose, what participation involves, and the recording procedures. The investigator explains confidentiality protections, the voluntary nature of participation, the right to decline any question or stop the interview at any time, the principal risk (professional discomfort if responses were attributed), and contact information for both the investigator and COUHES. The investigator's Chevron affiliation is disclosed at the start of every interview, with an explicit statement that the thesis is not sponsored by, reviewed by, or subject to approval by Chevron or any other oil company.

A.4 Data Handling

Interviews are audio-recorded with the participant's explicit consent. If a participant declines recording, the investigator takes written notes. Recordings are deleted after transcription is verified. Transcripts are retained until the thesis is submitted and approved, after which they are destroyed. All data is stored on the investigator's password-protected personal computer, with an encrypted backup maintained under the same access restrictions. No interview data is stored on shared drives, cloud services, or institutional servers. Participants are de-identified in all transcripts, notes, and thesis text. Names are replaced with alphanumeric codes (A1, A2, B1, B2, C1, C2). Organizational affiliations are generalized to prevent indirect identification.

A.5 Pre-specified Coding Categories

Interview transcripts are coded deductively against the following categories derived from the discrimination apparatus:

Capital constraint indicators: financing structure, capital expenditure trends, cash call compliance, contractor payment patterns, prioritized deferral sequence.

Fiscal regime indicators: PIA fiscal impact on investment, pre-PIA vs. post-PIA outcome comparison, operator-reported fiscal sensitivity.

Regulatory execution indicators: regulatory mandate content, regulatory staffing and capacity, inspection frequency and compliance actions, evidence of awareness without follow-through.

Regulatory design indicators: capability assessment criteria in approval framework, institutional self-diagnosis, framework reform content.

Coordination failure indicators: transition plan existence and quality, handover duration and scope, specific coordination breakdowns, staff mobilization timing.

Capability-related factor indicators: staff retention rates, knowledge transfer activities, pre-divestment deterioration, selective failure (routine vs. crisis), awareness (pre-failure procurement, organizational design, post-failure diagnosis), knowledge availability (handover, staff offer, documents, transition support), capability trajectory over time.

Political economy indicators: political connections (board, shareholders), oil theft rates pre/post transfer, security arrangements.

Emergent categories: new analytical categories that arise during coding are flagged as emergent and recorded separately from the pre-specified framework.

A.6 Pretest Design

The first two interviews serve as pretests (one from Group A or B, one from a different group). Pretests assess significance (whether questions produce action-based responses mapping to the discrimination apparatus), validity (whether role-specific probe tracks yield expected differentiation), and representativeness (whether questions elicit evidence across all five explanations). Core questions are not changed after the pretest; probe language may be refined. Pretest data enters the analytical dataset.

Appendix B

Discrimination Matrix

The full discrimination matrix, referenced in Section 2.6, maps observable evidence patterns to their implications for each of the five competing explanations and specifies the strength of each implication (Decisive or Suggestive). The empirical analysis (Chapter 4) applies it to each case through the five-step decision sequence specified in Section 3.4.

Strength indicators: D = Decisive (would reject or strongly confirm for that case). S = Suggestive (would weaken or strengthen but not conclusive alone). SYS = System-level condition (applies uniformly to all operators; cannot discriminate between individual outcomes).

Table B.1: The discrimination matrix: the strength-coded implication that each observable evidence pattern carries for the competing explanations.

Evidence Pattern	Capital Constraint	Fiscal Regime	Reg. Execution	Reg. Design (SYS)	Coordination	Capability-Related Factors	Political Economy
Well-funded operator with selective (not uniform) failure	D-REJECTS as sufficient	-	-	-	S-POSSIBLE	D-SUPPORTS	-
Demonstrably underfunded operator with cumulative degradation following deferral sequence	D-SUPPORTS	-	-	-	-	S-WEAKENS as sole cause	-
Failure predates PIA enactment	-	D-REJECTS for that case	-	-	-	S-SUPPORTS	-
Operator investment decisions explicitly attributed to PIA fiscal terms	-	D-SUPPORTS	-	-	-	-	-
Same regime, sharply different outcomes	-	S-WEAKENS	D-REJECTS exec as sufficient	Cannot discriminate (constant)	S-SUPPORTS	S-SUPPORTS	-
Routine success + crisis failure + buyer UN-AWARE of gap pre-crisis	S-WEAKENS	S-WEAKENS	-	-	S-WEAKENS (buyer should have known)	D-SUPPORTS	-
Routine success + crisis failure + buyer AWARE of gap pre-crisis	S-WEAKENS	S-WEAKENS	-	-	D-SUPPORTS	S-WEAKENS	-

Evidence Pattern	Capital Constraint	Fiscal Regime	Reg. Execution	Reg. Design (SYS)	Coordination	Capability-Related Factors	Political Economy
Pattern recurs across unrelated institutional contexts	-	-	-	SYS-SUPPORTS	S-WEAKENS independence	S-SUPPORTS	-
Knowledge inheritance correlates with outcome (confounding controlled)	-	-	-	-	-	D-SUPPORTS	-
Knowledge available (triangulated) but buyer fumbled reception	-	-	-	-	D-SUPPORTS	S-WEAKENS	-
Knowledge unavailable (triangulated) regardless of buyer quality	-	-	-	-	S-WEAKENS	D-SUPPORTS	-
Specific identifiable coordination gaps trace to failure	-	-	-	-	D-SUPPORTS	-	-
Regulator creates NEW framework (not enforces old)	-	-	D-REJECTS	SYS-SUPPORTS	-	-	-
Pre-failure: operator did not procure capability, function absent from org design	-	-	-	-	S-WEAKENS	S-SUPPORTS (unawareness)	-

Evidence Pattern	Capital Constraint	Fiscal Regime	Reg. Execution	Reg. Design (SYS)	Coordination	Capability-Related Factors	Political Economy
Pre-failure: operator attempted to procure but failed	S-POSSIBLE	-	-	-	D-SUPPORTS	S-WEAKENS	-
Operator survives despite low quality via connections	-	-	-	-	-	-	D-SUPPORTS
Capability-transferred operator shows high quality + production	-	-	-	-	-	D-SUPPORTS	D-EXTENDS
No knowledge inheritance + success including crisis	-	-	-	-	-	D-KILLS	-
No correlation between knowledge inheritance and outcome across cases	-	-	-	-	-	D-KILLS (core prediction fails)	-
Pre-divestment deterioration selective in knowledge-bearing functions	-	-	-	-	-	S-SUPPORTS Gap 1	-
Operator without connections succeeds on capability alone	-	-	-	-	-	S-SUPPORTS	S-WEAKENS

Table B.2: Decision protocol for Chapter 4: the five-step sequence for applying the discrimination matrix to each case.

Step	Procedure
1	Assess temporal and structural preconditions: which explanations are applicable to this case? Fiscal regime effects are excluded for any case where the transaction closed before the PIA enactment in August 2021. Regulatory design failure applies uniformly to all cases and is assessed at the population level.
2	Test falsification conditions: does the evidence contain any pattern that the matrix identifies as Decisive-REJECTS for an explanation in this case? Explanations that are decisively rejected are set aside with the rejecting evidence stated.
3	Assess support strength for explanations that survive falsification: does the evidence pattern decisively support, suggestively support, or leave the explanation unresolved?
4	Apply discrimination logic for jointly operative explanations: which mechanism accounts for a larger share of the observed evidence pattern, and what is the evidentiary basis for that judgment?
5	Report ambiguity where discrimination cannot be resolved, particularly the coordination-versus-capability distinction, which depends on the partially latent variable of knowledge availability.

The political economy explanation enters at a different stage: evaluated against commercial survival outcomes, not operational quality outcomes.

The governing evidential principle is that no single indicator is treated as decisive for attribution. Attribution rests on the convergence of multiple indicators consistent with one explanation and inconsistent with its competitors, drawn from at least two of four independent evidence source categories (seller testimony, buyer testimony, regulatory/observer testimony, observable operational outcomes).

Appendix C

Interview Guides

C.1 Group A Interview Guide (IOC Technical Staff and Management)

COUHES Protocol E-7658

Estimated Duration: 45-60 minutes

Role-Specific Probe Tracks

This guide contains standardized core questions asked identically to every Group A participant. Probes are marked with role indicators to help the interviewer follow the most productive knowledge path for each participant:

[T] = Technical/Operational roles (engineers, production managers, HSE managers, wellsite supervisors, operations supervisors). These probes ask about operational practices, equipment, procedures, and what happened at the asset level.

[M] = Management/Strategic roles (asset managers, country managers, business development managers, corporate strategy staff). These probes ask about corporate decisions, resource allocation, strategic rationale, and organizational design.

[BOTH] = Applicable to all participants regardless of role.

The interviewer should use all [BOTH] probes and the probes matching the participant's role. If a participant's role spans both categories, the interviewer uses judgment on which probes are within the participant's firsthand knowledge.

Before You Begin (5 minutes)

1. Introduce yourself and the study: 'I am Ikenna Ekekwe, a graduate student at MIT. I am conducting research on the operational outcomes of IOC asset divestments in Nigeria for my master's thesis. This research is not sponsored by or subject to approval by Chevron or any other oil company.'
2. Disclose your Chevron affiliation: 'In the interest of transparency, I should mention that I am currently employed by Chevron Nigeria. The views in this thesis are my own and independent of my employer.'
3. Confirm consent: 'Before we begin, I want to confirm that you received the study information, that you understand the purpose of this interview, and that you agree to

participate voluntarily. Do you consent to participate?’

4. Confirm recording preference: ‘Do you consent to audio recording of this interview, or would you prefer that I take written notes only?’
5. Remind participant: ‘You may decline to answer any question and may stop the interview at any time. Your responses will be de-identified. You will not be named in the thesis.’

Opening Question (5 minutes)

A1. Please describe your role and responsibilities during the period when [the specific asset/OML] was being operated and subsequently divested.

Probe A1.1 [BOTH]: What specific operations or functions were you responsible for?

Probe A1.2 [BOTH]: Over what time period were you involved?

Probe A1.3 [BOTH]: How many people reported to you or worked alongside you in this function?

Primary Questions (35-45 minutes)

Pre-Divestment Capability Architecture

A2. Before the divestment, what operational systems, processes, and personnel arrangements were in place to run [the specific asset]?

Probe A2.1 [T]: Walk me through what happened on a typical day at the flowstation or production facility.

Probe A2.2 [T]: When a well integrity issue or equipment failure arose, what was the response process? Who was involved?

Probe A2.3 [T]: How were new engineers or operators trained? What did the training look like in practice?

Probe A2.4 [BOTH]: Were there operational procedures, knowledge, or systems that existed only inside the organization, in people’s heads or in informal practices, and were not written down in documents that would go to a buyer?

Probe A2.5 [M]: How was the operational capability of this asset resourced and maintained at the organizational level? What investment, staffing, and management attention did it require to keep running at the standard the company maintained?

Pre-Divestment Deterioration

A3. In the years leading up to the divestment, and particularly between the announcement and the actual handover, did you observe any changes in how the asset was operated?

Probe A3.1 [BOTH]: When did you first notice a shift in the company’s commitment to the onshore portfolio? Was this before the formal divestment announcement?

Probe A3.2 [T]: Were staff reassigned to other operations (deepwater, LNG) before the transaction closed? Which functions lost people first?

Probe A3.3 [T]: Were maintenance programs, capital investments, or training activities deferred, reduced, or discontinued during this period? Were the cuts uniform across all operational functions, or were some categories, such as training, mentoring, or knowledge transfer activities, reduced earlier or more severely than production-related expenditure?

Probe A3.4 [T]: Were there wells, flowlines, or facilities that were left in a condition that would require significant intervention by the incoming operator?

Probe A3.5 [M]: Was the reduction in investment, staffing, or maintenance a deliberate corporate decision linked to the planned divestment, or was it driven by other factors such as cost-cutting or portfolio rebalancing? Who made these decisions, and what was the rationale?

Staff Decisions

A4. What happened to the technical and operational staff when the asset was divested?

Probe A4.1 [M]: Were staff offered employment by the acquiring company? If so, on what terms? If not, why not?

Probe A4.2 [M]: Did the IOC offer to second or loan staff to the buyer for a transition period? What was the buyer's response?

Probe A4.3 [T]: Which staff left before the transaction closed because they saw the divestment coming? Which functions were affected most?

Probe A4.4 [BOTH]: Of the staff who had the deepest operational knowledge of the asset, how many stayed with the new operator and how many left?

Transfer Process

A5. What specifically was included in the asset transfer when the transaction closed?

Probe A5.1 [T]: Were operating procedures, maintenance schedules, and HSE protocols transferred as documents? Were they complete?

Probe A5.2 [BOTH]: Was there a formal transition period during which IOC staff worked alongside the new operator? How long? What did it cover?

Probe A5.3 [BOTH]: Were there systems, processes, or institutional knowledge that you expected to be transferred but were not? What prevented the transfer?

Probe A5.4 [M]: What information did the buyer have access to about how the asset was actually operated, beyond the formal data room documents? Could the buyer assess the operational capability it was acquiring before the transaction closed?

Probe A5.5 [M]: Who decided what would be included in the data room and the transfer package? Were there categories of operational knowledge or organizational capability that were deliberately excluded? If so, what was the rationale?

Probe A5.6 [M]: Were there pressures within the company to ensure a responsible or safe transition, such as reputational concerns, regulatory expectations, or liability considerations? How were these balanced against the commercial priorities of the divestment?

Contractor and Vendor Ecosystem

A6. How did the relationships with service companies, drilling contractors, and maintenance vendors work, and what happened to those relationships after the divestment?

Probe A6.1 [T]: Which service companies were critical to the day-to-day operation of the asset? What did they provide?

Probe A6.2 [BOTH]: Were these contractor relationships tied to the IOC's corporate agreements, or were they asset-specific?

Probe A6.3 [M]: Would the same contractors extend the same terms, priority, and crew quality to the new operator? What would you expect to change?

Probe A6.4 [T]: Were there specialized capabilities (well control, subsea intervention, corrosion management) that depended entirely on contractor relationships?

Community and Stakeholder Architecture

A7. How did the relationships with host communities, security arrangements, and stakeholder agreements work, and what happened to them during the divestment?

Probe A7.1 [M]: Were there community agreements (Global Memoranda of Understanding, MOUs) that were tied to the IOC rather than the asset? Did they transfer?

Probe A7.2 [T]: Were there community liaison officers or other staff whose relationships with community leaders were critical to operational access?

Probe A7.3 [M]: How were security arrangements (military, community-based, private) organized, and did they transfer with the asset?

Probe A7.4 [BOTH]: Did you anticipate any community-related operational disruptions for the incoming operator?

Coordination from the Seller's Perspective

A8. How was the transition coordinated among the seller, the buyer, and the regulator?

Probe A8.1 [M]: Was there a formal transition plan? Who designed it? What did it cover?

Probe A8.2 [BOTH]: At what point did coordination break down, if it did: before the transaction was approved, during the handover period, or after the new operator took control?

Probe A8.3 [BOTH]: Did the regulator play any role in overseeing or verifying the transition? What specifically did the regulator do or not do?

Probe A8.4 [BOTH]: Looking back, what would have needed to be coordinated differently for the transition to produce a better operational outcome?

Probe A8.5 [M]: To your knowledge, what did the regulator assess about the acquiring company before approving the transaction? Were there aspects of operational readiness that you believe should have been assessed but were not?

Secondary Question: JV Governance

A9. How did the joint venture governance with NNPC work, and how would you expect it to change with a new operator?

Probe A9.1 [M]: How were cash calls, budget approvals, and work programs managed within the JV?

Probe A9.2 [BOTH]: Was NNPC's engagement with the IOC operator different from what you would expect with a domestic operator?

Framework Elicitation

A10. If you could redesign the divestment process to prevent the operational failures that have occurred, what would you change?

Probe A10.1 [BOTH]: What should the seller be required to do before or during the transfer?

Probe A10.2 [BOTH]: What should the buyer be required to demonstrate before being approved to acquire the asset?

Probe A10.3 [BOTH]: What should the regulator be required to verify before, during, and after the transaction?

Probe A10.4 [BOTH]: Are there examples of transfers that worked well? What made them different?

Closing (5 minutes)

1. 'Is there anything important about the divestment process or its consequences that my questions did not cover?'
2. 'Is there anyone else you would recommend I speak with who has direct knowledge of these events?'

3. 'Do you have any questions about the study?'
4. Thank the participant. Confirm their preferred contact method if follow-up questions arise.
5. Immediately after the call, complete expanded notes while the conversation is fresh. Note any statements the participant emphasized or that you judge to be analytically significant. Record the participant's role type (technical/operational or management/strategic) for analytical coding.

C.2 Group B Interview Guide (Domestic Oil Company Operators and Managers)

Group B: DOC Operators and Managers

COUHES Protocol E-7658

Estimated Duration: 45-60 minutes

Role-Specific Probe Tracks

This guide contains standardized core questions asked identically to every Group B participant. Probes are marked with role indicators:

[T] = Technical/Operational roles (production engineers, field operators, HSE staff, maintenance engineers). These probes ask about what happened operationally, what gaps were encountered, and what the participant experienced at the asset level.

[M] = Management/Strategic roles (managing directors, business development staff, acquisition team leaders, CFOs, operations directors). These probes ask about strategic decisions, financial positions, organizational design, and the rationale behind key choices.

[BOTH] = Applicable to all participants regardless of role.

The interviewer should use all [BOTH] probes and the probes matching the participant's role.

Before You Begin (5 minutes)

1. Introduce yourself and the study: 'I am Ikenna Ekekwe, a graduate student at MIT. I am conducting research on the operational outcomes of IOC asset divestments in Nigeria for my master's thesis. This research is not sponsored by or subject to approval by Chevron or any other oil company.'
2. Disclose your Chevron affiliation: 'In the interest of transparency, I should mention that I am currently employed by Chevron Nigeria. The views in this thesis are my own and independent of my employer.'
3. Confirm consent: 'Before we begin, I want to confirm that you received the study information, that you understand the purpose of this interview, and that you agree to participate voluntarily. Do you consent to participate?'
4. Confirm recording preference: 'Do you consent to audio recording of this interview, or would you prefer that I take written notes only?'

5. Remind participant: ‘You may decline to answer any question and may stop the interview at any time. Your responses will be de-identified. You will not be named in the thesis.’

Opening Question (5 minutes)

B1. Please describe your role and responsibilities during and after the acquisition of [the specific asset/OML].

Probe B1.1 [BOTH]: What specific operations or functions were you responsible for?

Probe B1.2 [BOTH]: Were you involved in the acquisition process itself, the transition, or only in post-acquisition operations?

Probe B1.3 [BOTH]: How long after the transaction closed did you remain involved with the asset?

Primary Questions (35-45 minutes)

Staffing Decisions and Organizational Preparation

B2. What technical and operational staff did your organization bring to the asset, and how were staffing decisions made?

Probe B2.1 [M]: Did your organization attempt to retain the IOC’s technical staff? Were you offered the option? If you declined, why?

Probe B2.2 [BOTH]: Where did your operational staff come from? Were they experienced in upstream operations at this scale and complexity?

Probe B2.3 [T]: Were there specific technical functions (well integrity, HSE, pipeline integrity, emergency response) where your team lacked experience?

Probe B2.4 [M]: Before the acquisition, what experience did your organization have in operating upstream assets at the scale and complexity of what you were acquiring?

Probe B2.5 [M]: What was your organization’s strategy for building the operational capability to run this asset? Was there a deliberate plan to develop capability internally, acquire it through hiring, or rely on the transition from the IOC? How was this decision made?

Probe B2.6 [M]: When your organization took over, what operational functions were established in your organizational structure? Were there functions that the IOC had maintained, such as well control, emergency response, or corrosion management, that your organization did not include in its initial structure?

Expectation vs. Reality

B3. What did you expect to find when you took over the asset, and how did the reality compare?

Probe B3.1 [M]: Based on due diligence and the data room, what was your understanding of the asset’s condition and operational requirements?

Probe B3.2 [T]: When you took over, what surprised you? What was different from what the pre-acquisition information led you to expect?

Probe B3.3 [T]: Were there specific problems (well conditions, pipeline integrity, environmental liabilities, staffing gaps) that were worse than disclosed?

Probe B3.4 [M]: Looking back, was there information you needed before the acquisition that you did not have and could not have obtained from the data room?

Post-Divestment Operational Reality

B4. After the transaction closed, what specific operational changes occurred in the first 6 to 12 months?

Probe B4.1 [T]: What was the first major operational challenge your organization encountered?

Probe B4.2 [T]: Were there specific functions or capabilities that your organization found it could not perform without outside assistance?

Probe B4.3 [M]: Did your organization bring in external contractors or consultants? For what functions? For how long? Who made this decision? Were external specialists needed to identify what had gone wrong, or did your organization know what the problem was and bring in specialists to help execute the response?

Probe B4.4 [T]: Were there differences between your organization's ability to handle day-to-day production operations and its ability to respond to emergencies, equipment failures, or well control situations? Can you give me specific examples of each?

Probe B4.5 [BOTH]: Were there any incidents, near-misses, or operational disruptions during this period? What happened, and how were they handled?

Probe B4.6 [M]: Before the first major operational incident or crisis, had your organization taken any steps to build capability in that specific area, such as hiring specialists, engaging contractors, or conducting training? If not, was the need recognized before the event occurred?

Probe B4.7 [BOTH]: Over the first two to three years of your operations, did your organization's capability in any specific area improve? If so, what caused the improvement: additional funding, hiring of experienced personnel, organizational changes, or something else? Can you identify specific changes that made a difference?

Probe B4.8 [M]: Looking at the period from the first year of operations to now, how has your organization's operational capability changed? Has it improved, stayed the same, or declined? If it improved, what drove the improvement? If it did not improve despite sustained production, why not?

Financial Resources

B5. What role, if any, did financial resources play in your organization's ability to operate the asset after the transaction?

Probe B5.1 [M]: Did your organization have the capital to maintain the asset's infrastructure, including wells, flowstations, pipelines, and export facilities?

Probe B5.2 [M]: Were there specific investments that were needed but could not be funded? What were they?

Probe B5.3 [T]: Were there cases where the money was available but the organization could not execute the work? What prevented execution?

Probe B5.4 [M]: Was the JV funding (NNPC cash calls) maintained throughout the period? Were there cash call disputes?

PIA Fiscal Effects

B6. Did the fiscal terms of operating the asset change after the Petroleum Industry Act of 2021, and if so, how did that affect operations?

Probe B6.1 [M]: Were operational decisions affected by the PIA's fiscal restructuring?

Probe B6.2 [BOTH]: For assets divested before 2021, were the challenges you observed related to the fiscal environment or to something else?

Probe B6.3 [M]: Did the fiscal terms make a difference in your assessment between operators who succeeded and those who did not?

Coordination from the Buyer's Perspective

B7. How were transition activities coordinated among the seller, the buyer, and the regulator?

Probe B7.1 [M]: Was there a formal transition plan? Who designed it? What did it cover?

Probe B7.2 [BOTH]: What transition support did you receive from the seller? What did you expect but did not receive?

Probe B7.3 [BOTH]: At what point did coordination break down, if it did: before the transaction was approved, during the handover period, or after your organization took full control?

Probe B7.4 [BOTH]: Did the regulator play any role in overseeing or verifying the transition? What specifically did the regulator do or not do?

Probe B7.5 [BOTH]: Looking back, what would have needed to be coordinated differently for the transition to produce a better outcome?

Secondary Questions

Security and Oil Theft

B8. Did the security situation or the level of oil theft change after the IOC departed, and if so, how did that affect your operations?

Probe B8.1 [BOTH]: Were there changes in community relations, pipeline vandalism, or militant activity after the operator changed?

Probe B8.2 [M]: Did the security arrangements that the IOC had in place transfer with the asset, or did your organization have to rebuild them?

Probe B8.3 [BOTH]: To what extent did security-related losses (theft, vandalism, shut-ins) account for production decline versus operational capability issues?

Post-Close Regulatory Engagement

B9. After the transaction closed, what role did the regulator play in overseeing your operations?

Probe B9.1 [M]: Were there regulatory inspections, compliance audits, or enforcement actions during the transition period?

Probe B9.2 [BOTH]: Did the regulator provide any guidance, support, or oversight during the first 12 months of your operations?

Probe B9.3 [T]: Was the regulatory environment different for your organization as a domestic operator compared to what the IOC had experienced?

JV/NNPC Relationship

B10. How did the joint venture governance with NNPC work after you took over as operator?

Probe B10.1 [M]: Were cash calls, budget approvals, and work programs managed smoothly, or were there disputes?

Probe B10.2 [BOTH]: Was NNPC's level of engagement or cooperation different from what the IOC had experienced?

Framework Elicitation

B11. If you could redesign the divestment process to prevent the operational failures that have occurred, what would you change?

Probe B11.1 [BOTH]: What should the seller be required to do before or during the transfer?

Probe B11.2 [BOTH]: What should the buyer be required to demonstrate before being approved to acquire the asset?

Probe B11.3 [BOTH]: What should the regulator be required to verify before, during, and after the transaction?

Probe B11.4 [BOTH]: Are there examples of transfers that worked well? What made them different?

Closing (5 minutes)

1. ‘Is there anything important about the divestment process or its consequences that my questions did not cover?’
2. ‘Is there anyone else you would recommend I speak with who has direct knowledge of these events?’
3. ‘Do you have any questions about the study?’
4. Thank the participant. Confirm their preferred contact method if follow-up questions arise.
5. Immediately after the call, complete expanded notes while the conversation is fresh. Note any statements the participant emphasized or that you judge to be analytically significant. Record the participant’s role type (technical/operational or management/strategic) for analytical coding.

C.3 Group C Interview Guide (Regulatory Officials and Industry Observers)

COUHES Protocol E-7658. Estimated Duration: 45-60 minutes.

Before You Begin (5 minutes): (1) Introduce yourself and the study. (2) Disclose Chevron affiliation with explicit statement that the thesis is not sponsored by or subject to approval by Chevron or any other oil company. (3) Confirm consent. (4) Confirm recording preference. (5) Remind participant of right to decline any question and stop the interview at any time.

C1. Please describe your role and responsibilities in relation to upstream petroleum operations and IOC asset divestments in Nigeria.

Probe C1.1: What specific regulatory, oversight, or advisory functions were you responsible for?

Probe C1.2: Over what time period were you involved in divestment-related activities?

Probe C1.3: Approximately how many divestment transactions were you involved in or had oversight of?

C2. When an IOC proposed to divest an upstream asset to a domestic company, what did the regulatory approval process involve?

Probe C2.1: What specific criteria were used to evaluate whether the acquiring company was qualified to take over the asset?

Probe C2.2: Was the acquiring company’s operational or technical capability assessed as part of the approval process? If so, how? If not, why not?

Probe C2.3: Was there a legal basis in the Petroleum Act, the PIA, or other legislation that would have allowed the regulator to require a demonstration of operational capability as a condition of approval?

Probe C2.4: Was there anything the regulator wanted to assess but lacked the legal authority or the institutional capacity to require? What was it, and what prevented it?

Probe C2.5: Was the buyer's financial capacity to maintain and operate the asset assessed as part of the approval process? In cases where the buyer subsequently experienced financial difficulties, was there any indication of this at the time of approval?

Probe C2.6: When assessing a buyer's qualification to operate the asset, was there a distinction between the buyer's formal qualifications, such as engineering credentials, corporate structure, and financial capacity, and the practical operational knowledge required to run the specific asset? Was the regulator in a position to assess whether the buyer could handle the operational realities of the specific asset, as opposed to upstream operations in general?

Probe C2.7: Was any proposed divestment transaction ever rejected because the acquiring company was judged to lack the operational capability to run the asset? If not, was that because all buyers were judged capable, or because operational capability was not a criterion on which a transaction could be rejected?

C3. Divestment approval involves ministerial consent under Section 29 of the Petroleum Act. How did the ministerial consent process interact with the regulatory assessment?

Probe C3.1: Were there cases where the technical or regulatory assessment raised concerns about a proposed buyer, but the transaction was approved at the ministerial level?

Probe C3.2: To what extent did considerations other than operational capability, such as indigenous ownership targets, political relationships, or commercial terms, influence divestment approvals?

Probe C3.3: In your experience, was there a tension between the policy objective of promoting indigenous ownership and the objective of ensuring operational continuity?

C4. After a divestment was approved and the new operator took over, what regulatory oversight was in place?

Probe C4.1: Were there scheduled inspections, compliance audits, or performance reviews of the new operator during the first 12 to 24 months?

Probe C4.2: When post-divestment operational problems emerged (production decline, safety incidents, environmental pollution), what enforcement tools were available to the regulator? Were they used?

Probe C4.3: Was there a formal mechanism for the regulator to intervene if the new operator's performance fell below acceptable standards?

Probe C4.4: Was the level of regulatory oversight different for newly divested assets compared to established operations?

C5. How has the Petroleum Industry Act of 2021 changed the regulatory framework for upstream asset divestments?

Probe C5.1: Did the PIA introduce new requirements for assessing buyer capability, or does the regulatory gap that existed under the Petroleum Act persist?

Probe C5.2: How has the creation of the NUPRC changed the institutional capacity for overseeing divestments compared to the former DPR?

Probe C5.3: In your assessment, did the PIA's fiscal restructuring contribute to the operational challenges observed among domestic operators, or were the challenges primarily

driven by other factors?

C6. How was the transition process coordinated among the seller, the buyer, and the regulator?

Probe C6.1: Did the regulator have a defined role in overseeing or facilitating the transition from the IOC to the new operator?

Probe C6.2: Were there cases where the regulator attempted to coordinate transition activities? What happened?

Probe C6.3: At what point in the process did coordination typically break down, if it did: before approval, during handover, or after close?

Probe C6.4: Was there a regulatory mechanism for ensuring that the operational capability to run the asset would survive the transaction?

C7. Across the divestment transactions you were involved in or observed, did you notice systematic differences in how acquiring companies performed after taking over?

Probe C7.1: Which acquisitions went well and which did not? In your assessment, what distinguished the successful ones from the unsuccessful ones?

Probe C7.2: Were there common patterns in the types of operational problems that emerged after divestments?

Probe C7.3: Were some types of buyers (those with prior operating experience, international partnerships, IOC staff retention) systematically more successful than others?

Probe C7.4: Does the pattern you observed suggest that the problem is primarily about individual operator quality, or about something structural in how divestments are designed and approved?

Probe C7.5: Did you observe parallels between the outcomes of the marginal field program (2003 onwards) and the subsequent IOC divestment outcomes? Were lessons from the marginal field experience considered when the IOC divestment framework was applied?

C8. The NUPRC announced a seven-pillar divestment assessment framework in 2024. Can you describe what prompted the development of this framework and what it is designed to address?

Probe C8.1: What evidence or experience led the NUPRC to conclude that a new framework was needed?

Probe C8.2: What specific gaps in the previous regulatory approach is the framework designed to close?

Probe C8.3: Does the framework include requirements for the seller to disclose operational capability, for the buyer to demonstrate readiness, or for the regulator to verify capability transfer?

Probe C8.4: How does the new framework compare to what existed before, and what do you expect its impact to be?

C9. If you could redesign the divestment process to prevent the operational failures that have occurred, what would you change?

Probe C9.1: What should the seller be required to do before or during the transfer?

Probe C9.2: What should the buyer be required to demonstrate before being approved to acquire the asset?

Probe C9.3: What should the regulator be required to verify before, during, and after the transaction?

Probe C9.4: Should capability transfer be a mandatory condition of divestment approval, or should it remain at the discretion of the transacting parties?

Closing (5 minutes): (1) Is there anything important about the divestment process or its consequences that my questions did not cover? (2) Is there anyone else you would recommend I speak with who has direct knowledge of these events? (3) Do you have any questions about the study? (4) Thank the participant. Confirm preferred contact method for follow-up. (5) Immediately after the call, complete expanded notes while the conversation is fresh.

Appendix D

Case-by-Case Evidence Table

The documentary evidence for each case appears below, organized operator by operator. It draws on desktop research across the formal cases and the marginal field program, and underlies the verdicts in the empirical analysis (Chapter 4) and the verdict traceability matrix (Appendix E).

D.1 Operator Evidence Inventory

Table D.1: Case-by-case evidence summary: each operator’s place in the discrimination logic and how it performed after acquisition. The detailed evidence for each operator follows below.

Case (OML)	Role in the thesis	Acquisition	Production performance	Operational integrity
Aiteo (OML 29)	Critical experiment: separates capability-related factors from the capital, fiscal, and regulatory explanations	Shell, 45%, \$1.7 billion (2015)	Tripled from 23,000 to 90,000 bpd by 2017	Santa Barbara blowout (2021): 33+ days, 200,000+ barrels, 41 communities; the well was undecommissioned since 2015; Boots and Coots was engaged only after the fact, and NOSDRA judged the incident beyond Aiteo’s capacity
Eroton (OML 18)	Literal replication; extends the pattern to systemic infrastructure cascade	Shell, 45% (2015)	Grew from 6,000 to 50,000 bpd in 24 months at the lowest technical cost per barrel, then collapsed to zero by the fourth quarter of 2021	The Aiteo-operated Nembe Creek Trunk Line entered force majeure and severed the export route; an NNPC forensic audit also found royalty non-payment and insider dealing
First Hydrocarbon, Afren (OML 26)	Regulator-verification component; the starkest buyer-assessment failure	Shell, 45% (2011), to an Afren special-purpose vehicle	Stalled; Afren collapsed before sustained development	The Afren chief executive and chief operating officer were convicted by the UK Serious Fraud Office, given concurrent custodial terms of up to six and five years, and the firm entered administration in 2015; capability was never tested, so capital was not the constraint for a then-\$2.6 billion company

Case (OML)	Role in the thesis	Acquisition	Production performance	Operational integrity
Seplat (OMLs 4, 38, 41)	Theoretical replication: success is predicted where the buyer holds absorptive capacity	Shell, 45% (2010); Exxon-Mobil shallow-water assets, \$1.28 billion (2024)	Peak 85,200 bpd; reserves nearly doubled by 2015; built two gas plants and the Amukpe-Escravos pipeline	Two million man-hours without a recordable incident; the outcome rests on pre-existing capability (the founding team, Maurel et Prom, and dual listing), not on seller-facilitated transfer
Shoreline, HEOSL (OML 30)	Within-OML natural experiment: the same asset and regime under different operators	Shell, Total, and Eni, 45%, \$850 million (2012); NPDC first operated	25,000 to 45,000 bopd under NPDC (2012 to 2017); grew to 80,000 bopd under HEOSL after 2017	Aggressive rehabilitation under HEOSL with no major incident reported; the cleanest available test of capability against regulation
NPDC (NEPL portfolio)	State subsidiary: even the government's own operator could not sustain IOC-era performance	55% in eight or more Shell-divested OMLs	180,000 bpd (2017); the 500,000 bpd target for 2020 was never met	Technical efficiency of 23%, the lowest of all operator categories (Adekunle et al. 2019); the Benin River explosion (2020) killed seven; the top spill contributor in 2016
Neconde (OML 42)	Confirms the NPDC underperformance-then-improvement pattern	Shell, 45%, \$585 million (2011)	NPDC declined from 2011 to 2015; Neconde grew output from 10,000 to 50,000 bpd after taking over in 2015	Improvement under a more capable operator following the NPDC period
Elcrest (OML 40)	Attribution boundary: the OML 40 explosion occurred under NPDC, not Elcrest	Shell, 45% (2012); Eland, later Seplat, 45%	Limited Elcrest-specific data	The Benin River fatal explosion occurred under NPDC operation; the case is primarily NPDC evidence rather than Elcrest evidence

Case (OML)	Role in the thesis	Acquisition	Production performance	Operational integrity
Renaissance (SPDC JV)	Future test: deliberate capability preservation applied from 2025, not yet confirmatory	100% of SPDC (18 OMLs), \$1.3 billion (2025)	Early figures of 200,000+ bpd, with all pipelines operational by April 2025	Former Shell staff were retained rather than replaced, a deliberate response to the documented failure pattern; the outcome is too recent to confirm
Shebah (FPSO)	Aged-infrastructure failure under an indigenous operator	Floating production, storage, and offloading operation	Trading-vessel service	A 2022 explosion on a nearly 50-year-old vessel carrying about 60,000 barrels left 10 crew dead or missing, illustrating infrastructure-age and maintenance failure

D.1.1 Aiteo, OML 29

Acquisition: Shell sold 45% for \$1.7 billion (2015). OML 29, onshore Niger Delta.

Production

Tripled from 23,000 bpd to 90,000 bpd by March 2017.

Crisis Failure

November 5, 2021: Santa Barbara Well 1 blowout. Duration 33+ days. Estimated 200,000+ barrels spilled. 41 communities affected. The well was non-producing since 2015, neither decommissioned by Shell before sale nor by Aiteo after acquisition. Prior spills from the same wellhead in 2018 and 2019, unremediated.

External Capability Engagement

Boots and Coots (Halliburton subsidiary) engaged for well control after the fact. The NOSDRA Director General stated the incident was ‘beyond the capacity of a single company like Aiteo.’ SDN documented that the operator ‘did not have the response strategy or expertise to plug the leak.’

Information Asymmetry

Aiteo v. Shell (FHC/ABJ/C8/738/2021): alleged Shell sold Kugbo West and Okiori wells already transferred to FGN, and did not disclose full asset condition.

Systemic Cascade

Aiteo’s inability to maintain the Nembe Creek Trunk Line (NCTL) directly caused the Eroton/OML 18 export failure and production shutdown.

Community Foresight

The Nembe community placed a ‘buyer beware’ decree on the Shell-to-Aiteo sale in 2015, before the transaction closed.

D.1.2 Eroton, OML 18

Acquisition: Shell sold 45% in 2015. OML 18, onshore Niger Delta.

Production Timeline

Inherited at 6,000 bpd. Grew to 50,000 bpd within 24 months. NNPC recognized Eroton as one of two operators with the lowest technical cost per barrel. Peak gross production: 45,000 (2018), 39,000 (2019), 21,100 delivered (2020), 4,400 (2021), zero from Q4 2021.

Collapse Mechanism (Dual Causation)

- 1) NCTL force majeure: the Aiteo-operated pipeline became unavailable, severing Eroton’s export route.
- 2) Internal governance failures: an NNPC forensic audit found non-payment of royalties, mismanagement, and insider dealing.

Funding

JV funding was maintained throughout the period.

Institutional Response

NNPC took over via NNPC Eighteen Operating Ltd (March 2023). EFCC investigation ongoing. Communities described the takeover as a ‘new lease of life.’

Investor Impact

San Leon Energy (listed, indirect interest) reported pipeline losses of 18-35% and an \$8.9 million loss in H1 2022.

D.1.3 First Hydrocarbon, Afren, OML 26

Acquisition: Shell sold 45% of OML 26 in 2011 to FHN, an SPV created by Afren Plc (FTSE 250, valued at \$2.6 billion at peak).

Corporate Fraud

CEO Osman Shahenshah and COO Shahid Ullah were convicted by the UK Serious Fraud Office for fraud and money laundering. Custodial terms: up to six and five years, served concurrently. Laundered \$45 million from a \$300 million deal. Both banned as directors for 14 years.

Financial Collapse

Afren entered administration July 2015. Nigerian banks lost N36.63 billion (\$185 million). Administrators attempted to redirect environmental cleanup funds to bondholders.

Institutional Rescue

NNPC formed an Asset Management Team (2016) to rescue OML 26. NPDC became operator.

Secondary-Source Observation

Cambridge Business and Human Rights Journal: ‘Several companies to which Shell has sold OML stakes have faced financial difficulties.’

D.1.4 Seplat, OMLs 4, 38, 41

Acquisition: Shell sold 45% of OMLs 4, 38, 41 in July 2010. Seplat acquired ExxonMobil’s Mobil Producing Nigeria (shallow water) for \$1.28 billion in 2024.

Performance

Peak production: 85,200 bpd. Nearly doubled reserves by end 2015. 2 million man-hours without a recordable incident. AfriSAFE Energy Company of the Year 2024. Built the Oben Gas Plant (465 MMscfd), Sapele Gas Plant (85 MMscfd), and Amukpe-Escravos pipeline (160,000 bpd capacity).

Absorptive Capacity

- 1) Founding team (Orjiako, Avuru) with decades of industry experience.
- 2) Maurel et Prom (French E&P) acquired 45% pre-IPO, providing international technical partnership.
- 3) Dual LSE/NSE listing imposed governance discipline.

Training Centre Provenance

The Transitional Training Centre (TTC) at Eket was Mobil Producing Nigeria’s program, established in 1995 and acquired through the 2024 ExxonMobil deal, not through the 2010 Shell transaction.

D.1.5 Shoreline, Heritage Oil, OML 30

Acquisition: Shell, Total, and Eni sold 45% for \$850 million (November 2012). NPDC held 55% and initially operated.

Performance Under NPDC (2012-2017)

Average production 25,000-45,000 bopd. NPDC operated on behalf of the JV.

Performance Under HEOSL (2017-present)

The government approved transfer of operatorship to Heritage Energy Operational Services Ltd (HEOSL) in 2017. HEOSL embarked on aggressive rehabilitation: equipment upgrade, gas compressor installation, pressure vessel inspection, wellhead and pipeline inspection, and well optimization. Gross production grew to 80,000 bopd, nearly doubling from NPDC-era levels.

D.1.6 NPDC, NEPL Portfolio

Portfolio: 55% equity in 8+ Shell-divested OMLs. 29 concessions including 22 OMLs and 7 OPLs.

Production Target Failure

2017: 180,000 bpd equity production. Target: 500,000 bpd by 2020. Reached 332,000 bpd average by late 2018. The 500,000 bpd target was never achieved. National production fell to record lows of approximately 1 million bpd by September 2022.

Technical Efficiency

Academic study (Adekunle et al. 2019): NNPC/NPDC technical efficiency rated at 23%, the lowest of all operator categories. JV/IOC operators: 75%. Marginal field operators: 67%.

Safety Incidents

BRVS explosion (July 7, 2020, OML 40): 7 fatalities during ladder installation; NPDC was operator; workers from contractors Weld Affairs and Flow Impact. October 2023: abandoned well explosion in OML 66 under NPDC operation. Top spillage contributor in 2016 (NOSDRA data).

Cross-OML Pattern

OML 30: production doubled after HEOSL took over from NPDC. OML 40: fatal explosion under NPDC operation. OML 42: Neconde alleged NPDC lacked capacity after production declined from 30,000 bpd.

D.1.7 Neconde, OML 42

Acquisition: Shell sold 45% for \$585 million (2011). Consortium: Nestoil, Aries Energy, VP Global, Kulczyk.

NPDC operated 2011-2015; production declined. Neconde took over operatorship May 2015 and grew production from 10,000 to 50,000 bpd.

D.1.8 Elcrest, OML 40

Acquisition: Shell sold 45% (2012). JV: Starcrest 55%, Eland Oil (UK) 45%. Seplat acquired Eland December 2019.

The BRVS fatal explosion occurred under NPDC operation, not Elcrest management; the case is primarily NPDC evidence.

D.1.9 Renaissance, SPDC JV

Future test, not confirmatory: capability preservation applied deliberately from 2025; the outcome is too recent to confirm and is treated as a future test and design contrast, not

as present evidence.

Acquisition: 100% of SPDC (18 OMLs) for \$1.3 billion (\$2.4 billion total). Completed March 2025.

Capability Preservation Mechanisms

- 1) Staff retention: SPDC staff continued employment under new ownership; leadership comprised former Shell staff; Renaissance explicitly chose not to replace IOC workers.
- 2) Shell provided \$1.2 billion in secured term loans plus \$1.3 billion for gas development and decommissioning.
- 3) Shell retained a supporting role in NLNG-related facility management.

Early Performance Indicators

Already at 200,000+ bpd. All pipelines operational by April 12, 2025. Contractors being paid on time.

Deliberate Learning from Failure

Renaissance leadership explicitly studied why previous buyers of IOC assets had failed. Africa Oil & Gas Report: the capability-preservation commitment ‘has never made it to any press release in 15 years of divestments.’

Institutional Context

First IOC divestment assessed under NUPRC’s new 7-pillar framework (2024). Ministerial consent required extensive regulatory review.

D.1.10 Shebah FPSO

2022 explosion at a nearly 50-year-old FPSO vessel operated by a DOC. ‘Most if not all of the big trading companies stopped using it several years ago.’ Exploded with approximately 60,000 barrels on board. 10 crew dead or missing. SDN: ‘DOCs could be content operating aged infrastructure to the point of environmental and human disaster.’

D.2 Marginal Field Program Evidence

The marginal field program is a parallel, independent case of the same institutional failure, predating the IOC divestment wave by a decade.

D.2.1 Program Overview

Conceived 2001. Objective: promote indigenous participation in the upstream sector, build technical capacity, and grow reserves and production. First round 2003: 24 fields awarded to 31 indigenous companies from the 116 identified marginal fields. Estimated total marginal field reserves: 2.3 billion barrels crude oil.

D.2.2 Performance Outcomes

By 2013 (10 years): less than 30% of awarded fields producing (Idigbe et al. 2013). By 2014 (11 years): approximately 65% of fields undeveloped (Sweet Crude Reports). By 2020 (17

years): only 9 of 30 awarded fields producing. 11 operators lost licenses for failure to meet government criteria. The program contributed only 2% to national oil and gas reserves. DPR acknowledged Nigeria ‘didn’t benefit from 2003 marginal field bid rounds.’

D.2.3 2020 Bid Round

57 fields awarded. 665 applications received, 540 pre-qualified, 161 shortlisted, 119 fully paid, 33 awardees failed to pay signature bonuses (approximately \$500 million total). The program was described as ‘at a crossroads’ with ‘legal ambiguities, financial bottlenecks, and infrastructure gaps.’

D.2.4 Academic Evidence on Marginal Field Capability

Oriaku et al. (2016): only 44% of 18 indigenous companies sampled scored the mandatory 60% minimum required for technical qualification in tubing inspection services. Adekunle et al. (2019): marginal field operators 67% technical efficiency vs. JV/IOC 75% vs. NNPC 23%. Danmadami et al. (2021): 50% developed and producing, 13% some progress, 37% undeveloped. Akinwale et al. (2022): collaboration between oil and gas firms and financial institutions, government agencies, and universities was low across all categories.

Appendix E

Verdict Traceability Matrix

One row per case-and-explanation. The matrix makes the falsification symmetry auditable: each explanation is rejected or supported on its own pre-specified evidence, not by residual. Supporting evidence, with full citations, is in Appendix D and the bibliography.

Verdict terms (Chapter 2, Section 2.6): most consistent with the evidence; consistent but not decisive; an enabling condition insufficient on its own; rejected as sufficient; rejected for specific cases. **Test status (not a verdict term):** where a case lacks the observable implication needed to test an explanation, the matrix records “not tested / not applicable” rather than forcing a verdict. **Confidence levels:** High = multiple independent sources converge and the inference is direct; Medium = sources converge but the inference requires interpretation; Low = single source or weak inference. **Evidence type:** audited/regulatory record (strongest), Tier B trade reporting, corporate profile, technical inference.

E.1 Aiteo (OML 29)

Table E.1: Verdict matrix for the Aiteo case (OML 29). Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Rejected as sufficient	Evidence that crisis-readiness funding was cut for financial reasons	~\$2.562B acquisition; audited growth 21.7 to 38.2 kb/d; capital available for production-directed spending; the 2022 collapse to ~4.9 kb/d followed the crisis and does not reopen capital	Audited	4.3.1	High
Fiscal regime effects	Rejected for this case	None; the timing is arithmetic	Transaction closed 2015; PIA enacted 16 Aug 2021; the capability gap (undecommissioned well, 2018-19 spills) predates the Act	Documentary	4.3.2	High
Regulatory inadequacy	An enabling condition insufficient on its own	Evidence that the framework screened for capability and that Aiteo passed	Section 29 specifies no capability criterion; NUPRC created a technical-capacity pillar in 2024, none before; same pre-2024 framework, different outcomes across operators	Documentary/regulatory	4.3.3	Medium
Political economy	Rejected as sufficient for operational quality	Evidence that political interference directly caused the operational failure, beyond enabling commercial survival	Connections could not have prevented or contained the blowout, which required well-control capability; may account for commercial survival only; board and shareholding data incomplete	Non-substitutability logic; corporate registry	4.3.4	Medium
Coordination failure	Rejected as independently caused	Evidence of a specific coordination breakdown independent of knowledge absence	Persistence at six years; domain-specific failure while routine production grew; unremediated 2018-19 incidents	Audited/documentary	4.3.5	Medium-High

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capability-related factors	Most consistent with the evidence for this case	An operator without inherited upstream knowledge handling a comparable crisis successfully	Selective failure signature; containment failure of about a month, NOSDRA attributing the delay to a capacity deficit (official JIV cause was third-party interference, so the finding rests on containment, not cause); after-the-fact reliance on external well-control specialists; six-year persistence; no documentary evidence of structured knowledge-transfer provisions; the founder's downstream-only background (secondary indicator)	Audited/regulatory; Tier B; corporate profile	4.3.6	Medium

E.2 Eroton (OML 18)

Table E.2: Verdict matrix for the Eroton case (OML 18). Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Rejected as sufficient	Evidence that the JV was underfunded during the decline and the funding statement was inaccurate	~\$1.1-1.2B acquisition; reported gross output ~39-45 kb/d before losses shows capital available for production-directed spending; NNPC stated JV partners met funding obligations through the decline; unpaid taxes indicate liquidity or governance stress, not JV capital scarcity; Eroton's rebuttal did not contest funding	Audited; regulatory statement	4.4.1	High
Fiscal regime effects	Rejected for this case	None; the conditions predate the Act	Acquired March 2015; PIA enacted 16 Aug 2021; the collapse predates the Act	Documentary	4.4.1	High
Regulatory inadequacy	An enabling condition insufficient on its own	Evidence that the framework screened buyer capability and Eroton passed	Same pre-2024 framework as Aiteo; screened no operational or organizational capability at transfer; cannot explain Eroton's specific collapse	Documentary/regulatory	4.4.1	Medium
Political economy	Rejected as sufficient for operational quality	Direct evidence that political access shaped the operational failure, beyond commercial or regulatory standing	Consortium background was business-development and financial (Midwestern, Mart, Sun-trust), not upstream technical; connections could not supply capability to execute the approved barge plan or meet fiscal obligations; board and shareholding detail incomplete	Corporate profile; non-substitutability logic	4.4.1	Low-Medium

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Coordination failure	Rejected as independently caused	Evidence of a specific transition-period coordination breakdown independent of capability	Collapse at 6 to 8 years, well beyond a short transition period; operator stabilized before failing; failure shape (delivery collapse) differs from Aiteo's domain-specific pattern, so persistence carries the inference	Audited/ documentary	4.4.1	Medium
Capability-related factors	Consistent but not decisive for this case	The internal organizational failures shown attributable to capital, liquidity, or political dynamics instead of capability	Internal pathway (FIRS office sealing 12+ months, NNPC forensic governance findings, unexecuted approved barge evacuation) consistent with an organizational-capability gap; the external NCTL force majeure, structurally theft-driven and inherited from the seller, is the common environment every line user faced and confounds the attribution rather than supporting it; consortium business-development background a secondary indicator	Audited/ documentary; corporate profile; Tier B	4.4.1	Medium-Low

E.3 First Hydrocarbon (OML 26)

Table E.3: Verdict matrix for the First Hydrocarbon case (OML 26). Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Rejected as sufficient	Evidence that a funding shortage, ahead of the fraud and collapse, drove the failure	Afren a ~\$2.6B FTSE 250 company; FHN funded by \$280M facilities; the failure was governance fraud, not funding scarcity	Corporate/financial record; UK criminal record	4.4.2	High
Fiscal regime effects	Rejected for this case	None; the timing is arithmetic	Agreement Oct 2010, completion 2011; Afren's collapse in 2015 and the fraud predate the PIA	Documentary	4.4.2	High
Regulatory inadequacy	An enabling condition insufficient on its own, most directly evidenced here	Evidence that the framework screened buyer fitness and that Afren passed a substantive test	Section 29 required ministerial consent but no fitness, capability, or governance test; the framework approved a buyer without an explicit governance-fitness test; subsequent fraud convictions and administration show why buyer integrity mattered	Documentary/regulatory; UK SFO and court record	4.4.2	Medium-High
Political economy	Not tested for this case	Not applicable; explanation not tested	The buyer was non-operating and the sponsor's fraud dominates the trajectory; the relevant state dynamic (NPDC's first-refusal operatorship) concerns the operator and is examined with NPDC	State-operator record; non-test	4.4.2	Not tested
Coordination failure	Not tested for this case	Not applicable; explanation not tested	Non-operating structure; joint asset-management team formed only in 2016; organizational features, not a tested transition breakdown	Documentary	4.4.2	Not tested

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capability-related factors	Not tested for this case	Not applicable; explanation not tested	The buyer did not operate the asset (NPDC operated) and failed through governance fraud rather than an operational event, so inherited operational capability cannot be assessed	Non-test (structural)	4.4.2	Not tested

E.4 Seplat (OMLs 4, 38, 41)

Table E.4: Verdict matrix for the Seplat case (OMLs 4/38/41). Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Rejected as sufficient	Evidence that Seplat's outcome rested on a capital advantage the failed operators lacked	Capital was present across both success and failure; the failed operators executed larger transactions still (Aiteo ~\$2.562B, Eroton ~\$1.1-1.2B, against Seplat's ~\$340M), so Seplat's outcome cannot rest on a funding condition the failures lacked	Corporate/financial record	4.5.1	High
Fiscal regime effects	Rejected for this case	Evidence of a fiscal treatment specific to Seplat that drove the outcome	Acquired 2010 and grew under the same pre-PIA regime as the failures	Documentary	4.5.1	High
Regulatory inadequacy	An enabling condition insufficient on its own	Evidence that Seplat faced a materially different regulatory regime or screen	Same pre-2024 framework as Aiteo and Eroton; a regime common to growth and collapse cannot explain divergence	Documentary/regulatory	4.5.1	Medium
Political economy	Rejected as sufficient for operational quality	Direct evidence that political connections, ahead of capability, produced the operational outcome	Sustained production and 11M+ man-hours without a lost-time injury rest on technical capability, which political access cannot supply	Non-substitutability logic; corporate profile	4.5.1	Medium
Coordination failure	Not tested for this case	Not applicable; explanation not tested	No operational failure occurred to which a transition coordination breakdown could be attributed	Non-test (no failure)	4.5.1	Not tested

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capability-related factors	Most consistent with the evidence for this case	An operator with comparable inherited capability failing under the same conditions	Founding team (Avuru, three decades of petroleum engineering) plus the Maurel et Prom partnership supplied absorptive capacity; metered 42.3 kb/d; integrity record; pattern recurs on a non-Shell asset (MPNU 2024)	Audited; corporate profile; safety/award record	4.5.1	Medium-High

E.5 OML 30: Within-Asset Operator Change

Table E.5: Verdict matrix for the OML 30 within-asset operator change. Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Rejected as sufficient	Evidence that new capital expenditure, ahead of the operator change, produced the recovery	Capital was present across the transition; the recovery came without drilling new wells on an unchanged physical asset, so new-drilling capital did not produce the increase; the remaining contrast turns on how the existing assets and rehabilitation spending were managed	Audited/ documentary	4.5.2	High
Fiscal regime effects	Rejected for this case	Evidence of a fiscal change coinciding with the operator transition	Acquired 2012; growth under the same pre-PIA regime	Documentary	4.5.2	High
Regulatory inadequacy	An enabling condition insufficient on its own	Evidence that a regulatory change coincided with the operator transition	The regulatory regime was identical across the operator transition; a regime that did not change cannot account for an outcome that did, and the same pre-2024 framework governed the failure cases	Documentary/ regulatory; within-asset comparison	4.5.2	Medium-High
Political economy	Rejected as sufficient for operational quality	Evidence that the recovery reflected political access ahead of operational capability	Rehabilitation of vessels, well-heads, and compression and the production growth are technical achievements political access cannot supply	Non-substitutability logic	4.5.2	Medium
Coordination failure	Not tested for this case	Not applicable; explanation not tested	No operational failure occurred to which a coordination breakdown could be attributed	Non-test (no failure)	4.5.2	Not tested

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capability-related factors	Most consistent with the evidence for this case	An operator change of comparable capability leaving the outcome unchanged, or a recovery traceable to new capital or a regime change	The principal change was operatorship, from a state company the regulator judged unsatisfactory to an indigenous team carrying direct Shell (Brown, 27 yrs SPDC) and NNPC (Ukpaka, 39 yrs upstream) experience; audited production recovered from a trough near 3,800 bpd to ~24,000-30,000 (2021-2022) and an operator-reported peak near 67,000 by 2025, without new drilling, isolating embedded capability from nationality though not from international technical support	Audited; operator-reported; corporate profile	4.5.2	High

E.6 NPDC

Table E.6: Verdict matrix for the NPDC case. Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Rejected as sufficient	Direct evidence of a binding funding shortage driving the underperformance	No case-specific evidence shows a binding funding shortage driving the underperformance; the sustained, structural deficit across licenses and years points to organizational and incentive problems rather than acute capital scarcity	State-entity structure; multi-year audited pattern	4.5.3	Medium
Fiscal regime effects	Rejected for this case	Evidence of a fiscal effect specific to NPDC	The underperformance spans and predates the PIA and is common to the regime all operators faced	Documentary	4.5.3	High
Regulatory inadequacy	An enabling condition insufficient on its own	Evidence that a regulatory difference, ahead of the operator, produced the outcome	The same framework that screened no capability governed NPDC; as a state entity it faced, if anything, weaker external discipline; the regime is common to all operators	Documentary/regulatory	4.5.3	Medium
Political economy	Most consistent with the evidence, on the commercial-survival dimension	Evidence that NPDC's survival rested on operational performance ahead of state backing	Technical efficiency far below the capable operators (23%) and underperformance against its own 500k-by-2020 target, yet continued operation across multiple licenses on state backing; the positive form of the test deferred at 4.3.4	Population-level; documentary; state-entity record	4.5.3	Medium-High

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Coordination failure	Rejected as independently caused	Evidence of a specific coordination breakdown independent of a standing capability gap	Sustained, structural underperformance across licenses and years, not a transition-period breakdown	Documentary	4.5.3	Medium
Capability-related factors	Most consistent with the evidence, on the operational-quality dimension	A state operator without embedded upstream capability matching the operational quality of the capable acquirers	A state operator with institutional history but without an embedded capability-development pathway shows the predicted deficit: 23% technical efficiency; OML 30 run to a trough and a formal unsatisfactory finding before the HEOSL recovery; OML 26 audited output falling to ~3,700 bpd metered and ~3,200 fiscalized in 2022 while the HEOSL-operated OML 30 on the same route rose, so the shortfall is operational, not stranding	Audited; population-level; regulatory finding	4.5.3	Medium

E.7 Marginal Field Program

Table E.7: Verdict matrix for the Marginal Field Program. Each explanation is assessed on its own pre-specified evidence, with the condition that would overturn its verdict.

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capital constraints	Consistent but not decisive for this cohort; rejected as sufficient for the well-funded divestment failures	Evidence that capital scarcity, not capability, drove the cohort underperformance even among adequately funded awardees	Small, underfunded awardees; the regulator attributes stalled 2020-round development partly to limited financial capacity; for small operators capital and capability are confounded	Regulatory statement; cohort structure	4.6	Medium
Fiscal regime effects	Rejected for this cohort	None; the awards predate the Act	All marginal-field outcomes (2003-2020) predate the PIA	Documentary	4.6	High
Regulatory inadequacy	An enabling condition insufficient on its own, most directly evidenced	Evidence that the allocation framework included a capability test the failed awardees passed	The allocation framework included no capability test; it permitted the failures without determining which operators would fail	Regulatory framework; contemporaneous attribution	4.6	Medium-High
Political economy	Not separately tested at the cohort level	Not applicable; not separately tested at the cohort level	Some 2001-2003 allocation was politically influenced, but cohort-level operator trajectories are not resolved by political access	Secondary	4.6	Not separately tested
Coordination failure	Rejected as independently caused	Evidence of a transition-period breakdown independent of a standing capability gap	Underdevelopment persisted over more than a decade, not a transition-period breakdown that eases as an operator settles	Documentary	4.6	Medium

Explanation	Verdict	What would change this verdict	Evidence	Evidence type	Verdict established	Confidence
Capability-related factors	Consistent but not decisive; structural corroboration	Evidence that capability-unassessed operators matched the cohort outcomes of assessed operators	Operators without assessed capability underperformed at the cohort level under a mechanism distinct from divestment; the Avuru/Okoloko divergence shows domain-specific capability scales; due-diligence failure traced to inadequate capability assessment	Cohort outcomes; biographical; academic attribution	4.6	Medium

E.8 Cross-References Not Scored as Separate Verdicts

Three transactions enter Section 4.7 without a per-case verdict row. Oando and ConocoPhillips (2014) and Oando and Eni-NAOC (2024) serve as seller-independence cross-references, where the largely non-operating interests test governance and seller identity in place of operational capability. Seplat and ExxonMobil MPNU provide a further seller-independence cross-reference, a capable buyer integrating from a different seller. Renaissance stands as a design contrast, with capability preservation structured into the transaction. Section 4.7 records their bearing on the cross-case reading; their interests or timing place them outside the per-case verdict scoring used above.

Appendix F

Survey Response Data

The structured survey instrument described in Section 3.5 was deployed as an alternative to interviews where scheduling prevented them, addressed to the three participant groups (Group A, IOC sellers; Group B, domestic operators; Group C, regulators). Of eight response records, three were completed. All three came from Group A, IOC sellers in technical or operational roles; the remaining five records comprised one test entry and four submissions abandoned at or below 14 percent progress. The three completed responses are reproduced here in full so that the summaries in Sections 4.7.1 and 4.8 can be read against the underlying evidence. Respondents are de-identified and labeled S1 to S3. Two of the three identified the transaction they were describing as the Shell or SPDC divestment to Renaissance; the third did not disclose a transaction.

Table F.1: Profile of the three completed respondents. All three are seller-side participants in technical or operational roles; two were describing the Renaissance transaction and one did not disclose a transaction.

Respondent	Group	Role	Transaction described	Basis of knowledge	Confidence in ranking
S1	IOC seller (Group A)	Technical / operational	Not disclosed	Professional knowledge	Fairly confident
S2	IOC seller (Group A)	Technical / operational	Shell to Renaissance	Directly observed	Fairly confident
S3	IOC seller (Group A)	Technical / operational	SPDC to Renaissance	Directly observed	Somewhat confident

Table F.2: Driver rankings by the three completed respondents, where 1 is the most important factor and the respondent’s top-ranked choice is shown in bold. No respondent ranked untransferred capability first; each named a different factor first, and capability placed second, third, and fifth across the three.

Factor	S1	S2	S3
Untransferred operational capability	5	2	3
Poor coordination during handover	4	3	5
Insufficient capital or financial resources	6	1	7
Unfavorable fiscal terms under the PIA	1	6	4
Weak regulatory oversight	7	5	2
Political interference, security, or oil theft	2	4	1
Asset already aging or in natural decline	3	7	6

Table F.3: Counterfactual, policy, and overall-adequacy responses. All three judged that a stronger capability transfer would have improved the outcome, and all three held that capability transfer should be a condition of divestment approval; a capital-only counterfactual drew weaker support.

Respondent	Stronger capability transfer would have improved the outcome?	More capital alone would have improved the outcome?	Capability transfer a condition of divestment approval?	Adequacy of the capability transfer observed
S1	Yes, somewhat	Yes, somewhat	Strongly recommended	Mostly adequate
S2	Yes, somewhat	No	Mandatory	Very adequate
S3	Yes, significantly	Yes, somewhat	Mandatory	Mostly adequate

The instrument's open-text fields were largely left blank. The one substantive free-text entry came from S2, who, having ranked capital first, explained the choice by reference to the borrowing power of the new owners relative to the financial reserves of the divesting major. The supporting-evidence, disconfirmation, and additional-requirement fields were not completed by any of the three respondents.

Appendix G

Capability Element Traceability Matrix

G.1 How the Matrix Is Built

The matrix below is the auditable bridge behind Table 5.3 and the transfer-design obligations in Section 5.4. For each element of operating capability it shows four things: the inter-organizational knowledge-transfer (IKT) factor that governs whether the element survives a handover, what the Chapter 4 cases reveal about that element, which existing Nigerian legal instrument already touches it, and which dimension of national value it protects. Read top to bottom, it traces the chapter’s central chain: a transfer-process failure degrades a specific capability element, the degraded element surfaces as a measurable outcome, and an existing legal hook can be re-specified to require that the element be disclosed, absorbed, and verified.

Two axes. The rows are the 20 elements of Risk-Based Process Safety (RBPS), grouped under its four pillars, supplying the integrity-management backbone; the governing-IKT-element column carries the transfer-process determinants, paired with each RBPS element as coded by Ferreira et al. Production performance, which Chapter 4 also treats as value at risk, is carried by the Production and subsurface operating-capability row in Table 5.3 and the V1 and V7 tags below, so the matrix protects production capability and national production resilience, not integrity alone.

Sources. RBPS pillars and elements: CCPS (2007). IKT elements, their criticality, and the IKT-to-RBPS associations: Ferreira et al. (2024, IEEE Transactions on Engineering Management) and the process-safety content analysis in Ferreira et al. (2026, Journal of Loss Prevention in the Process Industries), Tables 3 through 5. Chapter 4 findings: this thesis. Legal hooks: PIA 2021; NUPRC Assignment of Interests Regulations 2024 (Reg. 7); the NUPRC Upstream Asset Divestment and Exit Guidance Framework (the seven cardinal pillars); NOGICD 2010. This is an original synthesis; no source exhibit is reproduced.

Coverage and honesty note. The 2026 content analysis directly codes IKT-to-RBPS associations for 10 of the 20 RBPS elements. For those 10, the governing IKT element is the authors’ coded result. For the remaining 10, no direct coding exists; the governing IKT element is inferred from the nearest coded element and the association is carried by this thesis’s own cases and the legal hooks. Every such row is marked **[extended]** so the reader can separate the published bridge from the thesis’s extension of it.

G.2 Value-Dimension Legend

Internalized by the operator (demonstrated in Chapter 4): V1 Production performance · V2 Operational integrity and reliability.

Externalized onto others (warranted): V3 Safety and environmental harm to third parties · V4 Host-community welfare · V5 National capability development (local content) · V6 Fiscal and public-revenue protection · V7 Energy security and national production resilience · V8 Decommissioning and abandonment responsibility.

(Calibration carried from Table 5.1: **V1 and V2 are demonstrated** by Chapter 4, its two outcome dimensions; **V3 through V8 are warranted**; **V7 carries the external theft, sabotage, and price confound**; **V8 carries a financial-security mandate**. The dimensions are ordered by incidence, internalized then externalized: most of the value at risk is borne by parties outside the transaction. The matrix maps which capability element protects which dimension; it does not upgrade a warranted dimension to a demonstrated one.)

G.3 Criticality Tiers of the Transfer-Process Elements

Table G.1: Criticality tiers for the transfer-process elements and the design consequence of each.

Tier	IKT elements	Design consequence
Most critical (buyer cannot remedy alone)	Contractual characteristics; Decision-making characteristics; Transfer period; Availability; Proactivity; Provision of resources; Responsibility for the transfer; Access to databases	Mandatory seller and transfer-structure obligations
Manageable	Degree of similarity; Identification of training needs; Transfer objectives; Transfer scope; Evaluation mechanism and method; Definition of routines; Mechanisms used; Alignment	Proportionate buyer obligations
Facilitators	Previous experiences; Lessons learned; Due diligence; Conflict resolution	Encouraged; evidence of buyer absorptive capacity

Tier	IKT elements	Design consequence
Highest-leverage instrument	The assignment and transfer agreement (contractual characteristics map to six coded RBPS elements across three pillars)	Section 5.4 specifies the agreement's required content

Two takeaways from the criticality tiers carry into the design. **Process Knowledge Management** is the hub: seven IKT elements feed it, so most transfer-process failures converge there. **Contractual characteristics** is the broadest lever: it touches six coded RBPS elements across three pillars. Section 5.4 develops both.

G.4 Pillar 1: Commit to Process Safety

Table G.2: RBPS Pillar 1 (Commit to Process Safety): each element and the transfer factor that governs whether it survives a handover.

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Process Safety Culture	Shared commitment to safety as a priority	Contractual characteristics; Decision-making characteristics [extended]	Differentiated safety outcomes across acquirers under one regulatory regime are consistent with culture differences carried (or not) by retained teams	NUPRC Reg. 7(5) corporate competency and good standing	V2, V3
Compliance with Standards	Meeting internal and external requirements	Contractual characteristics (<i>coded</i>)	Differentiated regulatory and integrity performance, Seplat sustaining versus Aiteo's post-close incident record	PIA fit-and-proper; NUPRC Reg. 7(c) legal requirements; pillar 3	V2, V3
Process Safety Competency	Expertise to manage process hazards	Identification of training needs; Responsibility for the transfer [extended]	Aiteo's founder and CEO carried downstream-only experience; Seplat's leadership (Avuru) and HEOSL's technical operator (Salvic) carried upstream operating pedigree	NUPRC Reg. 7(5) technical knowledge and experience, and the director-conviction screen; NSTA director and controller fitness as the comparative analogue	V2, V3, V5

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Workforce Involvement	Engaging the operating workforce in safety	Contractual characteristics; Availability; Proactivity; Transfer objectives; Transfer scope (<i>coded</i>)	Seplat (Avuru with Maurel et Prom) and HEOSL with Salvic retained or brought equivalent people; Aiteo did not; the process-safety study found incidents rise when the operating team is not retained	NOGICD Sections 29-31 employment, training, and understudy succession; NUPRC Reg. 7(f) industrial relations and labour	V2, V3, V5
Stakeholder Outreach	Engaging external communities and parties	Decision-making characteristics [extended]	The externality lands on communities: forty-one communities affected by the Santa Barbara blowout; seven deaths in the Benin River explosion	PIA Chapter 3 Host Community Development Trusts; pillar 5	V4

G.5 Pillar 2: Understand Hazards and Risks

Table G.3: RBPS Pillar 2 (Understand Hazards and Risks): each element and the transfer factor that governs whether it survives a handover.

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Process Knowledge Management (hub)	Documented process and asset knowledge, retained and usable	Transfer period; Availability; Proactivity; Identification of training needs; Responsibility for the transfer; Evaluation mechanism and method; Access to databases (<i>coded; seven elements</i>)	The buyer could not assess embedded operating requirements from the data room (the Section 4.3.3 information asymmetry); absorbed operators retained data and the people who held it, de novo operators did not	NUPRC Reg. 7(g) data repatriation; pillar 7; NOGICD Sections 43-47 technology transfer	V1, V2, V7
Hazard Identification and Risk Analysis	Systematic identification and analysis of hazards	Contractual characteristics; Decision-making characteristics; Transfer objectives; Transfer scope; Definition of routines; Mechanisms used; Access to databases (<i>coded</i>)	Aiteo's absence of a blowout contingency plan implies the well-control scenario was not carried into a live hazard analysis at handover	NUPRC Reg. 7(a) technical capacity; pillar 1	V2, V3

G.6 Pillar 3: Manage Risk

Table G.4: RBPS Pillar 3 (Manage Risk): each element and the transfer factor that governs whether it survives a handover.

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Operating Procedures	Current, usable procedures for safe operation	Contractual characteristics; Transfer period; Degree of similarity (<i>coded</i>)	Absorbed operators ran inherited procedures with retained staff; the de novo acquirer lacked the operating knowledge the procedures assume	NUPRC Reg. 7(a); NOGICD Sections 43-47	V1, V2, V7
Asset Integrity and Reliability	Mechanical integrity of equipment and wells	Contractual characteristics; Decision-making characteristics; Degree of similarity (<i>coded</i>)	Aiteo inherited a suspended, undecommissioned well at Santa Barbara against documented pre-transfer underinvestment and integrity failures	NUPRC Reg. 7(d) decommissioning and abandonment; pillar 4; PIA Sections 232-233	V1, V2, V3, V8
Conduct of Operations	Disciplined execution of operating tasks	Decision-making characteristics; Availability; Proactivity; Provision of resources; Access to databases (<i>coded</i>)	Absorbed operators sustained operational control; the de novo acquirer's control gap surfaced in the post-close incident	NUPRC Reg. 7(a); Reg. 7(5) competency	V1, V2, V6, V7
Emergency Management	Plans, drills, and resources to respond to a crisis	Contractual characteristics (<i>coded</i>)	Aiteo well-control readiness evidence: no contingency plan, no standing well-control contract, Boots and Coots engaged only after the fact; NOSDRA found the incident exceeded Aiteo's internal capacity	PIA Section 103 environmental remediation; NOSDRA oil-spill contingency duties; NUPRC Reg. 7(5) competency	V2, V3, V4

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Training and Performance Assurance	Verifying personnel can perform their roles	Transfer objectives; Transfer scope; Evaluation mechanism and method; Definition of routines; Mechanisms used (<i>coded</i>)	The understudy logic: absorbed operators carried verified competency; the de novo acquirer’s gap was not closed before operator-ship	NOGICD Section 31 understudy succession (analogy); Sections 29-30 training plan	V2, V3, V5
Management of Change	Controlling changes to process, plant, and people	Availability; Proactivity (<i>coded</i>)	[extended] Direct documentary evidence on internal change control is limited (a Section 4.8 documentary boundary); inferred from the ownership-and-team turnover at handover	NUPRC Reg. 7(a)	V2, V3
Contractor Management	Selecting and overseeing contractors	Provision of resources; Responsibility for the transfer [extended]	Aiteo’s after-the-fact engagement of Boots and Coots is a contractor-management failure for crisis response; Shoreline’s use of Salvic as technical operator is contractor architecture done well	NUPRC Reg. 7(5) “own resources or those available within its group”; Reg. 7(a)	V2, V3
Safe Work Practices	Permit-to-work and hazardous-task controls	Definition of routines; Mechanisms used [extended]	[extended] Inferred from the operating-discipline differences between absorbed and de novo operators; not separately coded in the source	NUPRC Reg. 7(a)	V2, V3

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Operational Readiness	Confirming a facility is ready to operate or restart	Transfer period; Provision of resources [extended]	[extended] Supports the staged-operatorship lever: readiness must be demonstrated before operatorship transfers; the suspended Santa Barbara well is the unreadiness signal	NUPRC Reg. 7(4) site visit for competency evaluation	V1, V2, V7, V8

G.7 Pillar 4: Learn from Experience

Table G.5: RBPS Pillar 4 (Learn from Experience): each element and the transfer factor that governs whether it survives a handover. No Pillar 4 element is directly coded in the source content analysis; all are extended from this thesis's cases and the proposed post-transfer verification.

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Incident Investigation	Learning from incidents and near-misses	Lessons learned; Previous experiences [extended]	[extended] A de novo operator without retained institutional memory cannot investigate against prior incident history it never received	NOSDRA reporting duties; PIA Section 103	V2, V3

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Measurement and Metrics	Leading and lagging safety indicators	Evaluation mechanism and method [extended]	[extended] The framework's post-transfer verification instrument. Lagging verification: IOGP Report 456 Tier 1 and Tier 2 process-safety event rates, which show harm only after risk has materialized. Leading verification (gives the framework teeth before failure): readiness-drill completion, well-control-contract status, safety-critical maintenance backlog, overdue integrity actions, suspended-well monitoring, management-of-change compliance, near-miss reporting quality, and audit-closeout rate	Proposed post-transfer verification; IOGP Report 456 for the lagging metric	V2, V3
Auditing	Periodic checks that systems function as intended	Due diligence; Evaluation mechanism and method [extended]	[extended] The regulator-verification overlay rests here: Oriaku's NPDC finding that table-top assessment passed firms that site inspection failed is the warning that audit must be demonstrated, not documentary	NUPRC Reg. 7(4) site visit; proposed periodic post-transfer audit	V2, V3

RBPS element	What it is	Governing IKT element(s)	Chapter 4 finding	Legal hook	Value
Management Review and Continuous Improvement	Leadership review driving improvement	Decision-making characteristics [extended]	[extended] Inferred from the decision-rights and accountability differences between absorbed and de novo operators	Proposed; condition-setting at consent under PIA Section 95 and NUPRC Reg. 7	V2, V6

G.8 Reading the Bridge Backward

Two structural facts in the tables above drive Section 5.4. First, **Process Knowledge Management is the convergence point**: seven IKT elements feed it, so a single failure mode, losing the people and data that hold process knowledge, propagates into the hub of the capability system. This is the mechanism behind the Chapter 4 contrast, absorbed operators (Seplat, HEOSL with Salvic) retained the carriers; the de novo acquirer did not. Second, **contractual characteristics is the broadest lever**: it touches six coded RBPS elements across three pillars (a seventh extended), including Emergency Management, the element whose failure produced the Aiteo blowout. Because the contract is both the highest-leverage instrument and a seller-and-transfer-structure obligation the buyer cannot remedy alone, the framework's central prescription is to specify what the assignment and transfer agreement must contain, rather than to rely on buyer effort after close.

A third point governs verification. The post-transfer indicators in the Measurement and Metrics row split into two kinds. Tier 1 and Tier 2 process-safety event rates are **lagging**: they confirm capability only after an incident. The leading indicators in that row let the regulator verify readiness before harm occurs. The framework relies on both, so that verification has effect before failure and not only after it.

References

- Acha, S., J. Fadairo, D. Ogiesoba, I. Okeke, T. Emmanuel, and J. A. Brown (2019). “Achieving and Sustaining Impeccable HSE Performance in Brown Field Operations Comes at a Cost: OML 26 Blazes the Trail.” In: *SPE Nigeria Annual International Conference and Exhibition*. Lagos, Nigeria. DOI: [10.2118/198801-MS](https://doi.org/10.2118/198801-MS).
- Adekunle, I. J., I. O. Omowumi, E. C. Joseph, and A. O. Peter (Jan. 2019). “Upstream technical efficiency and its determinants: Evidence from non-parametric and parametric analysis of Nigeria exploration and production (E & P).” In: *Cogent Engineering* 6.1. Ed. by J. Guo, p. 1575638. ISSN: 2331-1916. DOI: [10.1080/23311916.2019.1575638](https://doi.org/10.1080/23311916.2019.1575638). URL: <https://www.tandfonline.com/doi/full/10.1080/23311916.2019.1575638> (visited on 06/18/2026).
- Africa Oil and Gas Report (2012). “NPDC Increases Production in OML 26.” In: *Africa Oil and Gas Report*. URL: <https://africaoilgasreport.com/2012/08/in-the-news/npdc-increases-production-in-oml-26/>.
- Africa Oil and Gas Report (2013). “Nigeria: The Marginal Field Experiment at 10.” In: *Africa Oil and Gas Report*. URL: <https://africaoilgasreport.com/2013/02/in-the-news/nigeria-the-marginal-field-experiment-at-10/>.
- Africa Oil and Gas Report (2014). “NPDC/Shoreline Make it to 50,000 bopd.” In: *Africa Oil and Gas Report*. URL: <https://africaoilgasreport.com/2014/01/in-the-news/npdcshoreline-make-it-to-50000bopd/>.
- Africa Oil and Gas Report (2024). “Nigerian Government Very Close to Granting Ministerial Consent for Shell-Renaissance Deal.” In: *Africa Oil and Gas Report*. URL: <https://africaoilgasreport.com/2024/12/farm-in-farm-out/nigerian-government-very-close-to-granting-ministerial-consent-for-shell-renaissance-deal/>.
- Africa Oil and Gas Report (2025). “Regulations Surrounding Commingling Production Need Be Addressed.” In: *Africa Oil and Gas Report*. URL: <https://africaoilgasreport.com/2025/08/in-the-news/regulations-surrounding-commingling-production-need-be-addressed/>.
- Akerlof, G. A. (1970). “The Market for “Lemons”: Quality Uncertainty and the Market Mechanism.” In: *The Quarterly Journal of Economics* 84.3, pp. 488–500. DOI: [10.2307/1879431](https://doi.org/10.2307/1879431).
- Akinwale, Y. O., J. B. Akarakiri, and J.-F. K. Akinbami (2022). “Nigerian Marginal Oil and Gas Field Development: A Collaborative Perspective.” In: *African Journal of Science Policy and Innovation Management* 3.1, pp. 106–117. DOI: [10.69798/37063086](https://doi.org/10.69798/37063086).
- Akpomera, E. (2015). “International Crude Oil Theft: Elite Predatory Tendencies in Nigeria.” In: *Review of African Political Economy* 42.143, pp. 156–165. DOI: [10.1080/03056244.2014.988696](https://doi.org/10.1080/03056244.2014.988696).

- Anumadu, S. U., A. Dosunmu, C. Anyanwu, E. Ekeinde, and B. Odagme (Aug. 5, 2014). “Evaluation of Safety Performance and Compliance of Workers in Selected Oil and Companies in Nigeria.” In: *SPE Nigeria Annual International Conference and Exhibition*. SPE Nigeria Annual International Conference and Exhibition. Lagos, Nigeria: SPE, SPE-172347-MS. DOI: [10.2118/172347-MS](https://doi.org/10.2118/172347-MS). URL: <https://onepetro.org/SPENAIC/proceedings/14NAIC/14NAIC/SPE-172347-MS/212537> (visited on 06/18/2026).
- Baxter, G. and I. Sommerville (2011). “Socio-technical Systems: From Design Methods to Systems Engineering.” In: *Interacting with Computers* 23.1, pp. 4–17. DOI: [10.1016/j.intcom.2010.07.003](https://doi.org/10.1016/j.intcom.2010.07.003).
- Beach, D. and R. B. Pedersen (2013). *Process-Tracing Methods: Foundations and Guidelines*. Ann Arbor: University of Michigan Press.
- BusinessDay (2025). “Why Renaissance hit the ground running after acquisition of SPDC.” In: *BusinessDay*. URL: <https://businessday.ng/life/article/why-renaissance-hit-the-ground-running-after-acquisition-of-spdc/>.
- Center for Chemical Process Safety (2007). *Guidelines for Risk Based Process Safety*. Hoboken, NJ: John Wiley & Sons.
- Coase, R. H. (1960). “The Problem of Social Cost.” In: *The Journal of Law and Economics* 3, pp. 1–44. DOI: [10.1086/466560](https://doi.org/10.1086/466560).
- Cohen, W. M. and D. A. Levinthal (1990). “Absorptive Capacity: A New Perspective on Learning and Innovation.” In: *Administrative Science Quarterly* 35.1, pp. 128–152. DOI: [10.2307/2393553](https://doi.org/10.2307/2393553).
- Court of Appeal (Criminal Division) (2019). *R v Shahenshah and Ullah: appeal against conviction dismissed*. Afren convictions upheld; confiscation order of £5.45 million followed. URL: <https://redlionchambers.co.uk/jamie-sharma-led-by-john-mcguinness-qc-succeeds-in-the-court-of-appeal-as-it-upholds-the-fraud-and-money-laundering-convictions-of-the-ex-ceo-of-the-ftse250-oil-and-gas-corporation-afren-plc/>.
- Danmadami, A., I. Iyalla, G. Oluyemi, and J. Andrawus (2021). “Challenges of Developing Marginal Fields in the Current Climate of the Oil and Gas Industry.” In: *SPE Nigeria Annual International Conference and Exhibition*. Lagos, Nigeria. DOI: [10.2118/207207-MS](https://doi.org/10.2118/207207-MS).
- Department of Petroleum Resources (2003). *Guidelines for the Award and Operation of Marginal Fields in Nigeria*. Department of Petroleum Resources, Nigeria.
- Energy Voice (2019). “Nigerian pipeline problems drive hunt for alternatives.” In: *Energy Voice*. URL: <https://www.energyvoice.com/oilandgas/africa/208803/nigerian-pipeline-problems-drive-hunt-for-alternatives/>.
- Energy Voice (2020). “SFO confiscates millions from ex-Afren execs.” In: *Energy Voice*. Corroborated by UK Serious Fraud Office (2018) and Court of Appeal (2019).
- Energy Voice (2022a). “San Leon eyes final straight in closing Nigeria deal.” In: *Energy Voice*. URL: <https://www.energyvoice.com/>.
- Energy Voice (2022b). “San Leon’s Fanning sets out the OML 18 plan.” In: *Energy Voice*. URL: <https://www.energyvoice.com/>.
- Federal High Court of Nigeria (2021). *Aiteo Eastern Exploration and Production Company v. Shell Petroleum Development Company of Nigeria*. Suit No. FHC/ABJ/C8/738/2021.
- Federal High Court of Nigeria (2024). *Opu Nembe Kingdom v Aiteo Eastern Exploration and Production Company*. Suit No. FHC/YN/CS/284/2024, Federal High Court, Yenagoa (pending).

- Federal Republic of Nigeria (1969). *Petroleum Act, 1969 (Cap. P10, Laws of the Federation of Nigeria)*. URL: <https://lawsofnigeria.placng.org/laws/P10.pdf>.
- Federal Republic of Nigeria (2010). *Nigerian Oil and Gas Industry Content Development Act, 2010*. URL: <https://www.ncdmb.gov.ng/NCACT.pdf>.
- Federal Republic of Nigeria (2021). *Petroleum Industry Act, 2021*. Act No. 6 of 2021.
- Ferreira, N. N., R. Dziedzic, A. Burcharth, and M. R. Martins (2024). “Challenges in Inter-organizational Knowledge Transfer for the Life Extension of Oil and Gas Facilities.” In: *IEEE Transactions on Engineering Management* 71, pp. 5559–5573. ISSN: 0018-9391, 1558-0040. DOI: [10.1109/TEM.2024.3361797](https://doi.org/10.1109/TEM.2024.3361797). URL: <https://ieeexplore.ieee.org/document/10419043/> (visited on 06/18/2026).
- Ferreira, N. N., R. Dziedzic, G. P. Monteiro, C. P. Migueles, and A. Burcharth (2026). “How failures in interorganizational knowledge transfer impact process safety: Insights from the case study of an ageing offshore oil & gas facility acquisition.” In: *Journal of Loss Prevention in the Process Industries* 99, p. 105779. ISSN: 0950-4230.
- Field, F. and R. de Neufville (2021). *Thesis Definition and Preparation: Some General Guidelines*. MIT Technology and Policy Program. URL: https://ardent.mit.edu/wp-content/uploads/2021/07/thesis_manual_2021-FINAL-2.pdf.
- Gaughran, A. and J. Wilde-Ramsing (June 2024). “Fossil Fuel Industry Divestment and the Energy Transition: Lessons and Red Flags from Shell and the Niger Delta.” In: *Business and Human Rights Journal* 9.2, pp. 301–307. ISSN: 2057-0198, 2057-0201. DOI: [10.1017/bhj.2024.15](https://doi.org/10.1017/bhj.2024.15). URL: https://www.cambridge.org/core/product/identifier/S2057019824000154/type/journal_article (visited on 06/18/2026).
- Grant, R. M. (1996). “Toward a Knowledge-Based Theory of the Firm.” In: *Strategic Management Journal* 17.S2, pp. 109–122. DOI: [10.1002/smj.4250171110](https://doi.org/10.1002/smj.4250171110).
- Health of Mother Earth Foundation (2021). *Woes of the Wellhead*. Field report on the OML 29 Nembe (Santa Barbara) oil spill, with Environmental Rights Action/Friends of the Earth International. Health of Mother Earth Foundation (HOMEF). URL: <https://homef.org/2021/12/22/aietos-oml-29-well-01-blowout-an-ecological-horror-tale/>.
- Heritage Oil (2020). *Our Operations: Nigeria (OML 30)*. URL: <https://www.heritageoiltd.com/our-operations/nigeria/>.
- Idigbe, K. I. and K. O. Bello (2013). “Sustainable Operation of Marginal Fields in Nigeria: Opportunities, Challenges and Best Practices.” In.
- Iwuoha, V. C. (2021). “Rethinking the ‘patron-client’ politics of oil block allocation, development and remittances in Nigeria.” In: *Review of African Political Economy*; JSTOR 48654582. URL: <https://www.jstor.org/stable/48654582>.
- Kogut, B. and U. Zander (1992). “Knowledge of the Firm, Combinative Capabilities, and the Replication of Technology.” In: *Organization Science* 3.3, pp. 383–397. DOI: [10.1287/orsc.3.3.383](https://doi.org/10.1287/orsc.3.3.383).
- Kogut, B. and U. Zander (1993). “Knowledge of the Firm and the Evolutionary Theory of the Multinational Corporation.” In: *Journal of International Business Studies* 24.4, pp. 625–645. DOI: [10.1057/palgrave.jibs.8490248](https://doi.org/10.1057/palgrave.jibs.8490248).
- Leveson, N. G. (2011). *Engineering a Safer World: Systems Thinking Applied to Safety*. Cambridge, MA: MIT Press.

- Mohamed et al. (2024). “A new progressive and efficient production sharing Contract for upstream oil and gas industry.” In: Energy Geoscience / ScienceDirect S2949891024001039. URL: <https://www.sciencedirect.com/science/article/pii/S2949891024001039>.
- Myers, S. C. and N. S. Majluf (1984). “Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have.” In: *Journal of Financial Economics* 13.2, pp. 187–221. DOI: [10.1016/0304-405X\(84\)90023-0](https://doi.org/10.1016/0304-405X(84)90023-0).
- National Oil Spill Detection and Response Agency (2021). *Statement on the Aiteo OML 29 (Santa Barbara) blowout*. Cited via SDN (2022); statement not individually archived.
- Nelson, R. R. and S. G. Winter (1982). *An Evolutionary Theory of Economic Change*. Cambridge, MA: Belknap Press of Harvard University Press.
- Nigeria Extractive Industries Transparency Initiative (2020). *Oil and Gas Industry Audit Report 2020*. NEITI. URL: https://eiti.org/sites/default/files/attachments/neiti_2020_oil_and_gas_report.pdf.
- Nigeria Extractive Industries Transparency Initiative (2021). *Oil and Gas Industry Audit Report 2021*. NEITI. URL: <https://eiti.org/sites/default/files/2024-01/NEITI-OGA-2021-Report.pdf>.
- Nigeria Extractive Industries Transparency Initiative (2022). *Oil and Gas Industry Audit Report 2022*. NEITI. URL: https://eiti.org/sites/default/files/2024-09/Final_Report_NEITI_OGA_2022_Final_26_Sept_2024.pdf.
- Nigeria Extractive Industries Transparency Initiative (2023). *Oil and Gas Industry Audit Report 2023*. NEITI. URL: https://eiti.org/sites/default/files/2024-09/Final_Report_NEITI_OGA_2023_Final_26_Sept_2024.pdf.
- Nigeria Extractive Industries Transparency Initiative (2024). *Statement on the Review of Oil and Gas Asset Divestments*. NEITI statement reported by Vanguard, March 2025. NEITI. URL: <https://www.vanguardngr.com/2025/03/neiti-to-investigate-6-033bn-iocs-divestments-in-oil-industry/>.
- Nigerian National Petroleum Corporation (2020). *Explosion at NPDC Benin River Valve Station (BRVS)*. Press release, 8 July 2020; reported via Rigzone.
- Nigerian National Petroleum Corporation (2023). *Removal of Eroton as operator of OML 18*. Press statement, 6 March 2023; reported via Nairametrics.
- Nigerian Upstream Petroleum Regulatory Commission (2022). *NUPRC cancels marginal oil field awards to 33 companies*. NUPRC. URL: <https://www.nuprc.gov.ng/nuprc-cancels-marginal-oil-field-awards-to-33-companies>.
- Nigerian Upstream Petroleum Regulatory Commission (2024). *Nigeria Upstream Petroleum (Assignment of Interest) Regulations 2024*. Abuja, Nigeria: Nigerian Upstream Petroleum Regulatory Commission.
- Nightingale, D. J. and D. H. Rhodes (2015). *Architecting the Future Enterprise*. Cambridge, MA: MIT Press.
- Nonaka, I. (1994). “A Dynamic Theory of Organizational Knowledge Creation.” In: *Organization Science* 5.1, pp. 14–37. DOI: [10.1287/orsc.5.1.14](https://doi.org/10.1287/orsc.5.1.14).
- North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge: Cambridge University Press.
- North Sea Transition Authority (2023). *Change of Control*. Licensing guidance implementing the UK Energy Act 2023 (section 300 and Schedule 21); prior consent and licence-revocation powers, including fitness of incoming controllers. URL: <https://www.nstauthority.co.uk>.

- Nwuke, K. (2021). *Nigeria's Petroleum Industry Act: Addressing old problems, creating new ones*. Brookings Institution. URL: <https://www.brookings.edu/articles/nigerias-petroleum-industry-act-addressing-old-problems-creating-new-ones/>.
- Okonkwo, O. (2019). "Knowledge transfer in collaborations between foreign and indigenous firms in the Nigerian oil industry: The role of partners' motivational characteristics." In: *Thunderbird International Business Review* 61.2, pp. 183–196. DOI: [10.1002/tie.22012](https://doi.org/10.1002/tie.22012).
- Organization of the Petroleum Exporting Countries (2006). *Annual Statistical Bulletin 2006*. OPEC. URL: https://www.opec.org/opec_web/en/publications/202.htm.
- Organization of the Petroleum Exporting Countries (2022). *Monthly Oil Market Report*. OPEC. URL: https://www.opec.org/opec_web/en/publications/338.htm.
- Oriaku, T. O., N. A. Udo, and E. A. Ifeduba (Aug. 2, 2016). "Technical Assessment of In-Country Capabilities in the Oil and Gas Operations in Nigeria." In: *SPE Nigeria Annual International Conference and Exhibition*. SPE Nigeria Annual International Conference and Exhibition. Lagos, Nigeria: SPE, SPE-184346-MS. DOI: [10.2118/184346-MS](https://doi.org/10.2118/184346-MS). URL: <https://onepetro.org/SPENAIC/proceedings/16NAIC/16NAIC/SPE-184346-MS/192734> (visited on 06/18/2026).
- Pfeffer, J. and G. R. Salancik (1978). *The External Control of Organizations: A Resource Dependence Perspective*. New York: Harper & Row.
- Pigou, A. C. (1920). *The Economics of Welfare*. London: Macmillan.
- Rexer, J. M. (2025). "Corruption as a Local Advantage: Evidence from the Indigenization of Nigerian Oil." In: *American Economic Review* 115.3, pp. 1019–1057. DOI: [10.1257/aer.20221085](https://doi.org/10.1257/aer.20221085).
- Rui, Z., K. Cui, X. Wang, J.-H. Chun, Y. Li, Z. Zhang, J. Lu, G. Chen, X. Zhou, and S. Patil (Sept. 2018). "A comprehensive investigation on performance of oil and gas development in Nigeria: Technical and non-technical analyses." In: *Energy* 158, pp. 666–680. ISSN: 03605442. DOI: [10.1016/j.energy.2018.06.027](https://doi.org/10.1016/j.energy.2018.06.027). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0360544218310867> (visited on 06/18/2026).
- San Leon Energy (2017). *Annual Report and Accounts 2017*. URL: https://www.annualreports.com/HostedData/AnnualReportArchive/s/LSE_SLE._2017.pdf.
- San Leon Energy (2019). *Annual Report and Accounts 2019*. URL: https://www.annualreports.com/HostedData/AnnualReportArchive/s/LSE_SLE._2019.pdf.
- San Leon Energy (2022). *Proposed Transactions (RNS announcement)*. URL: <https://www.investgate.co.uk/announcement/rns/san-leon-energy-cdi---sle/proposed-transactions/6921786>.
- San Leon Energy (2023). *Full-Year Results Announcement 2023*. URL: <https://markets.ft.com/data/announce/full?dockey=1323-15872709-0NLGDC5VQ00D5TLHBG9N6G63JI>.
- Seplat Energy (2024). *2024 Integrated Annual Report*. URL: https://www.seplatenergy.com/media/hu4nnkzt/seplat-2024-integrated-annual-report-10042025_final.pdf.
- Shell plc (2024). *Sale of The Shell Petroleum Development Company of Nigeria (SPDC) to Renaissance*. Announced 16 January 2024; transaction completed 13 March 2025. URL: <https://www.shell.com>.
- SOMO (2024). *Shell's Reckless Divestment from the Niger Delta*. SOMO (Centre for Research on Multinational Corporations). URL: <https://www.somo.nl/shells-reckless-divestment-from-niger-delta/>.

- Spiff, O. and B. A. Oriji (2022). “The Effect of Capital Structure on the Performance of Marginal Oil Fields in Nigeria.” In: *International Journal of Economics and Financial Management* 7.3.
- Stakeholder Democracy Network (2020). *Delta Divestments: Full Report*. Stakeholder Democracy Network. URL: <https://www.stakeholderdemocracy.org/wp-content/uploads/2021/11/Full-report-Delta-Divestments.pdf>.
- Stakeholder Democracy Network (2022). *Divestment, Decommissioning, and the Future of Oil in the Niger Delta*. Stakeholder Democracy Network. URL: <https://www.stakeholderdemocracy.org/>.
- Steiner, R. (2023). *The Unclean Getaway: How International Oil is Exiting the Niger Delta*. SOMO (Centre for Research on Multinational Corporations). URL: <https://www.somo.nl/the-unclean-getaway-how-international-oil-is-exiting-the-niger-delta/>.
- Stigler, G. J. (1971). “The Theory of Economic Regulation.” In: *The Bell Journal of Economics and Management Science* 2.1, pp. 3–21. DOI: [10.2307/3003160](https://doi.org/10.2307/3003160).
- Sweet Crude Reports (2020). *Why Nigeria didn't benefit from 2003 marginal field bid rounds, DPR*. URL: <https://sweetcrudereports.com/why-nigeria-didnt-benefit-from-2003-marginal-field-bid-rounds-dpr/>.
- Trist, E. L. and K. W. Bamforth (1951). “Some Social and Psychological Consequences of the Longwall Method of Coal-Getting.” In: *Human Relations* 4.1, pp. 3–38. DOI: [10.1177/001872675100400101](https://doi.org/10.1177/001872675100400101).
- UK Serious Fraud Office (2017). *SFO charges two former Afren executives with fraud and money laundering*. Charges announced 27 September 2017; case relating to a \$300m Afren-Oriental Energy transaction. URL: <https://www.sfo.gov.uk>.
- UK Serious Fraud Office (2018). *Osman Shahenshah and Shahid Ullah sentenced to 30 years for Afren fraud and money laundering*. Convicted 24 October 2018; sentenced 29 October 2018, Southwark Crown Court. URL: <https://www.sfo.gov.uk>.
- Vanguard (2015). “Nigerian banks risk losing N31.45bn to exposure on Afren’s financing.” In: *Vanguard*. URL: <https://www.vanguardngr.com/2015/08/nigerian-banks-risk-losing-n31-45bn-to-exposure-on-afrens-financing/>.
- Vanguard (2017). “NPDC poised to grow equity production to 500,000 bpd by 2020.” In: *Vanguard*. URL: <https://www.vanguardngr.com/2017/09/npdc-poised-grow-equity-production-500000-bpd-2020/>.
- Williamson, O. E. (1975). *Markets and Hierarchies: Analysis and Antitrust Implications*. New York: Free Press.
- Wood Mackenzie (2026). *Nigeria Upstream Summary*. Lens Upstream commercial database. Wood Mackenzie.
- Yin, R. K. (2009). *Case Study Research: Design and Methods*. 4th ed. Thousand Oaks, CA: Sage Publications.
- Zahra, S. A. and G. George (2002). “Absorptive Capacity: A Review, Reconceptualization, and Extension.” In: *Academy of Management Review* 27.2, pp. 185–203. DOI: [10.5465/amr.2002.6587995](https://doi.org/10.5465/amr.2002.6587995).
- Zander, U. and B. Kogut (1995). “Knowledge and the Speed of the Transfer and Imitation of Organizational Capabilities: An Empirical Test.” In: *Organization Science* 6.1, pp. 76–92. DOI: [10.1287/orsc.6.1.76](https://doi.org/10.1287/orsc.6.1.76).